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Suzanne Gilkes

1000985

A university mobile application allowing students to quickly access important and helpful information about services within the university

BSc (Hons) Computer Science and Software Engineering

Undergraduate Thesis Report

Department of Computer Science and Technology – The
University of Bedfordshire

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AY12/13

ABSTRACT

The rise and development of the Smartphone has brought power in portability and computing functionality that has attributed to the rapid adoption of the mobile application. Students have embraced this technology wholeheartedly into their mobile lifestyle that they also expect mobile computing to play a substantial role in their academic learning environment. This dominance in the usage of Smartphones in conjunction with the demand for mobile applications which focus on students' needs creates a unique opportunity for educational institutions to deploy such mobile applications.

The University of Bedfordshire has explored the potential of transforming students' learning experience through mobile learning in meeting its strategic plan by deploying the "BREO to Go" mobile application which supports formal learning. However research has indicated that informal learning has a stronger leverage over formal learning as part of a mobile learning method in creating a transformative learning experience.

Moreover, a mobile application which supports informal learning has the ability to bridge the gap between formal and informal learning. Although the university has also deployed two mobile applications which support informal learning, it currently lacks a mobile application which supports informal learning whilst capitalizing on this ability. In order to fulfil this gap, this project illustrates the development of a mobile application to this end and in doing so addresses a current student need to access important and helpful information about university services.

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special emphasis of the first year students who utilized the Peer Assisted Learning scheme that participated in my focus group sessions. As a result I would like to express my appreciation to these students for their input in fostering innovation, a new approach for resolving a current student need and creating an idea which could help the university in meeting its strategic plan.

DEDICATION

This dissertation is dedicated to my loving parents, Philip and Yvonne Gilkes for their unconditional emotional support and belief in my abilities throughout all of my academic pursuits.

KEY WORDS

Mobile learning, informal learning, mobile applications, informal mobile learning framework, Smartphones, Higher Education, Extreme Programming, Mobile Development System Development Lifecycle

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CHAPTER 1: INTRODUCTION

1.1 Background

The University of Bedfordshire's strategic plan which encompasses its mission, values and vision is built on providing an environment which enables students to benefit from a transformative learning experience (University of Bedfordshire, no date a). It is evident that the university has already explored the potential of transforming students' learning experience through mobile learning with its deployment of the "BREO To Go" mobile application (University of Bedfordshire, no date b). Mobile learning creates a transformative learning experience as it is defined as "the acquisition of any knowledge and skill through using a mobile device, anywhere, anytime that results in an alteration in behaviour" (Geddes, 2004, p. 214).

Mobile applications are designed to meet a particular user need (Initiative Media, 2010) and used for mobile learning (Botzer and Yerushalmy, 2007) because they run on mobile devices. They can support formal learning typically associated with in-class activities also referred to as formal mobile learning. In contrast, mobile applications can also support informal learning associated with outside-class activities (Khaddage and Knezek, 2011) which is referred to as informal mobile learning.

With the deployment of the "BREO To Go" mobile application which supports formal learning, the university recognized that this could negatively impact on its ability to follow its strategic plan. Research has indicated that formal learning is not as effective as informal learning because the latter is seen as the most valuable type of learning (Khaddage and Knezek, 2011). In order to utilize informal learning as part of a mobile learning method, the university has also deployed two mobile applications, "University of Bedfordshire" and the "UoB Learning Resources App" (University of Bedfordshire, no date d) which support informal learning.

Parsons (2013, p. 197) references (Cochrane, 2011) who in citing the works of (Laurillard, 2007), highlights there is the potential for informal mobile learning to “bridge pedagogically designed learning contexts, facilitate learner generated contexts, and content, both personal and collaborative while providing personalisation and ubiquitous social connectedness, that sets it apart from more traditional learning environments such as fixed classrooms and computer labs”. However the university currently lacks a mobile application which supports informal learning whilst bridging the gap between formal and informal learning.

It has been demonstrated that informal learning has a stronger influence over formal learning as part of a mobile learning method in creating a transformative learning experience. It is therefore reasonable to question why the university has not been influenced to create a mobile application that has a dual purpose in which it supports informal learning whilst bridging the gap between formal and informal learning.

1.2 Project Goal

The goal of this project is to develop a university mobile application which supports informal learning whilst bridging the gap between formal and informal learning.

1.3 Project Context

The concept of the Smartphone represents the emergence of a sophisticated technology that drives mobile computing. According to Scott Steinberg, editor at Digital Trends “a Smartphone is essentially a computer in your pocket. It is a cellular phone that does more than just make calls to the point that it can serve as a functional laptop or desktop replacement” (Digital Trends Inc, 2012). This is essentially the reason why “mobile computing has been described as the third digital revolution” (Educause, 2011, p.7). Smartphones have a wide range of functionality as these devices allow users to “access Google maps, make phone calls, take pictures, record audio and video, store data, music, movies, and interact with the internet” (oMbiel Limited, 2010, p.4).

All of these functionalities of the Smartphone have become so interwoven into student lifestyle (oMbiel Limited, 2010) that students' habits are changing as a result of their adoption of Smartphones as part of their mobile lifestyle. "From the moment they wake up to the moment they go to bed, students are constantly connected to their electronics, synching their Smartphones with all of the latest updates that came in while they caught their precious shuteye. 83% of young people sleep next to their cell phones; 35% boot-up apps before getting out of bed" (Melton and Kendall, p.3).

These statistics signify the dominance in usage of Smartphones by students as part of their mobile lifestyle and this promotes the ideal environment for the scope of this project because the proposed mobile application which supports informal learning will match the students' mobile lifestyle.

1.4 Project Objectives

The following project objectives have been modelled against the facilitating factors that affect the acceptance of a Smartphone application as indicated within the project inception. These objectives are also closely aligned with the best practices and insight for the development of a university mobile application as identified by the professional community.

- To capture requirements for the proposed system through the use of focus groups.
- To design a mobile application that is accepted by students, attracts positive student feedback and promotes a mobilely fluent campus thereby fostering continuous use by students.
- To develop a mobile application that has genuine useful functionality relevant to its target audience.
- To develop a mobile application that is attractive, very easy to use and intuitive.

- To provide personalised and highly relevant information which students like to have at their fingertips.
- To develop a mobile application embodying a direct alignment with the university's strategic plan.

1.5 Techniques for Realization

1.5.1 Research Methodology

The research process will be guided by the circulatory and evolutionary approaches, utilizing qualitative methods to realize the artefact. The use of qualitative methods will involve the adoption of techniques such as user surveys and focus groups.

The first phase involves a description of the background of study, definition for the basis of conducting the study as well as an outline of the project goal, context and objectives.

The second phase will establish the validity of the development of the proposed system. An investigation will be conducted based on the analysis of a case study about the different aspects affecting the acceptance of the artefact. This investigation will be conducted through the use of user surveys to identify the level of user acceptance of the proposed system. The results will be used in the literature review to justify the need for the project.

The third phase will be based on the implementation and evaluation of the artefact using focus groups and will aim at making a discovery. Best practices and insights identified by the professional community will also be utilized to provide a balanced and in depth look into developing a mobile application for an educational institution.

The final phase of the research process will draw conclusions based on the research findings and their alignment with the project goal and objectives outlined. It will also evaluate and highlight opportunities for further development of the artefact.

1.5.2 Development Methodology

This project will utilise a software development methodology because it structures the software development into a defined process model. This model can help support planning and control, provide progress visibility, divide the project into manageable chunks at each stage, serve as a means of evaluating the final system ensuring software quality as well as lead to better coding and documentation (Dawson, 2009).

There are several development methodologies a software engineer can choose from, with each having its own distinct advantages and disadvantages. Extreme Programming which is a relatively new methodology introduced in the 1990's has been selected as the best candidate amongst the various software development methodologies to develop the proposed system. It will be used in combination with the design phase of the Mobile Development SDLC (System Development) Lifecycle. The Mobile Development SDLC is dedicated for mobile development and can be used in a number of agile methodologies (Xamarin Inc, 2013).

Extreme Programming is a highly flexible, emerging customer focused development methodology which focuses on high speed incremental delivery. It is advantageous over other strict approaches such as the Waterfall approach which relies on heavy documentation without focusing on the product to be delivered. Utilization of the Waterfall approach can result in "documents that may be perfect but the actual product is flawed, broken or unusable" (Baird, 2003, no page). Moreover the Waterfall approach does not effectively handle dynamic requirements and any changes in requirements can result in significant re-planning.

Usage intention is one of the key factors that affect Smartphone application acceptance for practical mobile applications and it has a direct impact on recommendation intention. Usage intention is very much connected to actual use (Lee, Kim and Choi, 2012). It is therefore essential that regular feedback is received on the user experience with regards to the design and usability of the proposed system. Changing student habits will also reflect in and affect the creation of the proposed system. Therefore frequent consultation with the student body to ascertain the requirements for the mobile application will be vital. Extreme Programming

allows for such change during the project lifecycle as it accommodates constant feedback on the product and improves quality through the early resolution of defects (Baird, 2003).

Getting the minimal viable working piece of software produced to customers as soon as possible as opposed to developing a final product is one of Extreme Programming's top priorities (Baird, 2003). This project will capitalize on such features not only to meet the strict deadlines enforced in having a piece of working software but also to ensure accuracy of requirements, maintainability of project scope and robustness of the proposed system.

1.5.3 Development Creation

The mobile application will be developed via the Corona SDK development platform which utilises the Lua programming language. This mobile development platform is an excellent option for application developers to build high-quality mobile applications. It has been strategically selected for the development of the artefact for this project because it is easy to use.

Using Corona SDK will also boost my productivity and allow me to develop the mobile application faster within the short time frame of five weeks required for development. Moreover, it stands out as the “best solution for developing cross-platform mobile applications for all major platforms and devices” with the ability to “write once and build to devices such as Kindle Fire, Nook and iOS” (Corona Labs Inc, 2013, no page, a). It has the distinct advantage supporting cross platform app development as Corona “automatically chooses the right assets based on screen resolution and will scale your content, depending on the screen size of a user's device” (Corona Labs Inc, 2013, no page, a). Creating a cross-platform mobile application without additional development work would also be very beneficial should the university decide to undertake further work on the proposed mobile application and release it on several other platforms.

An integral feature of using Corona SDK that blends well with the type of mobile application created for this project, an educational mobile application, is that although Corona is “renowned for game creation, it is an equally powerful tool for other app types including business mobile apps, utility apps, educational apps and eBooks” (Corona Labs Inc, 2013, no page, a).

1.6 Structure of this Report

1.6.1 Introduction

The purpose of this introductory chapter is to set the scene for the project by shaping the theme and purpose of the project. The reader’s perspective is directed towards the organization under examination for the project. A thorough analysis of the background about the University of Bedfordshire and its current approaches used to achieve its strategic plan is presented in order to identify a critical gap. Specifically the current mobile learning methods adopted by the University of Bedfordshire will be reviewed to evaluate their effectiveness in meeting its strategic plan.

The identification of this gap will support the need for the project which is to develop a university mobile application which supports informal learning whilst bridging the gap between formal and informal learning. Related research in the field will be analyzed and a comprehensive review about the context in which the project takes place will be outlined.

The purpose of the project is clearly constructed through the formulation of a project goal and objectives used to measure the success of the project. This chapter also defines the development creation, the research and development methodologies utilized to realize the artefact as well as the structure of the report.

1.6.2 Literature Review

The second section is presented in a series of linked chapters that form the foundation of the project. It comprises a review of the literature on the subject matter

with emphasis on critical evaluation of the theories and debates on the subject. The literature review provides a rich vision and shapes the readers' expectations by establishing how this evaluation fits in a broader context and is employed to build a coherent argument with the use of investigations, in order to help construct a theory that is fully justified.

The first part of the literature review contains an introduction to its purpose. It also contains a synopsis on the following: - the rise and development of the Smartphone, the key drivers affecting growing interests of Smartphones amongst the student population as well as opportunities for universities to leverage mobile applications which support informal learning to their students. This approach is used to present a collection of interrelated scholarly supported judgements and conclusions that firmly justify the existence of the project.

1.6.3 Artefact Design and Development Strategy

This section contains the strategies used to plan the requirements, design, development, testing and evaluation aspects of the artefact.

1.6.4 Requirements Gathering

This chapter will provide the reader with a walk-through of all focus group sessions conducted with a Peer Assisted Learning (PAL) group in which requirements for the artefact were gathered. This will collectively include activities integrated in requirements gathering within the relevant Extreme Programming development lifecycle phases. In sequential order these phases and activities include Exploration and Planning where user stories are used to gather the customer requirements using the User Goal Oriented Design approach. Following this is the creation of the Release phase where user stories are written at a finer level of detail, estimation of the size of the release and number of iterations is conducted by the developer. The prioritization of user stories and writing of the initial release plan with the use of user interface prototypes by the customer is also established.

1.6.5 Artefact Development and Testing

This presents all the development activities undertaken in iterations to create the artefact. Based on the release plan, this section will comprise of work done on user stories based on their business value, highlight the breakdown of tasks for development work, selection of any failed acceptance tests as well as task estimation. This section will provide the reader with a breakdown of the development tasks required for each iteration which includes writing unit tests, coding and integration. Functional testing will also be executed by the developer for evaluating system functionality prior to integration.

It will also outline the process of executing the unit tests and having the customer write acceptance tests per each iteration. Within each iteration, tracking is conducted and the results of acceptance tests run by customers will be documented in which they will evaluate whether new functionality and usability aspects passes these tests. The evaluation of the system via acceptance testing marks the closure of the iteration.

1.6.6 Evaluation

The chapter provides a comprehensive evaluation of the artefact developed based on the acceptance testing performed by the customer. An analysis of these test results will be carried out to identify whether the artefact developed is fit-for-purpose, has met the requirements of the customer and has a high level of usability. The system will be reassessed for its level of acceptance by the customer. It will also be benchmarked against best practices in the industry for creating a mobile application for an educational institution.

1.6.7 Conclusions and Further Development

The final chapter presents an evaluation of the project with an emphasis on whether the initial aims and objectives have been met. The appropriateness of the development methodology and suitability of the programming languages utilized will also be discussed. This section will also feature lessons learned that will

identify what could be done differently and overall what I have learnt and experienced from completing the project. The concluding chapter will also determine whether the project is viable for further development. Recommendations for further work based on the artefact's comparison with best practices and insights identified by the professional community will also be highlighted.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction to the literature review

This study begins with a review of the relevant literature on the evolution of the Smartphone, the reasons behind its popularity amongst the student population and the opportunities universities have to leverage mobile applications to their students. A comprehensive critical evaluation of this literature sets the foundation for this project by defining clear linkages between the study's major themes and shapes the readers' expectation by establishing how this evaluation fits in a broader context. This strategic approach is adopted with the use of investigations conducted to build a rational and persuasive argument in order to strongly defend and fully justify the need for this project.

2.2 The rise and development of the Smartphone

From the introduction to the world's first media phone for internet access, the Nokia 7110 in 1999, to the emergence of 4G network Smartphones such as iPhones and Android, this illustrates that developments in mobile technology have evolved at an incredible rate. Over the past fifteen years, there have been rapid advances in mobile technology leading to the advancements of mobile phones and the evolution of the Smartphone (oMbiel Limited, 2010). Initiative Media (2010) referenced this fast growth of the Smartphone sector of the mobile phone market when citing the works of (Gartner, 2010) referenced by (Biz-news.com, 2010) which describe "Smartphones sales across the world showing a 49% increase in Q1 2010 compared to Q1 2009, resulting in a 17% rise in all mobile phone sales".

It is the Smartphone's power in portability and computing functionality that has attributed to the rapid adoption of the mobile application. People's views of mobile handsets have shifted from them being used for voice, text and email to one that is a catalyst for change through leveraging a platform, a mobile application, for launching a never ending array of new functionality that addresses their needs (Initiative Media, 2010). A study conducted in the United Kingdom has revealed that "downloading applications was the second fastest growing activity with 1 million new users in the third quarter to 4.1 million of the 6.2 million Smartphone users in the 4th quarter" (oMbiel Limited, 2010, pg. 2).

2.3 Key drivers affecting growing interests of Smartphones amongst the student population

As students become increasingly reliant on Smartphone devices, they "come to campus today with consumer devices and consumer expectations, and having embraced the convenience and utility of mobile devices in their non-academic lives, they want and expect mobile computing to play a similar role in their learning" (Educause, 2011, pg. 7). For this reason students are motivating the drive behind the adoption of mobile computing in higher education (Educause, 2011). "Today, students demand university services on the move so they can receive alerts and get the latest news and events whilst on the move" (oMbiel Limited, 2010 p. 4).

According to Hugh Griffiths, CEO at oMbiel, Birmingham, "universities have a lot of information to disseminate to students, some of which is general such as news, events and PC utilization and are using various means of trying to deliver this content using portals, Web sites and big TVs, but the most effective way to deliver this content is directly to the phone" (Crook, 2009). Research has indicated that "mobile applications focused on the needs of students are in the greatest demand" (Educause, 2011, p. 15).

2.4 Opportunities for universities to leverage mobile applications which support informal learning to their students

Informal learning is associated with “activities that support learning outside a dedicated learning environment and formal curriculum” (Naismith, Lonsdale, Vavoula and Sharples, 2004, p. 3). It can be described as learning that is authentic as it is “intentional, where the goals and processes of learning are explicitly defined in advance by the individual or unintentional where the goals and the learning processes develop as a learning opportunity arises” (Rahim, Chee Pee and Othman, 2012, p.43). Parsons (2013, p. 220) which references the works of (Kukulska-Hulme, Pettit, Bradley, Carvalho, Herrington, Kennedy and Walker, 2011) advises that it is important that learners be actively involved and shapers of their own learning. “They should be supported in using technologies of their own choice where appropriate” (JISC, 2009, p. 51). Having a mobile application which support informal learning may potentially place universities at the forefront of the pedagogical practice as today’s student look for engagement, involvement and flexibility to have options and choices in the way they learn (Khaddage and Knezek, 2011). Moreover according to Khaddage and Knezek (2011, p. 240), informal learning “accounts for over 75% of learning taking place in institutions today”.

Parsons (2013, p. 319) makes reference to the works of (Bradley and Holley, 2011) which cites the works of (Schuck, Aubusson, Keamey and Buden, 2010) who evaluates that “given the ubiquity of mobile devices, an imperative has arisen for educators in higher education to familiarise themselves with the affordances of mobile technologies for learning so that they are able to capitalise on their students’ usage of these devices for effective learning”.

Additionally, according to Parsons (2013, p. 33) in referencing (Pollara and Broussard, 2011) which cites the works of (Fetaji, 2008), “as non-traditional methods of education become more prevalent and thus, informal flexible learning environments become necessary for students in an ever-connected society, research of mobile learning will play a significant role in determining if institutions can support 21st century needs”.

It is therefore evident that there are clear indicators why universities should leverage mobile applications which support informal learning to their students because not only should universities support students' learning based on their choice of technologies such as mobile applications. Students' choice of mobile applications combined with the dominance of informal learning in educational institutions and the positive effect of this type of learning for both students and universities create a unique opportunity for universities to deploy mobile applications which support informal learning.

2.5 Justification for the development of a mobile application which supports informal learning for the University of Bedfordshire

The literature review has highlighted that the rise of the Smartphone has led to the rapid adoption of the mobile application. It has also indicated that the increased pressure the student population has put forward for mobile applications catering to their needs and the potential opportunities universities have to leverage mobile applications which support informal learning, has created a unique prospect. This prospect points to the need for the University of Bedfordshire to create a mobile application which addresses the students' needs with the support of informal learning. However the question remains why the university has not yet already implemented the proposed system. Perhaps there is uncertainty about its acceptance amongst students upon its release? Or maybe the university feels that there is no need for the proposed system?

The answers are found in the analysis of the User Acceptance Survey. A copy of this survey which can be found in **Appendix C-1** will be instrumental in meeting the second project objective. Within Extreme Programming, before requirements are gathered with user stories, a project inception must first be conducted so that enough information can be gathered to support the justification that the project should proceed. User surveys were used to discover current user practices, needs and attitudes to the proposed system in order to evaluate the user acceptance criteria of

the proposed system. According to Maguire and Bevan (2002, p. 5), “user surveys involve administering a set of written questions to a sample population of users.

Surveys can help determine the needs of users, current work practices and attitudes to new system ideas. Surveys are normally composed of a mix of 'closed' questions with fixed responses and 'open' questions, where the respondents are free to answer as they wish”. As a method of user needs identification, the use of user surveys allows for statistical analysis and is a quick method of determining preferences of large groups of users (Maguire and Bevan, 2002). However, Maguire and Bevan (2002), also reveals that a relative disadvantage of adopting user surveys is that they do not allow the capturing of in depth comments and may not permit follow-up.

Brockman, Stieglitz, Kmiecik and Diederich (2012, p. 863) provide a definition of user acceptance when citing the works of Kaasinen (2005) which can be defined as the “demonstrable willingness within a user group to employ information technology for the tasks it is designed to support”. Roslina, Fahmy and Haslinda (2011, p. 215) referenced the importance of meeting user acceptance criteria when citing the works of (Adams, Nelson and Todd, 1992) as “user acceptance is an important indicator in the success of a computer system”.

According to the research model of Lee, Kim and Choi (2012) which was hypothesized and empirically tested based on the Unified Theory of Acceptance and Use of Technology Model (UTAUT) to analyze the factors affecting the use of Smartphone applications, there are six factors that affect Smartphone application acceptance which include credibility, personalization, performance expectancy, effort expectancy, recommendation intention and usage intention. Credibility has not been taken into account as a factor in the acceptance criteria of the proposed system because it relates to the security of e-commerce transactions performed by the mobile application and this is not applicable to the nature of the system.

A copy of the User Acceptance Survey findings can be seen in **Appendix B**. Twenty-Five surveys comprising of ten questions were projected as anonymous and elicited to the student body. All surveys were completed. The gender makeup of students was proportionally balanced with twelve females and thirteen males completing the survey as shown in **Figure 1**. Additionally, a cross spectrum of responses ranging from the various subject areas was well-proportioned as shown in **Figure 2**. The majority of responses came from the undergraduate student population with fourteen undergraduate students and eleven postgraduate students completing the surveys as indicated in **Figure 4**. Furthermore the distribution of participants by student type was marginal with 44% home students and 56% overseas students. This is shown in **Figure 3**.

One of the key factors that affect mobile application acceptance for practical mobile applications is usage intention (Lee, Kim and Choi, (2012)) which is “the intention of continuous and regular use of the Smartphone application in the future” (Lee, Kim and Choi, 2012, p. 30). Usage intention also has a direct affect on recommendation intention which is the “degree of recommendation willingness of the Smartphone application” (Lee, Kim and Choi, 2012, p. 30).

According to the analysis of the survey, the proposed system already has a high chance of acceptance as 72% of respondents said they would prefer to regularly use the proposed system in the future rather than any other means available based on its description. This is indicated in **Figure 11**. Moreover 92% of participants have stated that based on its description they would recommend it to other students. This is indicated in **Figure 12**. Although this was observed and suggested a positive indicator for the acceptance of the mobile application, it is important to note that usage intention is strongly connected to the actual use of the mobile application (Lee, Kim and Choi, 2012).

In order to arrive at an accurate determination of usage intention and reassess recommendation intention, a survey will be administered to students after they have

used the developed mobile application. Personalization was also taken into account as the proposed mobile application will integrate the personalization of information, navigation and presentation. However, none of the participants indicated any additional features they would have liked to see implemented for personalization.

The majority of participants (36%) also highlighted that they would find the implementation of the proposed system as extremely useful as shown in **Figure 10**. This was strengthened by the fact that students already engage frequently and find very useful a number of informal mobile learning activities relating to the scope of the proposed system. This is indicated in **Figure 7 and Figure 8**. Moreover participants also felt that there were several perceived advantages of using the proposed system as indicated in **Figure 9**. Furthermore 72% of students felt that there was a real need for the proposed system. This is indicated in **Figure 11**. These answers were important because they addressed the project goal of the proposed system. If students' answers were not positive about the usefulness, did not perceive any advantages of the proposed system and avoided expressing a need for the proposed system, the development of the proposed system would be proven pointless.

It has been established that students already engage frequently and find very useful a number of informal mobile learning activities which are based on the informal mobile learning framework. The informal mobile learning framework is founded on the use of mobile applications to support informal learning and has several categories of informal mobile learning activities. These include activities that are referential which involve accessing materials via the mobile device, location aware used to interact with the environment, reflective which involves reflecting upon material, data collection used to recording data and information about the environment and maps, constructive in which learners create or construct knowledge and administrative used to support organization of the user's life (Clough, Jones, McAndrew and Scanlon, 2007).

Khaddage and Knezek (2011) references such mobile applications that can be used to empower informal learning which include social networking platforms such as Facebook; Mental Note to type, draw and add notes; iStudiez Pro for built-in planners to input schedules as well as tracking workload, InClass for organizing, sharing notes and saving Word documents; Audio Note to write notes, setup recording and audio recording with time stamp; Kindle for reading and downloading books; PenUltimate to store information in separate notes allowing students to email and share notes taken; Pages for word processing programs, synching documents or for creating new ones.

So why should the university create a mobile application which supports informal learning if mobile devices can be used to achieve this through other mobile applications already installed or mobile applications which run via the web? According to Lionbridge (2012, p. 4), “a native mobile application is built specifically for a particular device and its operating system. A native app is downloaded from a web store and installed on the device”. Whereas a “mobile web application is a web application formatted for Smartphones and tablets and accessed through the mobile device’s web browser” (Lionbridge, 2012, p. 3). In essence should the university employ the strategy of creating a mobile web application or a native mobile application?

What should influence the approach the university employs in creating a mobile native app is revealed by the case presented by John Lewis, chief software architect of the consulting group Unicon Inc, who argues that “a lot of users these days are not going to go find the mobile website for something they are interested in using. They are going to expect to be able to go to the app store or the Android market” (Roscorla, 2011). Such mobile applications take advantages of the best of both worlds as they can be used for mobile learning which can help the University of Bedfordshire follow its strategic plan whilst addressing the needs of its students to have access to university information on-the-go. It should also be distinguished that native mobile applications are faster and are more likely to work without internet connectivity.

According to Lionbridge (2012, p. 5) since “the application remains installed on the device from the original download, no internet connection is required. Users get peak performance at all times, with all graphics, images, scripts and data”. These types of applications also have a powerful influence over “front of mind” penetration because native mobile applications place a logo in the application list screen, providing visibility on a daily basis (Lionbridge, 2012). In addition, app stores remind users to upgrade apps, so these apps update frequently” (Lionbridge, 2012, p. 5) allowing for easier maintenance of the mobile application by the developer.

Moreover native mobile applications have a distinct advantage over web apps because they have the “ability to leverage device-specific hardware and software” (Lionbridge, 2012, p. 4) which helps to deliver a “richer, more compelling user experience” (Lionbridge, 2012, p. 5). Native mobile applications can “take advantage of location awareness and other native capabilities of phones and will allow the university to provide features that may not necessarily be available on a website or portal” (Roscorla, 2011, no page).

According to Lionbridge (2012, p. 5), “native apps can leverage the capabilities of the mobile device, including onboard hardware such as GPS, camera, graphics as well as software such as email, calendar, contacts, picture/ video gallery, file manager, and home screen widget areas”. The proposed mobile application will invest in such features by allowing students to use the camera for enhancing their student learning experience. It will also incorporate email to allow students to take notes via the mobile application and send these notes to their email account.

There is also an option to create a hybrid app which is a native app with a web app inside (Lionbridge, 2012). This will also be integrated in the proposed mobile application which will facilitate the viewing of up-to-date Twitter feeds on alerts from campus based services as well receiving up-to-date events hosted by the Student Union. As a result of this the complexity of the proposed mobile application would be reduced as it will “provide the look and feel of a native app but delivers the heart of the application within the browser” (Lionbridge, 2012, p.12).

It is also important to note that whilst in pursuit of a mobile development strategy, universities should also understand what devices their community is using if they go the app route (Domonell, 2012). It was highlighted that the level of ownership of Smartphones by students was extremely high as 96% of students stated that they owned a Smartphone as indicated in **Figure 5**. Moreover it was evident that the Android model of Smartphone was dominant in the participants' ownership with 40% of participants owning an Android Smartphone as indicated in **Figure 6**. It has already been established that the Smartphone has attributed the rapid adoption of the mobile application. These factors combined create the perfect environment for the university to create a native mobile application for the Android Smartphone device.

The development of a university native mobile application for the Android Smartphone which allows students to engage in informal learning activities in order to meet their needs to quickly access important and helpful information about services with university, will not only be established as a general purpose mobile application. However what will make this mobile application stand out is that it will also be used to support formal learning through informal mobile learning activities. According to Khaddage and Knezek (2011, p. 241), "by integrating mobile applications such as social networking platforms, students can be given a task to complete, so that they can use certain technologies and applications such as Facebook and Twitter. They may be asked to research a topic or discuss a specific area and find information, then bring it back to class to share with others. This method will teach students how to use their surroundings and their environment to generate ideas and grant them some flexibility and portability in the ways they learn".

Parsons (2013, p. 179) also highlights the benefits of using such collaborative types of informal mobile learning activities when referencing the works of (Nouri, Cerratto-Pargman, Eliasson and Ramberg, 2011) which cites (Mevarech and Kramarski, 2003; Nichols, 1996; Roseth, Johnson and Johnson, 2008) stating that "as a pedagogical technique, collaborative learning can be considered as an effective strategy to promote student achievement, higher order thinking, argumentation and

explanation skills, autonomy, interdependence, retention, problem-solving, self-regulation, and use of meta-cognitive strategies”.

The proposed mobile application will have a Facebook presence featuring social functionality through the use of Facebook group pages which allow students to share meaningful information and solve problems regarding their academic studies within the formal curriculum in order to enhance their understanding of the subject matter in a collaborative environment. Bradley and Holley (2010, p. 1) references that having “valuable insights into students’ practice and attitudes towards mobile learning, help us to design effective mobile learning activities that capitalize on what they already do, and can potentially do with their mobile phones”. The proposed mobile application will capitalize on informal mobile learning activities students already engage in based on User Acceptance Survey such as taking notes outside of class to support their own learning and using Facebook on their Smartphone to connect with other university students in order to share their knowledge.

A case study which explores how mobile phone cameras can support teaching and learning during a lesson which was referenced in Parsons (2013) when citing the works of Ekanayake and Wishart (2011) indicates that the camera function of mobile phones offers a wide variety of possible benefits as it “encourages student participation, collaboration and understanding of concepts”. “The use of the mobile camera not only provided a rationale for students to go into the field and capture the relevant images” (Parsons, 2013, p.174), “the images added new flavours to the learning activity but also provided additional opportunities for student participation” (Parsons, 2013, p.175).

According to Parsons (2013, p. 32) which cites the works of (Pollara and Broussard, 2011), “eleven studies that focused specifically on student learning outcomes and processes in order to better understand the direction of mobile learning in mainstream education were found to be positive and indicated several benefits of using mobile devices for learning which included an increase in achievement, productivity, engagement and motivation”.

The integration of the outlined mobile applications which support informal learning is fully supported by the works of Khaddage and Knezek (2011, p. 236) who acknowledges that “mobile application integration which promotes collaboration by informal learning applications can form a bridge between informal and formal learning”. Bridging this gap between informal and formal learning can help construct “an effective and solid educational infrastructure that can be acceptable by the majority of the millennial and digital students of today’s society” (Khaddage and Knezek, 2011, p. 238).

In conclusion, the University of Bedfordshire should not delay in developing a mobile application that has a dual purpose in which it supports informal learning whilst bridging the gap between formal and informal learning, as such a mobile application can be more effective in helping the university to meet its strategic plan. The implementation of this project will seek to achieve this in order to meet its final objective.

Moreover, it is already evident that the university has seen the need to sponsor these types of projects which further support the justification of the creation of the proposed mobile application as they are willing to invest in providing “students with a university app for Smartphones and tablets showing: - personalised timetables, campus maps and directions, key university data, email access and system access” (University of Bedfordshire, 2013, p.8).

CHAPTER 3: ARTEFACT DESIGN AND DEVELOPMENT STRATEGY

3.1 Requirements Strategy

This project will utilize the approach of gathering customers' requirements with user stories following the User Goal Oriented Design approach within focus group sessions. The success of the proposed system will be strengthened through the implementation of this qualitative method as it will allow customers to express in their own words the requirements for the system via oral communication. This is a powerful tool for a developer as it assists in having a clear understanding of the requirements of the proposed system which is a major challenge faced by software developers (Baird, 2003).

3.2 Design Strategy

The design phase of the Mobile Development SDLC will be used to create a draft artefact design. Extreme Programming does not have a design phase and acknowledges that software is a continuous state of design. This feature of Extreme Programming will be followed in the project.

3.3 Testing Strategy

“Central to Extreme Programming software development is the reversal of the test cycle where programmers write tests before they develop their source code” (Baird, 2003, no page). Based on this within the development of the proposed system, unit tests will be written before code for artefact is developed. These unit tests will be executed after coding. Functional testing will also be executed by the developer for evaluating system functionality.

Thereafter acceptance tests will be written by the customer and used to evaluate whether the artefact has a high level of usability and is being developed in accordance to the customer's requirements. Any failed acceptance tests will be fed into the next iteration cycle. Test plans will be used within unit tests, functional and

user acceptance tests. These test plans will identify what aspects of the program will be tested, inputs made, how the system is expected to perform and how it actually performs (Dawson, 2009).

3.4 Evaluation Strategy

The customer will evaluate the artefact by executing the acceptance tests they have written. The success of the proposed system will be determined by whether it is fit-for-purpose indicative of its ability to meet the requirements of the customer and has a high level of usability. A reassessment for the level of acceptance of the mobile application by the customer will also be conducted in order to support its success.

CHAPTER 4: REQUIREMENTS GATHERING

4.1 Overview of the Extreme Programming Development Lifecycle

The Extreme Programming (XP) development lifecycle has five phases which include: - Exploration, Planning, Iterations, Productionizing and Maintenance (Baird, 2003). The development of the mobile application for this project will be implemented using the Exploration, Planning and Iterations phases of the XP development Lifecycle. Moreover, fundamental to XP is its establishment of a unique set of values supported by principles, activities and practices and these will be appropriately correlated with the inner workings of each of the XP development phases employed in this project.

4.2 The Exploration Phase

4.2.1 Overview of the Exploration Phase

Exploration is the first phase of the XP development lifecycle and it is within this phase that the requirements for the artefact are gathered. The requirements gathering process involves “customers and developers working together to explore technical

options, define requirements, and complete a list of user stories” (Baird, 2003, no page).

4.2.2 The importance of project inception for requirements gathering

The project inception conducted in the literature review is a fundamental predecessor of requirements gathering forming part of the Exploration phase, as it is where the customer provided meaningful and critical input with regards to the “discovery of the system vision statement” (Baird, 2003, no page). A vision statement “is a high-level statement that describes the goal of the system” (Baird, 2003, no page) which in the context of this project is to produce a university mobile application allowing students to quickly access important and helpful information about services within the university. An evaluation of this statement by students within the project inception revealed that there is a need for the system thereby justifying the existence of a business case which drives the reason for the project (Baird, 2003). Moreover the system has a potentially high level of acceptance which further enhances its feasibility.

The vision statement is a critical component of development because it progresses into the formulation of a system metaphor. The system metaphor forms part of one of XP practices and it refers to “how the developer conceptualizes the system and is typically written in the language that is relevant to the business domain” (Baird, 2003, no page). The system metaphor for this project is to develop a mobile application which supports informal learning whilst bridging the gap between formal and informal learning.

With the customer defining the value of the system through the vision statement, this statement is used to help guide the customer with the creation of high-level user stories. According to (Baird, 2003, no page), a user story “is the tool that captures user requirements. User stories are similar to use cases and are written by the customer in nontechnical language”.

4.2.3 The adoption of focus groups to capture the customer requirements

Unlike other development lifecycles which make use of documentation such as user surveys for capturing requirements, XP emphasizes the adherence to its value of communication in order for developers to have an accurate understanding of user requirements. This focus on communication places significant prominence on the XP practice of active listening where “quality verbal communication is key” (Baird, 2003, no page).

Based on the fact that communication is a core value of XP (Baird, 2003), focus groups were appropriately selected and conducted as the qualitative method used to capture the user requirements thereby meeting the first project objective. A copy of the first focus group session which support following the Exploration phase can be found in **Appendix D-2**. A thorough analysis of the focus group session can be seen in **Appendix D-1**.

According to Eliot and Associates (2005, p. 1), a focus group can be defined as a “small group of six to ten people led through an open discussion by a skilled moderator. The group needs to be large enough to generate rich discussion but not so large that some participants are left out”. Focus groups are best used for requirements gathering because “it brings together a cross-section of stakeholders in a discussion group format” (Maguire and Bevan, 2002, p. 5) and allows “the analyst to rapidly obtain a wide variety of user views and possibly a consensus” (Maguire and Bevan, 2002, p. 10). The general idea of a focus group “is that each participant can act to stimulate ideas in the other people present, and that by a process of discussion, the collective view becomes established which is greater than the individual parts” (Maguire and Bevan, 2002, p. 5).

All focus groups conducted utilized a consistent sample size of fifteen participants and followed the guidelines set out by (Dzija, Foster, Hernandez, Nardi, Theriault

and Wynne, 2005). Adherence to such guidelines was instrumental in strengthening the validity of my qualitative research.

4.2.4 Selection of the focus group target audience

A section of the student population who attended Peer Assisted Learning sessions undertaking computer science degree programmes, were selected as participants in the focus group sessions. The Peer Assisted Learning (PAL) initiative is a scheme managed by the Centre for Learning Excellence which promotes cross-year support amongst students on the same course (University of Bedfordshire, no date c). The scheme “encourage students to support each other and learn co-operatively under the guidance of students from the year above” (University of Bedfordshire, no date c).

Although using a small section of the student population with a technical background may create a potential bias within the requirements gathering process, this was compensated by the fact that the mobile application will not only address students’ needs but also capitalize on the benefits of mobile learning which are in direct alignment of the aims of the PAL scheme.

These aims include allowing students to “adjust quickly to their university life, improve study skills and adjust their study habits to meet the requirements of higher education and enhance their understanding of the subject matter of their course through collaborative group discussion” (University of Bedfordshire, no date c). As it has already been acknowledged from the literature review that informal learning has leverage over formal learning, the mobile application which supports informal learning will take advantage of this and in doing so the mobile application will offer the following benefits which include:- helping students to adjust quickly to university life (University of Bedfordshire, no date c) as mobile learning “is good for just in time review of content” (Corbeil and Valdes-Corbeil, 2007, p. 54) that students often need at hand such as important information on university services.

Another benefit of mobile learning delivered by the mobile application is that it will “facilitate collaboration amongst students” (Corbeil and Valdes-Corbeil, 2007, p. 54) and this can assist with their understanding of subject material in a collaborative environment (University of Bedfordshire, no date c). Finally mobile learning “can enhance student-centred learning” (Corbeil and Valdes-Corbeil, 2007, p. 54) as well as “support a differentiation of student learning needs and personalized learning” referenced by (Corbeil and Valdes-Corbeil, 2007, p. 54) who cites the works of (Kukulska-Hulme, 2005), which can help improve their study skills and adjust their study habits in meeting the demands of higher education (University of Bedfordshire, no date c).

Moreover the developed mobile application will also display similarities and benefits to other mobile applications which are targeted at first year students and are designed to improve their student learning experience. When compared to NUI Maynooth’s proposed Android application for first year students such similarities and benefits comprise of the mobile application serving as a “one-stop-shop” to access important information about the university in order to help students in their transition to university (National University of Ireland, 2013). In comparison to the Android application “First 7 Weeks” for the University of Limerick, similarities and benefits were identical to those of NUI’s Maynooth’s proposed Android application (Burke, 2012).

4.2.5 The representation, role and involvement of the customer

Students participating in focus group sessions represented the customer during the development of the mobile application. In XP, the customer is defined as “the person or department that purchases the software. They define the functional requirements of the system. They are not concerned about usability and to a lesser extent, performance” (Baird, 2003, no page). On the other hand, the user “is the person who directly interacts with the software product. The end user has concerns about functionality, robustness, performance, usability and readability” (Baird,

2003, no page). Although both roles have marked differentiations, it is important to note that the customer could include users (Baird, 2003).

XP takes customer involvement to the extreme by mandating that an on-site customer be on the project full-time and be located on site with the developer as one of its practices (Baird, 2003). Although this practice could not be fully applied within this project, XP does allow flexibility in this practice if it is not possible for it to be properly executed by suggesting frequent meetings with the customer (Baird, 2003), in order to successfully manage changing requirements and design this approach was taken in the project.

4.2.6 Overview of the User Goal Oriented Design approach

The fundamental premise of the User Goal Oriented Design approach is that in order “to develop useful and usable systems we have to understand user needs and contexts, represent them in user requirements and maintain a focus on user requirements throughout development” as according to (Kauppinen and Rekola, 2002, p. 1) who cites the works of (Coble, Karat and Kahn, 1997). A central element of the user story is the name of the user story as it “captures the user’s goal in a few words” (Baird, 2003, no page). The concept of user needs and user goals are intertwined as “user needs refer both to the difference between users’ goals and the present condition, which is manifested by user problems and possibilities, and the context of use, which includes the characteristics of the intended users, users’ present tasks and environment” as highlighted by (Kauppinen and Rekola, 2002, p.1) who cites the works of (Rombach, 1990).

By extension of this “requirements are the more formal descriptions of user needs” as according to (Kauppinen and Rekola, 2002, p.1) who cites the works of (Rombach, 1990). User requirements are written from the user point of view as referenced by (Kauppinen and Rekola, 2002, p.1) who cites the works of (Abbott,

1986) and “describe how a future product can help users achieve their goals effectively, efficiently and with satisfaction in their context of use” (Kauppinen and Rekola, 2002, p.1) who cites the works of (ISO 13407, 1999).

4.2.7 Capturing the customer requirements with user stories following the User Goal Oriented Design approach

XP only advises the contents of a well-structured user story and does not provide a concrete technique for capturing user stories via an interview process such as a focus group. According to Baird (2003, no page), “understanding the customer’s requirements is vital to the success of any project. Instead of forcing the customer to explicitly define requirements upfront, XP allows for evolution. Who does the actual writing of the user story is less important than who owns the resulting stories. All stories have to be an expression of what is in the mind of the customer”.

In recognition of this, the implementation of the User Goal Oriented Design approach was used because not only did it fit in parallel with the goal of writing a user story, but also aided in fulfilling the third project objective to develop a mobile application that has genuine useful functionality relevant to its target audience. There are three steps outlined in following the User Goal Oriented Design approach. The first step involves figuring out what users want to do with the mobile application (Lehtimäki, 2012). Lehtimäki (2012) emphasizes the positive impact of having a clear understanding of what users want to do with the mobile application in increasing its chances of being successful. According to Lehtimäki (2012, no page), “if you have an abstract definition of what your app can be used for, designing the actual app without a bloated and confused user interface is much easier and much more likely to succeed”.

In following the first step outlined in the User Goal Oriented Design approach, the aim of the initial focus group session was to have a comprehensive understanding of what users wanted to achieve through using the mobile application. With this

perspective, user scenarios were integrated within the focus group session because users may not always be fully aware of what they would want to achieve upfront and this would be an excellent tool in guiding them. According to Maguire and Bevan (2002, p.5), user scenarios “give detailed realistic examples of how users may carry out their tasks in a specified context with the future system. The primary aim of scenario building is to provide examples of future use as an aid to understanding and clarifying user requirements. The method also promotes developer buy-in and encourages a human-centred design approach”. In essence, user scenarios are an “effective way of thinking about future system use in context” (Maguire and Bevan, 2002, p.11).

User scenarios also have integrated one of the core features of a user story, which is a persona. According to Maguire and Bevan (2002, p.11), “personas can bring user needs to life”. Maguire and Bevan (2002, p. 5) which cites the works of (Cooper, 1999) defines a persona as a “caricature which is created with a name, personality and picture, to represent each of the most important user groups. Potential design solutions can then be evaluated against the needs of a particular persona and the tasks they are expected to perform. Personas are used by innovative design groups to stimulate creativity rather than refine a design solution”.

Three case studies implemented by (Bradley and Holley, 2010) in an empirical research to find out how students in Higher Education use their mobile phone for learning were utilized to extract user scenarios that illustrated students’ mobile learning activities and their attitude towards it based on the project scope. Questions regarding the following areas were administered to students: - the key university services students want to access and find most useful, what information they require when accessing these services and their attitude towards using their Smartphone for learning. A re-evaluation of the informal mobile learning activities students engage in relating to the project scope which were compiled from the project inception was also conducted.

As highlighted in the literature review, students should be actively involved and shapers of their own learning and be able select a technology of their preference. Moreover student practices regarding informal mobile learning activities are likely to change as result of their mobile lifestyle. This re-evaluation of current student practices regarding the informal mobile learning activities in which they engage and an assessment of their attitude towards using their Smartphone for learning was essential as both are subject to change thereby affecting the requirements and scope for the system.

The second step of the User Goal Oriented Design Approach as highlighted by (Lehtimäki, 2012), is to have a “clear picture of the limits of your app by understanding the actual content. Mobile applications that focus to a smaller set of goals are generally much better and have greater user acceptance”. Although the concept of the use case and user story are interchangeable as both exist to enable the achievement of a user goal (Sehlhorst, 2009), use cases were utilised in the initial focus group session as they were seen as the best tool in “bridging the gap between the technical world of the designers and the professional world of the users” as documented by (Kauppinen and Rekola, 2002, p.1) who cites the works of (Kynge, 1995).

A use case driven approach is a popular solution to help with the problem of gathering and representing requirements within software engineering (Kauppinen and Rekola, 2002). A use case “describes the possible sequences of interactions between the system and one or more actors in response to some initial stimulus by one of the actors” as referenced by (Kauppinen and Rekola, 2002, p. 2) who cites the works of (Rumbaugh, 1994). Essentially use cases assists in supporting one of the purposes of XP which is to help designers to decrease the gap between user needs and user requirements (Kauppinen and Rekola, 2002) as “they help designers to gain a coherent view of the product” (Kauppinen and Rekola, 2002, p.5) and are instrumental in allowing “user needs and the user point of view to be stated explicitly in requirements definition” (Kauppinen and Rekola, 2002, p.5). A copy of

the use cases drafted in collaboration with the customer based on the feedback from the user scenarios and questions can be found in **Appendix E**.

The final step in the User Goal Oriented Design approach is to derive the features of the mobile application from the user goals captured. According to Lehtimäki (2012, no page), “once you understand what your app is going to do in abstract terms you're ready to jump into design. What you need to do is to take your set of user goals and figure out which features your app needs. Every single one of your app's features must be derived from one or more of the user goals you decided to support”. As Extreme Programming does not advocate a specific phase for design and recommends that software is a continuous state of design, the first stage of the Mobile Development SDLC design phase, the User Experience Design stage, was incorporated because it makes use of design prototypes such as the User Experience or UX wireframes as well as User Interface designs.

In order to have a good idea of what to design for the mobile application and gather additional feedback from users, UX wireframes were also used because low-fi sketch wireframes allow the developer and client to collaborate and focus design conversations on functionality (Balsamiq Studios, 2013). According to Maguire and Bevan (2002), a prototype is an excellent tool for validating and refining user requirements as it allows user feedback to be obtained. During prototyping “designers create paper or software-based simulations of user interface elements (menus, buttons, icons, windows, dialogue sequences) in a static or dynamic way” (Maguire and Bevan, 2002, p. 7).

User Experience Designs were developed with the use of wire frames or mock-ups using tools such as Balsamiq Mockups which allow developers to quickly solve and design the user experience whilst adhering to platform-specific guidelines, hardware factors and form factors without having to worry about the actual User Interface Design, in order to ensure a high level of usability of the system (Xamarin Inc, 2013). The use of User Experience Designs support the XP feature of evolutionary

design where “XP developers postpone design decisions until they are required as the dependence on simplicity and feedback ensure that the resulting system is not cobbled together” (Baird, 2003, no page).

The use of such low fidelity paper prototypes in the focus group session allowed for “requirements to be clarified as well as enable draft interaction designs and screen designs to be rapidly simulated and tested” (Maguire and Bevan, 2002, p.14). A copy of the User Experience Designs can be found in **Appendix F**. The Android Design Guidelines which were adhered to whilst creating these designs can be found in **Appendix G**. This approach adopted complemented the XP value of feedback. According to (Baird, 2003, no page), “feedback is vital to the success of any software project”.

4.2.8 Representation of the customer requirements in the form of user stories

The initial user stories captured were compiled with the assistance of the UX designs and use cases gathered thereby forming a set of initial requirements for the system.

According to Baird (2003), all user stories must have a name as part of its element, which is a brief description of what the user wants to achieve by using the mobile application. It usually has a verb-object construction for the title as this is easiest to understand. The initial user stories are provided below:

4.2.9 User stories containing the basic element of a name

- Access Student Support Services
- View Events
- View Clubs and Societies
- Access Services within the community

- View Local Maps
- View Campus Alerts
- Personalize Alerts
- Access Student Profile
- Access Calendar
- Access UoB MOBILE on Facebook
- Access Note Taker

According to Baird (2003, no page), “the Exploration phase is mostly about discovery. Customers discover what they want and developers discover their probable approach”. The formulation of high-level user stories by the customer marks the end of the Exploration phase.

4.3 The Planning Phase

4.3.1 Overview of the Planning Phase

During planning customers and developers agree on the features of the system for the first release (Baird, 2003). Planning follows a unique pattern in which the “customer defines requirements, the developer estimates effort and the customer prioritizes the work and a plan is agreed upon” (Baird, 2003, no page). There are also unique responsibilities assigned to both the developer and customer during planning as follows: - “the customer defines the scope of the release, the developer estimates how long each user story will take, the customer defines the order of delivery, the developer communicates technical impacts implementing requirements, the customer sets dates and times of the release and the developer breaks down the user stories into tasks and allocates work” (Baird, 2003, no page).

It is critical to have a comprehensive plan before developing any software and the objectives of the Planning phase involves the prioritization of work, breakdown into releases and creation of the first release plan (Baird, 2003). This plan should be

created quickly with the premise that planning is not an upfront activity but that this plan will change as the need arises (Baird, 2003). This approach forms part of XP's many core practices such as the Planning Game. The release plan contains a formal list of user stories thereby representing the requirements for the system to be developed. Unlike the Exploration phase, the Planning phase is governed by a structured and guided process with three stages to follow. These stages include Scoping, Release and Iterations (Baird, 2003). A copy of the second focus group session which support following these stages can be found in **Appendix D-4**. A thorough analysis of the focus group session can be seen in **Appendix D-3**.

4.3.2 Creating the Release Plan

The following provides an in depth description of how the three stages in the Planning phase were followed to create the release plan for the system.

4.3.3 The Scoping Stage

This stage has already been completed as it overlaps with the processes of the Exploration phase in which the decision to carry the project forward was conducted (Baird, 2003).

4.3.4 The Release Stage

According to XP “breaking the project down into smaller, bite-sized pieces enables the customer and technical team to make corrections in midcourse. The Release in XP is akin to a product-release cycle and is driven by business issues” (Baird, 2003, no page). In order to develop the software the customer expects, XP adopts short release cycles in which work developed on the system is produced incrementally.

Broadly speaking, XP projects involve “breaking down the customer's product vision into releases and the segmenting those releases into iterations” (Baird, 2003, no page). Releases should be short with the aim of delivering business value to the customer (Baird, 2003). The developer “uses each release as a checkpoint where they measure estimation accuracy” (Baird, 2003, no page). This technique draws upon one of XP’s integral practices of small releases and incremental change in which problem resolution is applied through “a series of small changes rather than one release” (Baird, 2003, no page).

4.3.5 The Iterations Stage

During this stage, an iteration plan is formulated with the stories forming the release plan. Each release will comprise of a “number of iterations that slowly builds towards the release” (Baird, 2003, no page). An iteration is where development of the user stories occurs (Baird, 2003). It is important to note that “iterations are time boxed with each time-box having a fixed duration typically of one to four weeks. The delivery date is not moveable as some working product must be delivered on the appointed date”. The customer is responsible for setting the dates of each iteration with “each iteration delivering business value that continues to grow with time” (Baird, 2003, no page).

4.3.6 Refining the customer requirements with the use of prototypes

There are a number of activities included in creating the release plan within the Release and Iterations stages. The first activity involves writing the user stories for the release with a finer level of detail. Customers are not usually comfortable with proceeding without some kind of visual sign off before development proceeds. In order for the customer to have a clearer idea of how they would want to use the mobile application based on the initial requirements formulated, a non-working prototype was created in collaboration with the customer.

Baird (2003, no page) highlights that “development using requirements necessitates a lot of questions and answers when work begins. User stories will not work if the customer or developer has a sign off mindset where he will not move forward until requirements have been locked down”. Specifically during this stage, in order to manage changing requirements, a software prototype created on the Android Smartphone was utilized because not only did it “provide a greater level of realism than is normally possible with simple paper mockups” but according to Maguire and Bevan (2002, p. 15) “to ensure a successful outcome, the developer must satisfy the needs and wants of the user when the development is complete. To achieve this, user needs should not only be elicited by techniques such as surveys, focus groups, interviews etc., but they should also be reflected back to users via simulations in order to prototype the user requirements. The requirements will of course then evolve as the system develops and more formal user evaluation takes place”.

This rapid prototype was also used to establish an acceptable design for the user (Maguire and Bevan, 2002) which would evolve into the final system completed. It is this use of evolutionary prototyping that supports one of XP’s activities of designing in which it is accepted that “design activities are ongoing and XP approaches it as a fluid responsive nature of design” (Baird, 2003, no page). The second stage of the Mobile Development SDLC design phase was utilized in creating the software prototype. Within the User Interface Design stage “while User Experience Designs are typically just black and white mock-ups, the User Interface Design stage is where colours, graphics, are introduced and finalized. Spending time on good User Interface design is important and generally, the most popular apps have a professional design”.

All User Interface Designs were created with Fluid UI Mobile prototypes in which high-fidelity prototypes (Fluid Software, 2013) were developed in collaboration with the customer. These prototypes served as an excellent tool for involving customers as well as for producing fully linked mock-ups presentations and demos to the customer in the process of refining the requirements. The Android Design Guidelines found in **Appendix G** were also utilized in creating the interface

prototypes. Furthermore the adoption of user design as well as interface prototypes was vital in meeting the fourth project objective. A copy of the User Interface designs can be found in **Appendix H**.

4.3.7 Writing well-formed user stories for the Release Plan

In collaboration with the customer and the User Interface Designs created, user stories initially gathered were refined and written at a finer level in which changes to the initial customer requirements occurred. A set of guidelines provided by (Sehlhorst, 2009) and (Baird, 2003) were adhered to in helping the customer to formulate well-written user stories. According to (Sehlhorst, 2009) there are three processes which apply to writing a well-written user story.

All well-written user stories have the following format:

“As a [role], I want to [do something] [with some frequency] so that I can/will [achieve some goal] where the [role] that starts a user story is the persona. The [do something] you identify is the meat of the user story – what the persona is going to do. The [with some frequency] element tells your team how often a particular user story will happen. This can be represented as an attribute of the user story. The [achieve some goal] component of the user story provides the context for why a user will be performing an action. The user will be performing the action to achieve a user goal” (Sehlhorst, 2009, no page).

This structure of a well-written user story outlined by (Sehlhorst, 2009) was followed in combination with the principles and elements of a user story outlined by (Baird, 2003). According to Baird (2003, no page), “all user stories should follow the below principles:

- “They should be understood by the customer.
- They should be one or two sentences long.
- They should represent a concept and not be too detailed”.

Moreover Baird (2003) also highlights there are two main elements of a user story. These include its name and text. In complying with these techniques, the following user stories were captured thereby forming the release plan.

Access Help

Allows students to access the help feature which will provide advice on how to use the mobile application.

Access Student Support Services

Enables students to access important and helpful information about student support services such as a summary on the university service and the ability to call them directly, which help students to adjust quickly to university life.

View Events

View a list of events provided by the Student Union in order to adjust quickly to university life.

View Campus Alerts

View alerts on university services for improved management of student life.

Access Note Taker

Allows students to create notes on the go for their lectures to assist with their learning.

Access Camera

Enables students to take pictures of learning materials to assist with their learning.

Access Pocket Guide

Students can access common videos on support services to help them adjust quickly to university life and for improved management of student life.

Access UoB MOBILE on Facebook

Students can share information in a collaborative environment regarding their course work via a Facebook group page.

Personalize Alerts

Enables students to select campus alerts they would like to view for improved management of student life.

Access About Us

Students can access the purpose and functionalities of the mobile application and submit ratings based on their user experience with the option to provide additional feedback.

4.3.8 Estimating the size of the release

The second activity in creating the release plan is for the developer to determine the size of the release by “estimating how long the user stories will take to implement using ideal programming time” (Baird, 2003, no page). “Ideal programming time is how long the story would take if there were no distractions” (Baird, 2003, no page). Based on the ideal programming time, the size of the release was estimated at five weeks in duration.

4.3.9 Estimating the number of iterations

“After the size of the release has been decided we need to divide it into iterations” (Baird, 2003, no page). The objective of this activity is to “deliver business value to the customer with each iteration” (Baird, 2003, no page). Based on the length of the release plan of a total of five weeks, it was estimated that there would be five iterations, with each iteration lasting one week, with the exception of the first iteration.

“The initial iteration focuses on the architecture components which are required to provide a baseline system. This can be called a zero business value iteration because the user stories are purely technical and have no direct business benefit to the customer. Subsequent iterations should deliver business value” (Baird, 2003, no page). The first iteration referred to as a zero business value iteration was integrated in order to establish the infrastructure for the working environment. XP recommends that this purely technical iteration be shorter in length compared to others (Baird, 2003) and should be mainly “narrow in its focus to build the minimum to establish the working environment” (Baird, 2003, no page). As a result of this the first iteration was set to last two days whilst the other iterations were estimated to last five days.

4.3.10 Prioritizing the release features

This is the final step before the release plan is formulated and agreed upon. During prioritization the customer analyzes all of the user stories to determine which ones will be developed first. However there are two sides in this ordering process in which the customer determines which stories will “give the greatest business benefit” (Baird, 2003, no page). On the other hand, the developer determines ordering from the “angle of risk” and communicates to the customer technical aspects with implementing the requirements.

In ordering the user stories in the release plan, those that delivered the greatest business impact were positioned at the top of the list, gradually decrementing to those which offered lesser business value. Moreover possible challenges with implementing features such as an integrated camera and note taking feature which were relayed by the developer were also taken into consideration by the customer whilst prioritizing the release features.

4.3.11 The Release Plan for the development of the artefact

“The output of release planning is a schedule that outlines what stories will be in the release and the order of delivery” (Baird, 2003, no page). The release plan for the development of the mobile application can be found below in **Figure 1**.

Before the release plan is agreed upon, the customer also sets dates and times for each release. The expected dates for the delivery of a working piece of software per each iteration were as follows:

Iteration 0: January 31st, 2013. **Iteration 1:** February 7th, 2013.

Iteration 2: February 14th, 2013. **Iteration 3:** February 21st, 2013.

Iteration 4: February 28th, 2013.

User Story	Effort	Iteration 0	Iteration 1	Iteration 2	Iteration 3	Iteration 4
Test framework setup	1 day	▪				
Creation of database ER diagram	1 day	▪				
Access Student Support Services	3 days		▪			
View Campus Alerts	2 days		▪			
View Events	2 days			▪		
Access Note Taker	1 day			▪		
Access Camera	1 day			▪		
Access UoB MOBILE on Facebook	1 day			▪		
Access Pocket Guide	3 days				▪	
Access About Us	2 days				▪	
Personalize Alerts	3 days					▪
Access Help	2 days					▪

Figure 1: The release plan

CHAPTER 5: ARTEFACT DEVELOPMENT AND TESTING

5.1 The Iterations Phase

5.1.1 Overview of the Iterations Phase

This was the final phase followed for the implementation of the artefact. During release planning it was noted that an iteration is “a time box between one to four weeks and is the building block that the customer uses to implement their software. The developer and customer gain confidence that the software is on track by breaking down releases into short iterations” (Baird, 2003, no page). Moreover, the adoption of iterations is vital for managing changing requirements enabling the

customer to direct the developer on how to deliver business value, whilst measuring and adjusting progress through short feedback loops. Adopting this approach supports one of XP principles of rapid feedback, where “developers use short feedback loops to quickly learn if their product is meeting customers’ needs” (Baird, 2003, no page). It also supports the XP value of simplicity where the developer “only works on what the customer actually needs not what they expect they might need and implements the simplest thing that could possibly work” (Baird, 2003, no page).

With the creation of the release plan containing the user requirements for the system, it is within the iterations phase that these user requirements are implemented as “software development happens inside the iteration” (Baird, 2003, no page). XP provides an iteration cycle which is used during this process (Baird, 2003). This iteration cycle embodies a set of organized activities which include the following. The iteration commences with planning, proceeds into development, and then is released to the customer for acceptance testing” (Baird, 2003, no page). A copy of the third, fourth, fifth, sixth and seventh focus group sessions which support following the Iterations phase can be found in **Appendix D-5 to D-9**.

5.1.2 Planning of the first iteration

The planning of the first iteration involved the selection of user stories to develop, the breakdown of these stories into tasks as well as task estimation and assignment. The first set of user stories which were implemented below were selected based on the priority given by the customer.

5.1.3 Selection of user stories for the first iteration

- Test framework setup
- Creation of database

5.1.4 Breakdown of user stories into tasks for the first iteration

The activities undertaken in this step are self-explanatory with the concept that there is a division of user stories into tasks by the developer that describe discrete development activities (Baird, 2003). By breaking user stories into manageable chunks before development work is carried out, the developer is in a better position to provide an accurate estimate of how long each user story should take to complete.

User story: Test framework setup

1. Download and install the test framework, Lunatest.
2. Copy appropriate files, lunatest.lua into main directory containing the code to execute unit testing.

User story: Creation of the database ER Diagram

1. Create the Entity Relationship Diagram.

5.1.5 Task estimation and assignment

With a concrete understanding on the development tasks for the user story, the developer can progress from high-level estimates made in release planning to provide a refined and accurate estimation of the how many user stories should be completed in the iteration. The developer uses a velocity for task estimation. Velocity is referred to as “the number of user stories a developer can complete in one sprint” (Duarte, 2008, no page) and is all about measuring how fast the developer should be working compared to how fast they are actually working.

Velocity is calculated using a load factor as an estimation benchmark. A component of the load factor is the actual time to complete where the developer “looks at the previous iterations or similar work and reflect on how long the tasks will actually take” (Baird, 2003, no page). Velocity is also used for tracking the iteration as steering the iteration is about maximising throughput through the cycle (Baird, 2003).

$$\text{Load Factor} = \text{Actual time to complete} \div \text{Estimated time to complete}$$

$$\text{Velocity} = \text{Iteration Length} * \text{Number of Developers} \div \text{Load Factor}$$

The following load factor was determined:

$$2/2 = 1 \text{ day}$$

The velocity was calculated at:

$$1 * 1/1 = 1 \text{ user story}$$

Signing up for tasks where a developer accepts responsibility for the completion of a user story, usually occurs in a development team of two or more before task estimation. However, as the development work for the system was autonomous in nature, this was not applied to the project.

5.1.6 Development of the artefact for the first iteration

Coding, testing and continuous integration are activities executed during the actual development of the system. With the exception of tracking and the customer writing acceptance tests, these and other remaining activities within the iteration cycle could not be applied to this type of iteration as it did not involve the customer evaluating the system or the development of source code. This iteration was used to allow customers to write acceptance tests for evaluating functionalities and usability.

5.1.7 Usage of a SQLite database

In order to meet the fifth project objective, a database was created in order to implement personalization within the mobile application. “SQLite is a software library that implements a self-contained, server-less, zero-configuration, transactional SQL database engine” (SQLite, no date a, no page). SQLite was used because it uses SQL commands and this complemented my previous familiarity with SQL.

Moreover “SQLite is appropriate for embedded devices and applications. It also requires little or no administration and is a good choice for devices or services that must work unattended and without human support. SQLite is a good fit for use in cell phones, PDAs, set-top boxes, and/or appliances” (SQLite, no date b, no page).

5.1.8 Entity Relationship Diagram

The ER diagram for the SQLite database illustrating the logical structure of the database which contains one table, is seen below in **Figure 1**.



Figure 1: Entity relationship diagram

5.1.9 Tracking of the first iteration

A selection of key indicators was used to track each iteration so both the customer and developer are aware of the progress that is being made with the development of the system (Baird, 2003).

Tracking was done in the following four broad areas:

“Resources - Planned versus actual resource usage: number of developers, number of customer team members, hardware, and other resource related metrics.

Scope - Number of stories: total completed and new ones in the pipeline.

Quality - Number of acceptance tests passed which changes over time.

Time - User stories planned against actual on the release plan” (Baird, 2003, no page).

Resources, scope and time were all well managed during this iteration with the planned versus actual resource usage remaining in tandem such as availability of the customer and Corona SDK software development platform. Moreover in tracking the scope, two user stories were completed as per the release plan with ten remaining for completion.

Furthermore, the actual time required to complete development activities for each user story in this iteration was on target with the timeline set out in the release plan. The level of quality in this iteration could not be assessed as it did not involve the customer testing the system. Finally the measurement and controlling of the velocity was well above target as two user stories were completed as opposed to one user story. This reflected that the calculation of my velocity was under estimated.

5.1.10 The writing of user acceptance tests by the customer

As part of Quality Assurance and Testing, in order to meet a critical aspect of one of my project objectives to ensure that the mobile application was accepted by students, a form of Black-box testing called User Acceptance Testing was utilized. Black-box testing takes the approach that the “customer or developer knows what the program is supposed to do, not how it is programmed to do it” (Dawson, 2009, p. 138).

A core feature of XP is for the customer to make the final decision whether the software developed meets their requirements (Baird, 2003) which are both implicitly and explicitly stated. “Closing the gap by getting customer feedback on progress to date is vital for accuracy and quality” (Baird, 2003, no page). XP advocates that the best way for the developer to achieve this is to allow the customer to write and run user acceptance tests which should be done as soon as possible.

According to JISC infoNet (2012, p.1) “acceptance testing is a key feature of projects to implement new systems or processes. It is the formal means by which we ensure that the new system or process does actually meet the essential user requirements”. For this project, customers wrote the acceptance tests for all iterations after the release plan. These acceptance tests were created for the sole comparison of actual test results against expected results.

The objective of the first category of acceptance tests evaluated explicit requirements and was used to confirm the use of the system in performing the functionality required by the customer. These tests were used to providing comprehensive and effective coverage of all aspects of functionality in order to ensure that the “customers functional requirements have been met and the system is acceptable” (Wells, 1999). The second category of acceptance tests evaluated the implicit requirements which are “requirements that indicate characteristics that are common to all systems in a domain” (Prosperity Heights Software, no date). These

tests were designed to ensure that the system had a high degree of usability based on the Android Design Principles and all final outcomes of these tests were satisfactory by the judgement of the customer. Any of the acceptance tests which failed were fed back into the next iteration.

The following user acceptance tests were written by the customer based on the user stories in the release plan. The customer collaborated with the developer to ensure that the acceptance tests seen in 5.1.12 comprised of a well-constructed test plan by adhering to the following XP Acceptance Test outline as highlighted by (Baird, 2003) seen in 5.1.11.

5.1.11 Adherence to a user acceptance test outline

“Initialize - Known conditions and/or data are staged into the system to prepare it for testing.

Execute - The tester walks through the test script, following the actions verbatim and recording the results.

Verify - The output or behavior of the test is verified against the expected result” (Baird, 2003, no page).

5.1.12 User acceptance tests for evaluating system functionalities

Test 1: Acceptance Test for Access Student Support Services

Initialize: The customer is at the main menu and wants to access important and helpful information such as a summary on the university services available and have the ability to call them directly.

Execute: The customer executes the Campus Life menu option, followed by Student Support Services menu option and either one of the four Student Support Services

menu options. The customer proceeds to access the university service by calling them directly using the phone icon.

Verify: The customer can access important and helpful information about student support services. The mobile application should enable the users to easily read the information and quickly make calls to access these services.

Test 2: Acceptance Test for View Campus Alerts

Initialize: The customer is at the main menu and wants to view alerts for a particular university service.

Execute: The customer executes the Campus Alerts menu option, and selects to view either one of the four Campus Alerts menu options.

Verify: The customer can easily read the alerts displayed from the university service's Twitter feed and has access to the screen via a scroll object to read the information comfortably.

Test 3: Acceptance Test for View Events

Initialize: The customer is at the main menu and wants to view events provided by the Student Union.

Execute: The customer executes the Campus Life menu option, followed by the Events menu option to view a list of current events hosted by the Student Union.

Verify: The customer can view a list of up-to-date events provided by the Student Union which also allows for different views of these events.

Test 4: Acceptance Test for Access Note Taker

Initialize: The customer is at the main menu and wants to take notes via the mobile application on the go for their lectures.

Execute: The customer executes the Student Tools menu option, followed by the Note Taker menu option. They use the onscreen keyboard to enter a note followed by the Email button to email the note to an email account of their choice. They then automatically return to the note once it has been emailed and use the back button to return to the Student Tools menu. They then re-execute the Note Taker menu option

to verify that the note was also saved when it was emailed as it appears on the note screen.

Verify: Notes are easily taken, can be saved on the phone and emailed within the mobile application.

Test 5: Acceptance Test for Access Camera

Initialize: The customer is at the main menu and wants to take pictures of learning materials.

Execute: The customer executes the Student Tools menu option, followed by the Access Camera menu option. The customer takes a picture by pressing the capture button in the inbuilt camera. The customer access the Gallery on their phone to ensure the picture taken is found.

Verify: Pictures of learning materials are saved on the phone and accessed via the Photo Gallery of the phone. Taking pictures using the mobile application does not close the mobile application.

Test 6: Acceptance Test for Access UoB MOBILE on Facebook

Initialize: The customer is at the main menu and wants to share information in a collaborative environment regarding their course work via a Facebook group page.

Execute: The customer executes the Student Tools menu option, followed by the UoB on Facebook menu option. The customer signs in to Facebook if necessary and makes a post to the group page.

Verify: Posts can be made in the interactive environment which allows the customer to collaborate on information regarding their coursework.

Test 7: Acceptance Test for Access Pocket Guide

Initialize: The customer is at the main menu and wants to access common videos on Student Support services.

Execute: The customer executes the Campus Life menu option, followed by the Pocket Guide menu option. The customer then selects video to watch based on the

four menu options provided. They view the video transcript which is scrollable to the top of the screen. They then click the play button to watch the video.

Verify: Students can access a video on support services. Access to the video transcript is available for those who are hearing impaired.

Test 8: Acceptance Test for Access About Us

Initialize: The customer is at the main menu and wants to view the purpose of the app and rate the mobile application by giving a recommendation level based on their user experience.

Execute: The customer executes screen overflow option menu option, followed by the About Us menu option. The customer then selects a recommendation level by pressing one of the rating buttons of their choice. The customer then sends the recommendation level via email to the developer and has the option to provide additional feedback.

Verify: The customer can select a recommendation level which is then sent to the developer via an email.

Test 9: Acceptance Test for Personalize Alerts

Initialize: The customer is at the Campus Alerts menu and wants to personalize which campus alerts they want to view within this screen.

Execute: The customer executes the screen overflow option menu option, followed by the Personalize Alerts menu option. The customer then selects which campus alerts they want to receive via checkbox and returns to Campus Alerts menu via the caret button to view the updated menu list.

Verify: The customer can personalize the campus alerts they want to receive.

Personalization is made easy with use of checkboxes, choices selected can be varied and are saved for future use.

Test 10: Acceptance Test for Access Help

Initialize: The customer is at the main menu and wants to access help features before using the mobile application.

Execute: The customer executes screen overflow option menu option, followed by the Help menu option.

Verify: The customer can quickly view help information on how to use the mobile application.

5.1.13 User acceptance tests for evaluating usability

The following acceptance tests which evaluated the level of usability based on the Android Design Principles forming part of the Android Design Guidelines, found in **Appendix G**, were created in collaboration with the customer. These tests were also formulated when acceptance tests were created to evaluate the functionalities of the mobile application.

Test 11: Screen Navigation For All Screens

Navigation between screens is consistent. Transitions and tabs are used to show relationships among screens so the user always knows their way around. The ability to scroll through any of the menu options is clearly made aware to the user.

Test 12: Screen Content For All Screens

Only information relevant to the user needs is presented. Tasks and information is broken down into small digestible chunks to avoid the user being overwhelmed.

Test 13: Feedback on actions for all screens

Complex tasks are broken into smaller steps that can be easily accomplished. Feedback on actions is provided for all screens using a subtle glow.

Test 14: Intuitive and Ease of Learning of All Screens

The mobile application is easy to learn as it leverages visual patterns such as the caret and screen overflow for navigation from other Android mobile applications.

Test 15: Decision Making for Pocket Guide Screen when any of the menu options are selected.

The user is presented with the activity, watching the video, rather than being asked for confirmation on whether they would like to execute the activity.

Test 16: Decision Making for Access Student Services when any of the menu options are selected.

The user is presented with the activity, making a phone call, rather than being asked for confirmation on whether they would like to execute the activity.

Test 17: Personalized settings for campus alerts are saved

Personalized campus alerts settings the user created are stored and remembered by the mobile application which adds a personal touch to the user experience.

Test 18: Splash Screen for User Experience

A well-timed sound effect in the splash screen has a positive effect on the user experience.

5.1.14 Planning of the second iteration

In continuation of following the XP iteration cycle, this section will highlight conducting the activities previously mentioned during the planning of the first iteration.

5.1.15 Selection of user stories for the second iteration

- Access Student Support Services
- View Campus Alerts

5.1.16 Breakdown of user stories into tasks for the second iteration

User story: Access Student Support Services

1. Create the splash screen and the ability to go to the main menu screen by touching this screen.
2. Create the interface for the main menu screen, excluding the action menu options.
3. Incorporate the menu option for Campus Life and the ability to go to the Campus Life screen by selecting this menu option.
4. Create the interface for the Campus Life screen, excluding the menu options.
5. Incorporate a menu option in the Campus Life screen for Student Support Services and the ability to go to the Student Support Services screen by selecting this menu option.
6. Create the interface for the Student Support Services screen, excluding the menu options.
7. Include four menu options in the Student Support Services screen which include Library Services, Careers and Employment Services, International Student Support Services and Student Information Desk.
8. Create the interface for each of the menu options, with each having their own screen. Incorporate the functionality to go to each corresponding screen when the relevant menu option is selected.
9. Include a summary, a picture for each university service screen and the ability to directly call the service directly by pressing the phone icon.
10. Incorporate all navigational features for the caret and split action bar tabs for all screens created for this user story.
11. Incorporate any other additional features relating to usability based on the Android Design Guidelines.

User story: View Campus Alerts

1. Incorporate an action menu option for Campus Alerts in the main menu screen and the ability to go to the Campus Alerts screen by selecting this menu option.
2. Create the user interface for the Campus Alerts screen, excluding the action menu options.

3. Include four menu options in the Campus Alerts screen which include Library Services Alerts, Careers and Employment Services Alerts, International Student Support Services Alerts and Student Information Desk Alerts.
4. Create the interface for each of the menu options, with each having their own screen. Incorporate the functionality to go to each corresponding screen when the relevant menu option is selected.
5. Incorporate the web app view to display up-to-date Twitter feeds from the relevant campus service for all of the four menu options individual screens.
6. Incorporate all navigational features for the caret and split action bar tabs for all screens created for this user story.
7. Incorporate any other additional features relating to usability based on the Android Design Guidelines.

5.1.17 Task Estimation

Velocity was used to calculate how many user stories should be implemented within the second iteration.

The following load factor was determined:

$$5 / 5 = 1 \text{ day}$$

The velocity was calculated at:

$$1 * 1 / 1 = 1 \text{ user story}$$

5.1.18 Development of the artefact for the second iteration

The actual development of the artefact which included coding, testing and integration commenced within this activity of the iteration cycle as part of the second iteration. The XP practice of Pair Programming is commonly utilized during coding and testing of the system in which “developers work in pairs to do software development. Essentially, two developers share the same workstation, taking turns

using the keyboard. All code is reviewed constantly with one developer looking at detail while the other is looking at the conceptual” (Baird, 2003, no page). However due to the autonomous nature of this project, this practice could not be conducted.

5.1.19 Writing and running unit tests for the second iteration

Unit testing is a form of White-box testing (Yadav, 2007) in which its primary goal “is to take the smallest piece of testable software in the application, isolate it from the remainder of the code, and determine whether it behaves exactly as you expect” According to Microsoft (2013, no page), white-box testing is a form of dynamic testing which uses “program source code as the basis for designing tests and test cases and is concerned with testing the implementation of the program. The intent of this testing is not to exercise all the different input or output conditions (although that may be a by-product) but to exercise the different programming structures and data structures used in the program” (Learn SAD, no date). “White-box testing means the software developer knows how code is programmed and structured so that they can test our potentially error-prone parts of the system” (Dawson, 2009, p. 138).

Testing refers to the “testing of the program to see if it works or has any bugs or errors within it” (Dawson, 2009, p. 137). Overall testing is a fundamental step through the XP development lifecycle as it supports one of its principles of ensuring quality work because “it elevates the importance of code and testing with test-first programming so that the quality of work can never be compromised” (Baird, 2003, no page).

One of XP’s core practices is Test Driven Development where the developer “writes the unit tests which test each method and everything that could possibly break. When all tests are written for a component, the developer writes only enough code to pass the test” (Baird, 2003, no page). All of the unit tests created ensured all of

the user interface features for the mobile application which allowed use of the system were loaded successfully.

Lunatest is a randomized testing framework for Lua which supports Test Driven Development (Warden, 2012) and can be used within the Corona SDK development platform. This framework was used to quickly execute the unit tests. The first step was to write unit tests and then write just enough code for the tests using the Lua programming language to pass these tests during the development of the two user stories.

A copy of the executed test plan for this iteration can be found in **Appendix I-1**.

A screenshot of the number of unit tests passed for this iteration can be found in **Appendix I-2**.

5.1.20 Adoption of functional testing for the second iteration

In addition to testing the loading of interface components, it was also important to ensure that the functionalities for the interface were fully tested. In order to achieve this, the combination of Test Driven Development and functional testing, a type of black box testing was conducted. “Functional testing verifies that each function of the software application operates in conformance with the requirement specification. This testing mainly involves black box testing and it is not concerned about the source code of the application. Each and every functionality of the system is tested by providing appropriate input, verifying the output and comparing the actual results with the expected results” (Guru99, no date, no page).

A copy of the executed test plan for this iteration can be found in **Appendix J-1**.

This test plan formed part of following the functionality testing process as outlined by (Guru 99, no date).

In addition to using dynamic testing, as it is important “to test code both dynamically and statically” (Dawson, 2009, p. 138). Static testing was conducted throughout all iterations in which this technique was used to “assess the content, level of comments in the program, the scope and declaration of variables, the overall structure of the program and the design and layout of the code” (Dawson, 2009, p. 138). This type of testing is an important part of complying with coding standards outlined within XP which aim “to produce software that has a consistent style, independent of the author and is easy to understand and maintain” (Baird, 2003, no page).

5.1.21 Integration of the system

Continuous integration forms one of XP core practices in which “components of the system are built or integrated many times a day as the developer integrates new code” (Baird, 2003, no page). “Integration is the process of assembling the team’s individual components and building them into the software product” (Baird, 2003, no page). Building the software everyday is crucial to the success of a project as its aim is to get the product visible that creates business value for the customer (Baird, 2003). Small releases aimed at creating business value for the customer facilitates frequent builds therefore “development tasks are small as the longer the time period between integrations the more conflicts arise and this creates difficulty with integration” (Baird, 2003, no page).

Writing and running unit tests form part of the first and second steps within continuous integration. In completing the third step of continuous integration, after successfully running the complete test suite which is a collection of similar test cases, the source files are integrated on the local machine. The daily build of the product “is one of the most important diagnostic tools for the health of the software” (Baird, 2003, no page). During this step the build must pass a series of verification tests within the Lua SDK framework to call the build good. The following four steps for the criteria for a successful build outlined by (Baird, 2003) were adapted when

applied to integration of the system. According to Baird (2003, no page), “you can devise your own criteria for a successful build”.

The steps for the criteria for a successful build included:

1. “All working source files are checked into the source code repository.
2. All source files are compiled.
3. Object or intermediate files are linked to create deployment binaries such as executables, JAR files.
4. The system is verified against a series of tests, this smoke test ensures testability of the system” (Baird, 2003, no page).

My adaptation of this criterion included following steps two and three in conducting daily builds per each iteration.

After source files are integrated on the local machine, integration must take place on a build machine which is usually on a remote server. Forming part of the third step of continuous integration, the daily build within XP usually involves the use of automated scripts as part of build automation where “automated scripts are run to extract source from the version control system, compile, link and then build. Normally at the end of the build some kind of smoke test is run to check the build status” (Baird, 2003, no page).

Currently Corona SDK automated build tool features are only available for PRO subscription users and I only had access to the university’s Academic Site License subscription. Although I had installed EasyMercurial, “a simple interface for the Mercurial distributed version control” (EasyMercurial, no date) which contains a source control repository so that the latest versions of the source could be extracted, proceeding to use this would be pointless. This was due to the fact that even if I could connect a continuous integration server such as Jenkins to the repository, Corona lacks automated build and testing tools under my subscription. Such build

and testing tools would be required by Jenkins after files are extracted from source control repository in order to conduct automated builds and tests as outlined in the continuous integration in action diagram highlighted by (Baird, 2003, no page).

As a result, the final step for continuous integration which involves migrating the source to the integration machine could not be completed due to these limitations. However I was still able to build the software manually in accordance to the steps outlined for a good build. In completing the first and second step all source files were checked for compilation in which no errors were recorded in the Corona Simulator Output Screen. A screenshot of the Corona Simulator Output screen showing the compilation of the source code with no errors can be found in **Appendix I-2**.

Following the second step, an .apk file was created using Corona SDK build feature which builds the source into a deployment binary after it uses the files which were added to the source control repository. Whilst the system is built, it undergoes a series of tests to ensure the build is good. The results of the manual build indicated that the build was good as it passed verification tests within the Lua SDK framework. A screenshot showing the successful build message on the local machine can be found in **Appendix K-1**. According to Baird (2003, no page), “the build can be called good or successful if the system passes verification tests. It is then ready for testing or review by the customer”.

5.1.22 Execution of user acceptance tests by the customer

The next activity in the iteration phase is for the customer to execute acceptance tests on the implemented user stories.

The following documents the results of the final stage within the acceptance testing workflow as according to (Baird, 2003). The execution of the user acceptance tests by the customer will determine whether the tests results indicate success or failure thereby marking the closure of the iteration.

Acceptance tests were used to test the system functionalities behaved as the customer intended. Additional acceptance tests were utilized to ensure that the system had a high level of usability based on the Android Design Guidelines. A copy of these test results can be found in **Appendix L-1**. Acceptance tests which evaluated usability were executed per each iteration within the release plan. It was recorded that one of these acceptance tests failed within this iteration.

5.1.23 Tracking of the second iteration

Resources, scope and time were all also well managed during this iteration with the planned versus actual resource usage remaining in tandem such as the availability of the customer and the Corona SDK software development platform. Moreover in tracking the scope, four user stories were completed as per the release plan with eight remaining for completion. Furthermore, at the end of this iteration, the actual time required to complete the development activities for each user story in this iteration was on target with the timeline set out in the release plan.

Moreover, there was a high level of quality of the mobile application within this iteration as the customer assessed that with the exception of one test, nine acceptance tests passed successfully. Finally the measurement and controlling of the velocity was well above target as two user stories were completed as opposed to one user story. This reflected that the calculation of my velocity was under estimated.

5.1.24 Acceptance of the implemented user stories for the second iteration

For this iteration, all final outcomes of the tests evaluating the system functionalities and the majority testing its usability were satisfactory by the customer. The failed acceptance test reported was carried forward into the third iteration.

5.1.25 Planning of the third iteration

This section outlines the tasks undertaken during planning activities of the iteration cycle for the third iteration.

5.1.26 Selection of user stories for the third iteration

- View Events
- Access Note Taker
- Access Camera
- Access UoB MOBILE on Facebook

5.1.27 Breakdown of user stories into tasks for the third iteration

User story: View Events

1. Incorporate a menu option for Events in the Campus Life screen and the ability to go to the Events Screen when this menu option has been selected
2. Create the interface for the Events screen.
3. Create a web app view to display the web page showing the events hosted by the Student Union.
4. Incorporate all navigational features for the caret and split action bar tabs for all screens created for this user story.
5. Incorporate any other additional features relating to usability based on the Android Design Guidelines

User story: Access Note Taker

8. Incorporate a menu option for Student Tools in the main menu screen and the ability to go to the Student Tools screen when this menu option has been selected.
9. Create the interface for the Student Tools screen, excluding the menu options.
10. Include a menu option for Note Taker in the Student Tools screen and the ability to go the Note Taker screen when this menu option has been selected.
11. Create the interface for the Note Taker screen.
12. Incorporate the functionality for creating the note using on-screen keyboard, emailing and saving of the note for the Email button.
13. Incorporate the functionality for the Back button.
14. Incorporate any other additional features relating to usability based on the Android Design Guidelines.

User story: Access Camera

1. Incorporate a menu option for Camera in the Student Tools screen.
2. Incorporate the functionality for the in-built camera on the phone to appear when the Camera menu option is selected.
3. Incorporate any other additional features relating to usability based on the Android Design Guidelines.

User story: Access UoB MOBILE on Facebook

1. Incorporate a menu option for UoB MOBILE in the Student Tools screen and the ability to go the UoB MOBILE when this menu option has been selected.
2. Create the interface for the UoB MOBILE on FB screen.
3. Create a web app view to display the web page for the UoB MOBILE Facebook group page.
4. Incorporate the functionality for the Back button.
5. Incorporate any other additional features relating to usability based on the Android Design Guidelines.
6. Resolve the issue regarding navigation as discovered in the failed acceptance test.

5.1.28 Task Estimation

Velocity was used to calculate how many user stories should be implemented within the third iteration.

The following load factor was determined:

$$5 / 5 = 1 \text{ day}$$

The velocity was calculated at:

$$1 * 1 / 1 = 1 \text{ user story}$$

5.1.29 Development of the artefact for the third iteration

This section outlines the tasks undertaken during development activities of the iteration cycle for the third iteration.

5.1.30 Writing and running of unit tests for the third iteration

A copy of this executed test plan for the iteration can be found in **Appendix I-1**.

A screenshot of the number of unit tests passed for this iteration can be found in **Appendix I-3**.

5.1.31 Adoption of functional testing for the third iteration

A copy of this executed test plan for the iteration can be found in **Appendix J-2**.

5.1.32 Integration of the system

A screenshot of the Corona Simulator Output screen showing the compilation of the source code with no errors can be found in **Appendix I-3**.

A screen shot of the successful build on the local machine can be seen in **Appendix K-1**.

5.1.33 Execution of user acceptance tests by the customer

A copy of these test results can be found in **Appendix L-2**. It was recorded that one of these acceptance tests failed within this iteration.

5.1.34 Tracking of the third iteration

Similar to the second iteration, resources, scope and time were all also well managed during the third iteration with the planned versus actual resource usage remaining in tandem such as the availability of the customer and Corona SDK software development platform. Moreover in tracking the scope, eight user stories were completed as per the release plan with four remaining for completion. Furthermore, at the end of this iteration, the actual time required to complete the development activities for each user story in this iteration was on target with the timeline set out in the release plan.

There was also a high level of quality of the mobile application within this iteration as the customer assessed that with the exception of one test, eleven acceptance tests passed successfully. Finally the measurement and controlling of the velocity was well above target as two user stories were completed as opposed to one user story. This reflected that the calculation of my velocity was under estimated.

5.1.35 Acceptance of the implemented user stories for the third iteration

For this iteration, all final outcomes of the tests testing the system functionalities and the majority testing its usability were satisfactory by the customer. The failed acceptance test reported was carried forward into the fourth iteration.

5.1.36 Planning of the fourth iteration

This section outlines the tasks undertaken during planning activities of the iteration cycle for the fourth iteration.

5.1.37 Selection of user stories for the fourth iteration

- Access Pocket Guide
- Access About Us
- Access Camera

5.1.38 Breakdown of user stories into tasks for the fourth iteration

User story: Access Pocket Guide

1. Incorporate the menu option for Pocket Guide in the Campus Life screen and the ability to go the Pocket Guide screen when this menu option has been selected.
2. Create the interface for the Pocket Guide screen, excluding the menu options.
3. Include four menu options in the Pocket Guide screen which include Literature Searching, Print Credit Machines, Referencing and Self Issue Machines. Incorporate the functionality to go to each corresponding screen when the relevant menu option is selected.
4. Create the interface for each of the menu options, with each having their own screen.
12. Include a summary for the video, a transcript that is scrollable, and the ability to play the video when the play icon is pressed for each of these screens.
13. Incorporate all navigational features for the caret and split action bar tabs for all screens created for this user story.
14. Incorporate any other additional features relating to usability based on the Android Design Guidelines.

User story: Access About Us

1. Include functionality for the screen overflow button in the main screen to include to a menu option for About Us. Incorporate the ability to go to the About Us screen when this menu option has been selected.
2. Create the interface for the About Us screen.
3. Include relevant information for the About Us section and four buttons allowing users to rate the mobile application.
4. Incorporate the functionality which allows users to rate the mobile application when the button is selected and which also sends this rating the developer of the mobile application via email.
5. Incorporate the functionality to go to each corresponding screen when the relevant menu option is selected.
6. Incorporate any other additional features relating to usability based on the Android Design Guidelines.

User story: Access Camera

1. Resolve the issue discovered in the failed acceptance tests where pictures of learning materials are saved on the phone and accessed via the Photo Gallery of the phone. However, taking pictures using the mobile application closes the mobile application.

5.1.39 Task Estimation

Velocity was used to calculate how many user stories should be implemented within the fourth iteration.

The following load factor was determined:

$$5 / 5 = 1 \text{ day}$$

The velocity was calculated at:

$$1 * 1 / 1 = 1 \text{ user story}$$

5.1.40 Development of the artefact for the fourth iteration

This section outlines the tasks undertaken during development activities of the iteration cycle for the fourth iteration.

5.1.41 Writing and running unit tests for the fourth iteration

A copy of this executed test plan for the iteration can be found in **Appendix I-1**.

A screenshot of the number of unit tests passed for this iteration can be found in **Appendix I-4**.

5.1.31 Adoption of functional testing for the fourth iteration

A copy of this executed test plan for the iteration can be found in **Appendix J-3**.

5.1.32 Integration of the system

A screenshot of the Corona Simulator Output screen showing the compilation of the source code with no errors can be found in **Appendix I-4**.

A screen shot of the successful build on the local machine can be seen in **Appendix K-1**.

5.1.33 Execution of user acceptance tests by the customer

A copy of these test results can be found in **Appendix L-3**.

5.1.34 Tracking of the fourth iteration

Similar to the third iteration, resources, scope and time were all also well managed during the third iteration with the planned versus actual resource usage remaining in tandem such as the availability of the customer, Corona SDK software development platform. Moreover in tracking the scope, ten user stories were completed as per the release plan with two remaining for completion. Furthermore, at the end of this iteration, the actual time required to complete the development activities for each user story in this iteration was on target with the timeline set out in the release plan.

There was also a high level of quality of the mobile application within this iteration as the customer assessed that eleven acceptance tests passed successfully. Finally the measurement and controlling of the velocity was well above target as two user stories were completed as opposed to one user story. This reflected that the calculation of my velocity was under estimated.

5.1.35 Acceptance of the implemented user stories for the fourth iteration

All final outcomes of the tests testing the system functionalities and its usability were satisfactory by the customer.

5.1.36 Planning of the fifth iteration

This section outlines the tasks undertaken during planning activities of the iteration cycle for the fifth iteration.

5.1.37 Selection of user stories for the fifth iteration

- Personalize Alerts
- Access Help

5.1.38 Breakdown of user stories into tasks for the fifth iteration

User story: Personalize Alerts

1. Include functionality for the screen overflow button in the Campus Alerts screen to include to a menu option for Personalize Alerts. Incorporate the ability to go the Personalize Alerts screen when this menu option has been selected
2. Create the interface for the Personalize Alerts screen.
3. Create the SQLite database using Lua code as well as creation and population of an SQL table to record which alerts have been selected.
4. Incorporate the functionality for the checkboxes so that they check and uncheck when selected or deselected. This entry is saved in the database table.
5. Ensure that database entries are retrieved for each checkbox and based on whether the checkbox is selected or deselected, displays the appropriate campus alert.
6. Incorporate all navigational features for the caret and split action bar tabs for all screens created for this user story.
7. Incorporate any other additional features relating to usability based on the Android Design Guidelines.

User story: Access Help

1. Include the functionality for the screen overflow button in the main screen to include to a menu option for Help.
2. Incorporate the ability to go the Help screen when this menu option has been selected.
3. Create the interface for the Help screen.
4. Include relevant information for the Help section such as an advice tip.
5. Incorporate all navigational features for the caret and split action bar tabs for all screens created for this user story.
6. Incorporate any other additional features relating to usability based on the Android Design Guidelines.

5.1.39 Task Estimation

Velocity was used to calculate how many user stories should be implemented with the fifth iteration.

The following load factor was determined:

$$5 / 5 = 1 \text{ day}$$

The velocity was calculated at:

$$1 * 1 / 1 = 1 \text{ user story}$$

5.1.40 Development of the artefact for the fifth iteration

This section outlines the tasks undertaken during development activities of the iteration cycle for the fifth iteration.

5.1.41 Writing and running unit tests for the fifth iteration

A copy of this executed test plan for the iteration can be found in **Appendix I-1**.

A screenshot of the number of unit tests passed for this iteration can be found in **Appendix I-5**.

5.1.42 Adoption of functional testing for the fifth iteration

A copy of this executed test plan for the iteration can be found in **Appendix J-4**.

5.1.43 Integration of the system

A screenshot of the Corona Simulator Output screen showing the compilation of the source code with no errors can be found in **Appendix I-5**.

A screen shot of the successful build on the local machine can be seen in **Appendix K-1**.

5.1.44 Execution of user acceptance tests by the customer

A copy of these test results can be found in **Appendix L-4**.

5.1.45 Tracking of the fifth iteration

Similar to the fourth iteration, resources, scope and time were all also well managed during the third iteration with the planned versus actual resource usage remaining in tandem such as the availability of the customer, Corona SDK software. Moreover in tracking the scope, all twelve user stories were completed as per the release plan. Furthermore, at the end of this iteration, the actual time required to complete the development activities for each user story in this iteration was on target with the timeline set out in the release plan.

There was also a high level of quality of the mobile application within this iteration as the customer assessed that ten acceptance tests passed successfully. Finally the measurement and controlling of the velocity was well above target as two user stories were completed as opposed to one user story. This reflected that the calculation of my velocity was under estimated.

5.1.46 Acceptance of the implemented user stories for the fifth iteration

All final outcomes of the tests testing the system functionalities and its usability were satisfactory by the customer.

CHAPTER 6: EVALUATION

6.1 Analysis of user acceptance test results

Testing was utilized to allow the target audience to test the mobile application before “accepting” the system. Students constructed these tests and executed them to assess whether the system being delivered had a high level of usability and met their user requirements to which the system must conform, thereby making it fit-for-purpose. In analysing the outcomes of the acceptance testing performed, it was derived that the mobile application met the end user’s satisfaction levels as the test results were positive and favourable with all eighteen of the acceptance tests written and executed by the end user passing successfully.

6.2 Assessment of the developed system based on best practices in the industry

As part of the evaluation of the artefact within my role as a developer, I benchmarked my artefact based on the seven best practices and insight identified by professionals who have developed mobile applications for universities as highlighted by Eustice (2011). This assessment revealed that my development practices employed displayed notable conformity to those in the industry. The first best practice includes considering short term solutions when developing a wider digital strategy (Eustice, 2011).

The university has already explored short term solutions such as delivering prospectus style content through a mobile application for the iPhone and a mobile application for the Android Smartphone which allows quick access to library services from anywhere (University of Bedfordshire, no date d) that both utilized informal learning. With its added leverage of bridging the gap between formal and informal learning, the development of this mobile application introduces an innovative and improved approach in meeting the university’s strategic plan via informal mobile learning.

Another best practice advocated is to use development tools that allow you to develop the mobile application cheaply and with no real programming knowledge (Eustice, 2011). Choosing Corona SDK for mobile development helped me to meet this best practice as Corona's ability through the use of the programming language Lua allowed me to develop my mobile application quickly with very little coding required (Corona Labs Inc, 2013 b).

The third best practice indicates that a "good idea needs to fuse with usability, design and marketing" (Eustice, 2011, no page). The developed mobile application is fit-for-purpose and this coincides with the high degree of acceptance it has ascertained based on an evaluation of its recommendation intention levels by students. A survey which allowed students to rate the mobile application indicated that the mobile application has a high level of recommendation intention being observed in the results of recommendation level ratings.

Sixteen surveys comprising of three questions were projected as anonymous and elicited to the student body. All surveys were completed. The gender makeup of participants completing the survey comprised of fifteen males and one female. Analysis of the results from the survey indicated that 63% of participants gave the mobile application a recommendation level/intention of Quite Recommended based on their actual use of the mobile application. This can be seen in **Figure 1**. This signifies a possibility for a significant number of downloads of the mobile application as recommendation intention is based on usage of the mobile application. It also served as an accurate determination of the recommendation intention and usage intention of the mobile application as referenced in the literature review. According to Lee, Kim and Choi (2012, p. 33), "recommendation intention was found as high when practical apps users had usage intention, and this usage intention is highly likely connected to actual use".

With regards to usability and design, designing of the mobile application has adhered to the Android Design Guidelines, and its ease of use is reflected in the

results of the acceptance tests which evaluated usability seen in **Appendix L**. With a high level of acceptance the university could use posters to market the mobile application to its students as part of its marketing strategy.

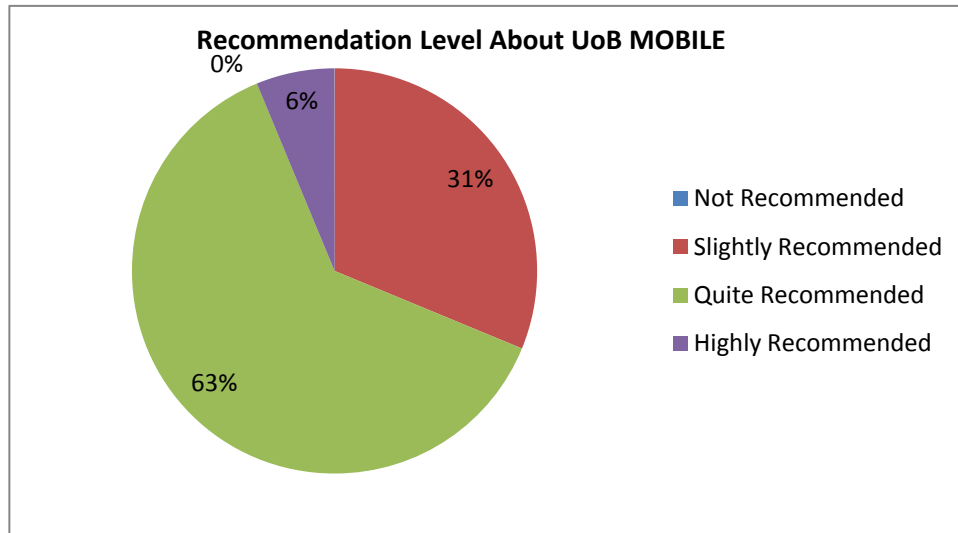


Figure 1

Another key best practice acknowledged is to “consider the benefits of developing tools that are device agnostic” Device agnostic mobile applications can work across different platforms and are therefore “compatible with most operating systems and may work on different types of devices” (Rouse, 2011, no page). Although the developed mobile application was created for the Android Smartphone, the Corona SDK development platform easily allows for deployment to other devices such as the iPhone (Corona Labs Inc, 2013 a) thereby signifying the mobile application’s potential cross platform ability.

“Universities should also consider the user experience in the delivery of new mobile application products” (Eustice, 2011, no page). During the development of the mobile application, the user experience was consistently included as a factor during its design with the use of User Experience and User Interface Designs.

The integration of the mobile application with other internal systems should also be implemented as emphasized by Dough Poole, Digital Media Officer at the University of Southampton, as that is when the user will experience genuinely useful functionality (Eustice, 2011). This was a limitation during the development of the mobile application as there was an inability to connect with other internal university systems. As a result this practice was not followed.

Finally, it was also noted that “stakeholder engagement is important and the mobile application should be developed with a wider institutional strategy” (Eustice, 2011, no page). This mobile application was developed in the line with the university’s strategic plan as it effectively used an informal mobile learning method to transform the students’ learning experience.

CHAPTER 7: CONCLUSIONS AND FURTHER DEVELOPMENT

6.1 Adherence to the project goal and objectives

In completion of this project, it has been established that the main goal to develop a university mobile application which supports informal learning whilst bridging the gap between formal and informal learning has been achieved. Moreover the project’s six objectives have also been met.

The forthcoming paragraphs demonstrate how these objectives were met. The requirements for the proposed system were captured through the use of focus groups as full documentation of the focus groups conducted can be found in **Appendix D**. The mobile application has a high level of acceptance amongst students as it is fit-for-purpose and all of the factors outlined in the literature review that affect Smartphone application acceptance were favourably met as seen in **Appendix B**. One of these critical factors highlighting the success of the mobile application is

recommendation intention which is based on the actual use of the mobile application (Lee, Kim and Choi, 2012). 63% of students gave the mobile application a recommendation level of “Quite Recommended” which indicated that the mobile application was accepted. This signified that it attracted positive student feedback, thereby potentially promoting a mobilely fluent campus by fostering continuous use.

This project made use of user stories to gather requirements allowing the customer to clearly state the functionality being delivered. The customer consistently provided direction on how to deliver business value throughout the lifecycle of the project (Baird, 2003). This was vital in my role as the developer to ensure the mobile application had genuine useful functionality relevant to its target audience.

The fourth objective was met as the acceptance tests which evaluated the usability of the mobile application passed successfully. These tests included assessment criteria such as attractiveness, ease of use and level of intuitivism based on the Android Design Guidelines. The fifth project objective was met as the mobile application contains personalization features in which it personalises information students like to have at hand such as campus alerts. The final objective has been met as the mobile application was developed in line with the university’s strategic plan as it successfully used an informal mobile learning method to transform the students’ learning experience.

6.2 Evaluation of the appropriateness of the development methodology

Gauging the level of appropriateness of the development methodology can be viewed from the perspective of a developer and that of a graduate seeking employment in the industry. From the perspective of a developer, it was moderately appropriate to use Extreme Programming to develop the artefact. This was derived as some of XP’s core principles could not be applied during development whilst other phases of the development lifecycle were excessive for the scope of the

project. However there were several benefits ascertained from using XP such as effectively managing risks in keeping the project on track as evident in the project plan found in **Appendix A-2**.

By capitalizing on XP's principles and values, I was able to mitigate three major risks in the following ways. With regards to over-running time estimates, I was able to handle any slippages to releases that were more difficult which appeared later in the release plan by focusing on delivering system features that offered the highest business value appearing at the top of the list. This allowed me to work on the more difficult tasks with minimal impact to the customer in order to avoid over-running time estimates for these tasks.

Another non-technical risk I overcame with XP included the need to have an on-site customer. This was mitigated by ensuring that the customer was actively involved throughout the development process and selected the functionality being delivered within short feedback loops. A thorough project inception which was conducted to evaluate the need for the system helped mitigate the risk of building an excellent mobile application that is not wanted or required by the customer. Moreover the adoption of XP enabled me to create a quality artefact incorporating key attributes of a high degree of usability, functionality and extendibility with the use of XP practices such as the system metaphor, small releases and simple design (Baird, 2003).

From the standpoint of a computer science graduate, XP was very suitable for this project because utilizing it has reflected my growth as a software engineer to develop a piece of software that will benefit a real client. Moreover it will help boost my employability skills as it has allowed me to demonstrate use of an agile methodology in my project. XP is a highly flexible, emerging customer focused agile development methodology which the industry is currently keen about. It is considered as one of the best practices in the industry with its adoption of continuous integration and test-driven development.

6.3 Suitability of the programming language utilized

The decision to use the Lua programming language to develop the mobile application was an excellent choice. My previous knowledge and experience using Lua allowed me to resolve challenges with coding easily thereby enabling me to focus on other important factors of the mobile application such as the User Experience. I also had the support of the Department of Computer Science, which provided tablets to test my code and could contact my previous lecturer in Mobile Applications who directed me to appropriate resources and technical help on the language. Moreover it allowed for fast development of the mobile application and with the integration of the Corona SDK development platform, I was able to develop a high quality educational mobile application.

6.4 Lessons learned

One of the key lessons I learned from undertaking this project is that using XP activities such as incremental planning where the release plan was broken down into releases and iterations, allows the customer to alter the direction of the development as work progressed. This was vital in managing changing requirements and the scope of the project which is central to one of XP's principles of embracing change. Towards the completion of integration, I discovered the importance of continuous integration during the development of the system as it is critical for monitoring changes which occur to code, making rollbacks to code after committing changes and integrating code daily in monitoring the health of the software.

As a result of my failed efforts in implementing continuous integration, I found the experience very insightful as I now have a deeper understanding of its inner workings which is essential as a graduate in software engineering. As automated build and testing is a critical component of continuous integration, in doing things differently I would have carefully selected a development language such as Java which has the support of build automation software like Ant. I also recognized that the implementation of continuous integration would have been more applicable in a

team environment where several programmers build their code and migrate it to one integration machine.

Moreover, I could have investigated development tools used to automate my acceptance and unit testing as this is a core part of XP. Exploring the inner workings of pair programming has also been a valuable lesson from this project experience. Although not implemented in the project, I placed emphasis on understanding this XP practice and have realized the benefits of this practice in development. Essentially when “two developers share the same machine and work side-by-side to solve and produce” (Baird, 2003, no page), this can result in better quality code as the result of continuous code review as well as shorten development time.

Additionally, a notable discovery I made from the completion of this project was my efficient use of project management techniques which helped me to control the project and ensure deadlines for deliverables were met. A copy of the project plan outlining these techniques can be found in **Appendix A**.

Although the mobile application was developed in the line with the university’s strategic plan as it effectively used an informal mobile learning method to transform the students’ learning experience, I would have integrated active stake holder engagement in my evaluation of the artefact. Such an evaluation would have contributed considerably to the project’s credibility regarding the quality level and return of investment for the stakeholder of the mobile application.

According to Power (2010) a stakeholder is predominately limited to customers, end users or project sponsors who have a significant influence over product development activities. Within XP, stakeholders also include developers, so in particular I would have sought out feedback from the members of Boards of Governors who would be responsible for sponsoring the project as well as developers who have already created and distributed app in-house on behalf of the university. In so doing I would have also incorporated a fundamental agile modelling core practice of active stakeholder participation.

6.5 Recommendations for further development

Based on the high acceptance and the need expressed by students for the developed mobile application, this project is extremely viable for the opportunity for further development. The university should initiate inter-department collaborative research between the Centre for Learning Excellence and the department responsible for distributing in-house apps at the university in developing the mobile application further. Such development can include activities aimed at targeting the best practices identified by the professional community that have not been met in this project. These include making the mobile application available across other platforms such as the iPhone. Within the survey which allowed students to rate the mobile application, when asked what features or tools they would like to have, the majority of students wanted to have access to university related data, in particular their timetabling and interactive campus maps.

Integrating the mobile application with other internal systems should therefore be explored in order to improve the genuine functionality of the mobile application and include such features. In implementing features such as interactive campus maps, the university should take note of “a master's degree student at University of Limerick (UL) who has introduced an app that will help first-year students find their way around the 332-acre campus by scanning QR codes to access indoor maps of each of its 40 buildings”(Burke, 2012, no page). Research findings used to develop the mobile application was conducted primarily at the Luton campus. However in order to improve the marketability of the mobile application and its outreach to the wider student population, the university should consider including information on university services at the Bedford campus as well. Additionally the personalization of navigation and presentation could be integrated in improving the user experience. If Extreme Programming is selected as the development methodology in improving the mobile application, the university should seriously consider implementing the last two phases of XP lifecycle which are Productionizing and Maintenance. Productionizing allows the university to certify and verify the deployment the mobile application to the customers' environment, with the end result to release the software so it is available for download. Maintenance would also be conducted for distributing patches and enhancements in upgrading the system.

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APPENDIX A: PROJECT PLAN

A series of project management techniques outlined in Dawson (2009) was utilized during the development of the project plan which identifies the work to be completed and provides a timeline of deadlines to ensure the project was kept on track.

Work Breakdown Structure

This project planning technique was employed to give a pictorial overview of the hierarchy of work that needed to be performed. A copy of the Work Breakdown Structure can be seen in **Appendix A-1**.

Risk Management

A complete risk management process was adopted to identify, assess, alleviate and control risks as work on the project progressed. A copy of the Risk Management document can be seen in **Appendix A-2**.

Time Estimates

These estimates will give a projection on how long the project will take to complete. Time estimates will provide an outline of the individual effort, estimated in weeks, which is required to complete individual tasks that make up an activity. The Time Estimates can be seen in **Appendix A-3**.

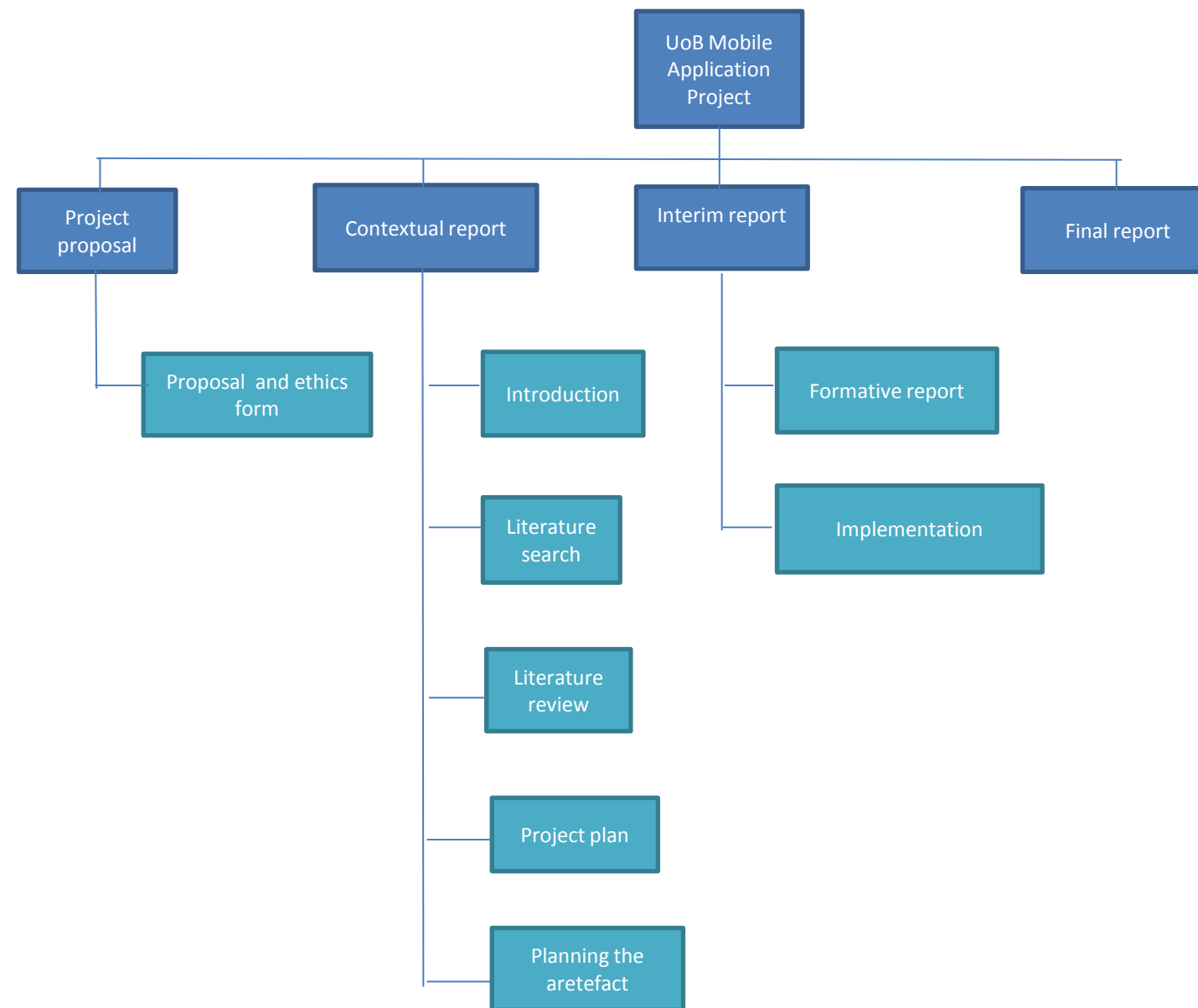
Milestone Identification

The purpose of milestone identification is to provide intermediate reference points which highlight the significant steps towards the completion of the project. The Milestone Identification can be seen in **Appendix A-4**.

Activity Sequencing

This project utilized Activity sequencing found in **Appendix A-5** as a guide on the order in which the project work should be performed. Stemming from this, a Gantt chart seen in **Appendix A-6** was also used for keeping track of the duration of activities and monitoring simultaneous tasks which were required to be performed.

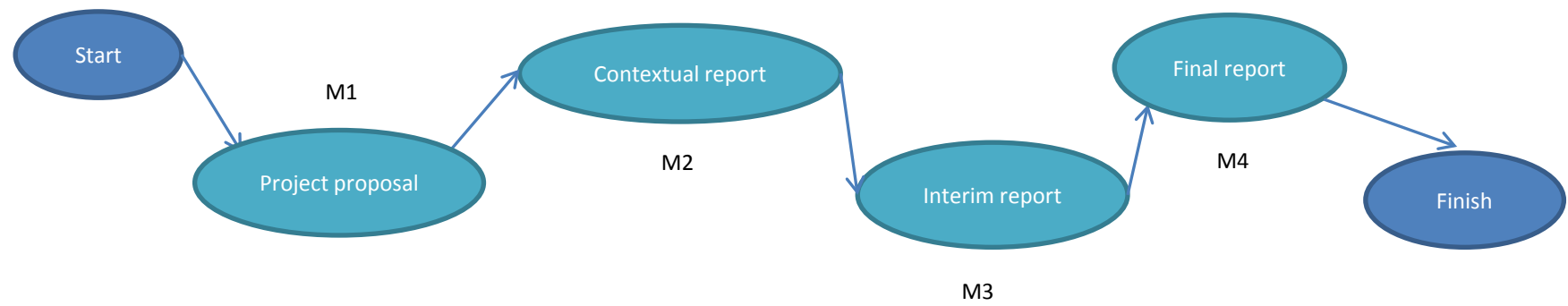
A-1: Work Breakdown Structure



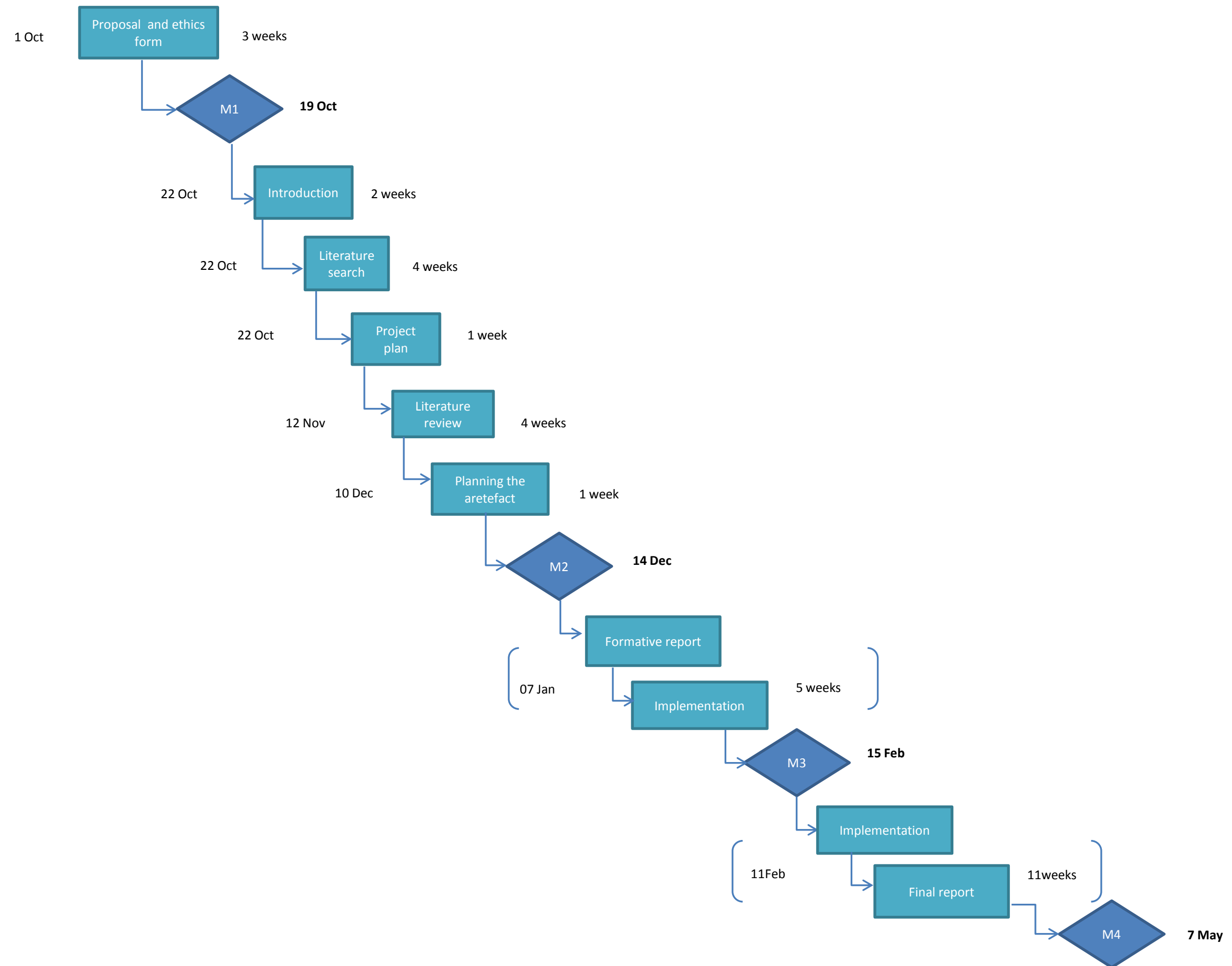
A-2: Risk Management												
Risk type	Checkpoint 1:			Checkpoint 2:			Checkpoint 3:			Checkpoint 4:		
Technical	Risk impact	Risk classification	Risk management	Risk impact	Risk classification	Risk management	Risk impact	Risk classification	Risk management	Risk impact	Risk classification	Risk management
Newness to technology as there is unfamiliarity with the development platform.	2 * 4 = 8		Reduction	2 * 3 = 6		Reduction	1 * 1 = 1		Reduction	1 * 1 = 1		Reduction
Unfamiliarity with the unit testing and building tools.	2 * 4 = 8		Reduction	2 * 3 = 6		Reduction	1 * 1 = 1		Reduction	1 * 1 = 1		Reduction
High possibility that requirements will change according to customer's needs for the mobile application.	3 * 4 = 12		Contingency	3 * 4 = 12		Contingency	1 * 2 = 2		Contingency	1 * 1 = 1		Contingency
Inability to integrate mobile application with existing university internal systems.	3 * 3 = 9		Acceptance	3 * 3 = 9		Acceptance	3 * 3 = 9		Acceptance	3 * 3 = 9		Acceptance
Possibility loss of data due to failure of laptop.	1 * 5 = 5		Avoidance	1 * 4 = 4		Avoidance	1 * 4 = 4		Avoidance	1 * 4 = 4		Avoidance
Not creating a cross platform mobile application.	2 * 4 = 8		Acceptance	1 * 5 = 5		Acceptance	1 * 5 = 5		Acceptance	1 * 3 = 3		Acceptance
Handling the complexity of the mobile application to be built based on its features.	2 * 3 = 6		Contingency	2 * 3 = 6		Contingency	1 * 4 = 4		Contingency	1 * 5 = 5		Contingency
Difficulty implementing build automation and automated testing.	3 * 4 = 12		Acceptance	3 * 4 = 12		Acceptance	3 * 4 = 12		Acceptance	3 * 4 = 12		Acceptance
Non-technical												
Implementation of pair programming.	3 * 4 = 12		Acceptance	3 * 4 = 12		Acceptance	3 * 4 = 12		Acceptance	3 * 4 = 12		Acceptance
Difficulty in having an onsite customer.	2 * 3 = 6		Avoidance	2 * 2 = 4		Avoidance	2 * 1 = 2		Avoidance	2 * 1 = 2		Avoidance
Over-running time estimates.	2 * 4 = 8		Reduction	2 * 4 = 8		Reduction	2 * 2 = 4		Reduction	1 * 1 = 1		Reduction
Difficulty arranging regular meetings with the customer.	2 * 4 = 8		Prevention	2 * 4 = 8		Prevention	1 * 2 = 2		Prevention	1 * 1 = 1		Prevention
Building an excellent mobile application that is not wanted or required by the customer.	2 * 5 = 10		Reduction	2 * 1 = 2		Reduction	1 * 1 = 1		Reduction	1 * 1 = 1		Reduction
Ensuring content of mobile application is dynamic and up-to-date.	2 * 5 = 10		Acceptance	2 * 3 = 6		Reduction	1 * 1 = 1		Reduction	1 * 1 = 1		Reduction
Implementation of continuous integration.	3 * 4 = 12		Acceptance	3 * 4 = 12		Acceptance	3 * 4 = 12		Acceptance	3 * 4 = 12		Acceptance
Checkpoint 1: Upon completion of the project proposal.												
Checkpoint 2: Upon completion of the contextual report.												
Checkpoint 3: Upon completion of the interim report.												
Checkpoint 4: Upon completion of the final report.												
Risk impact = likelihood * consequence												
1 to 5 - Green Risk												
6 to 10 - Amber Risk												
Greater than 10 - Red Risk												
Risk Likelihood	Score											
Low	1											
Medium	2											
High	3											
Risk Consequence	Score											
Very Low	1											
Low	2											
Medium	3											
High	4											
Very High	5											
Risk management techniques employed												
Avoidance: Identifying and implementing alternative procedures to eliminate the risk.												
Contingency: Having a pre-determined plan of action to come into force as and when the risk occurs.												
Prevention: Employing countermeasures to stop a problem from occurring.												
Reduction: Taking action to minimise either likelihood of the risk developing or its effect.												
Transferrance: Transferring the risk to a third party.												
Acceptance: Tolerating the risk when its likelihood and impact are relatively minor or when it would be too expensive to mitigate.												

A-3: Time Estimates	
Activity	Estimated Duration (in weeks)
Proposal and ethics form	3
Introduction	2
Literature search	4
Literature review	4
Project plan	1
Planning the artefact	1
Formative report and implementation	5
Final report and implementation	11
Total Effort	31

A-4: Milestone Identification



A-5: Activity Sequencing



APPENDIX B: USER ACCEPTANCE SURVEY FINDINGS

Distribution of participants according to gender

The gender makeup of participants was proportionally balanced with twelve females and thirteen males completing the survey.

Distribution of participants across age group

The majority of participants came from the 18-25 and 26-35 age group as indicated in **Figure 1**.

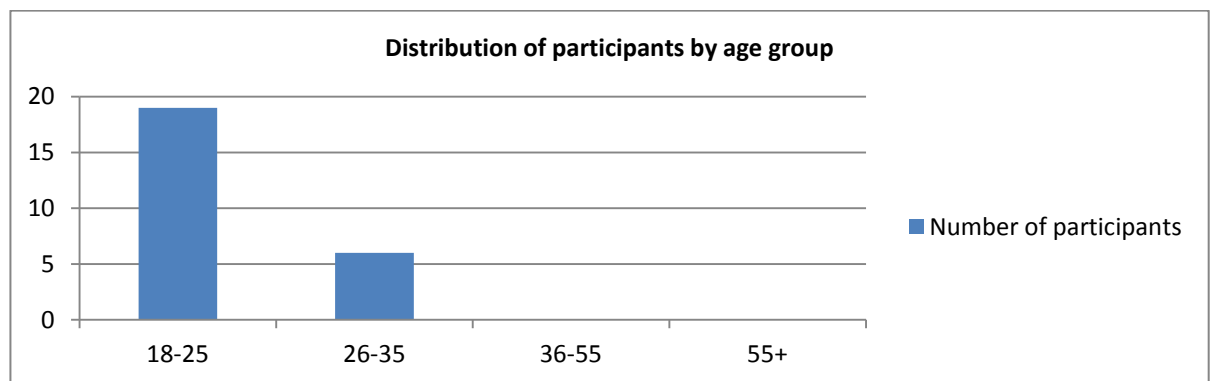


Figure 1

Distribution of participants across the various subject areas

A cross spectrum of responses ranging from the various subject areas was well-proportioned as shown in **Figure 2**.

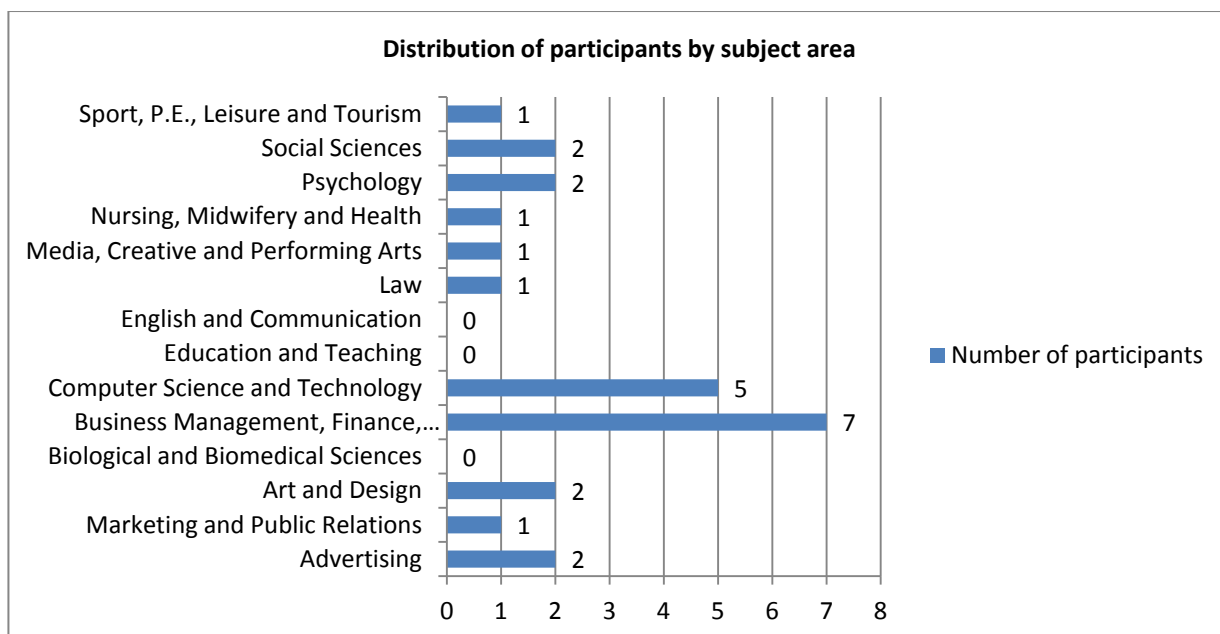


Figure 2

Distribution of respondents by student type

It was evident that the distribution of participants by student type was balanced with 44% home students and 56% overseas students. This is shown in **Figure 3**.

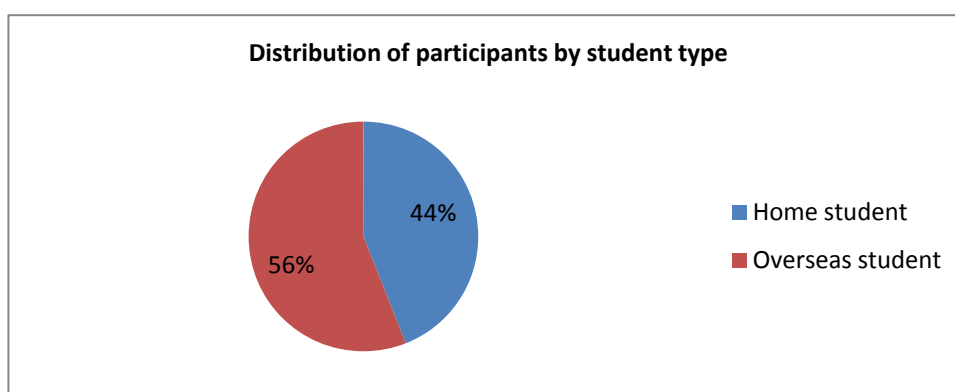


Figure 3

Distribution of participants by study level

An assessment of the study level of participants indicated that the response level came mainly from the undergraduate and postgraduate study level. These responses were well-proportioned with fourteen undergraduate students and eleven postgraduate students completing the surveys as indicated in **Figure 4**.

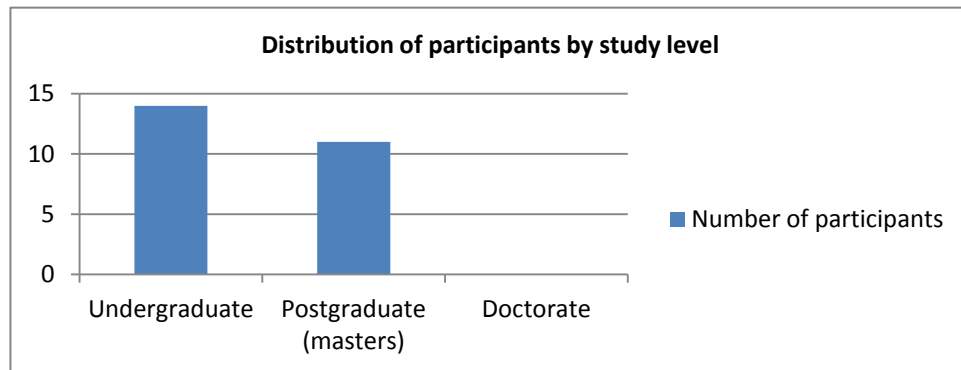


Figure 4

Ownership of Smartphones by participants

The level of ownership of Smartphones by participants was extremely high as 96% of participants stated that they owned a Smartphone. Only 4% cited that they did not own a Smartphone. This is clearly noted below in **Figure 5**.

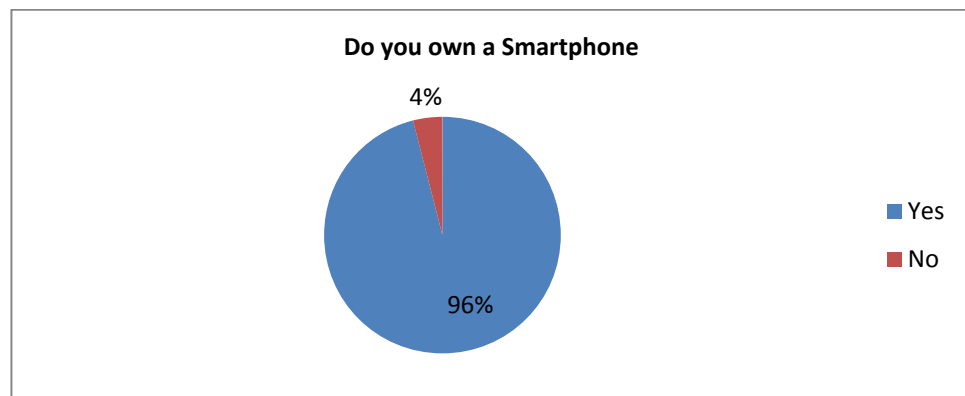


Figure 5

Models of Smartphones participants own

It was evident that the Android model of Smartphone was dominant in the participants' ownership with 40% of participants owning an Android Smartphone. This was followed by closely by the iPhone (28%), Blackberry (24%) and Nokia (4%) as indicated in **Figure 6**.

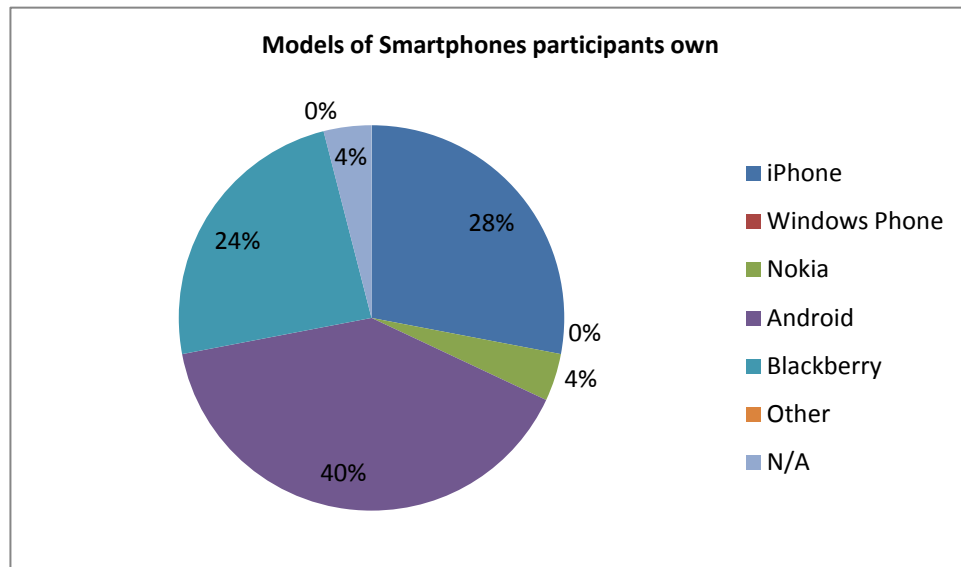


Figure 6

Frequencies of informal mobile learning activities in which participants engage

As noted in **Figure 7**, the most popular informal mobile learning activities amongst participants were those that they engaged in either on a weekly or daily basis. These included: - planning studies and managing their time by recording appointments using the calendar tool, take notes outside of class to support their own individual learning purposes, connect with other university students through Facebook, Twitter to share their knowledge, access up-to-date feeds on university services such as SID, library services and career services and search for sights, activities, restaurants and entertainment spots using Google Maps with GPS.

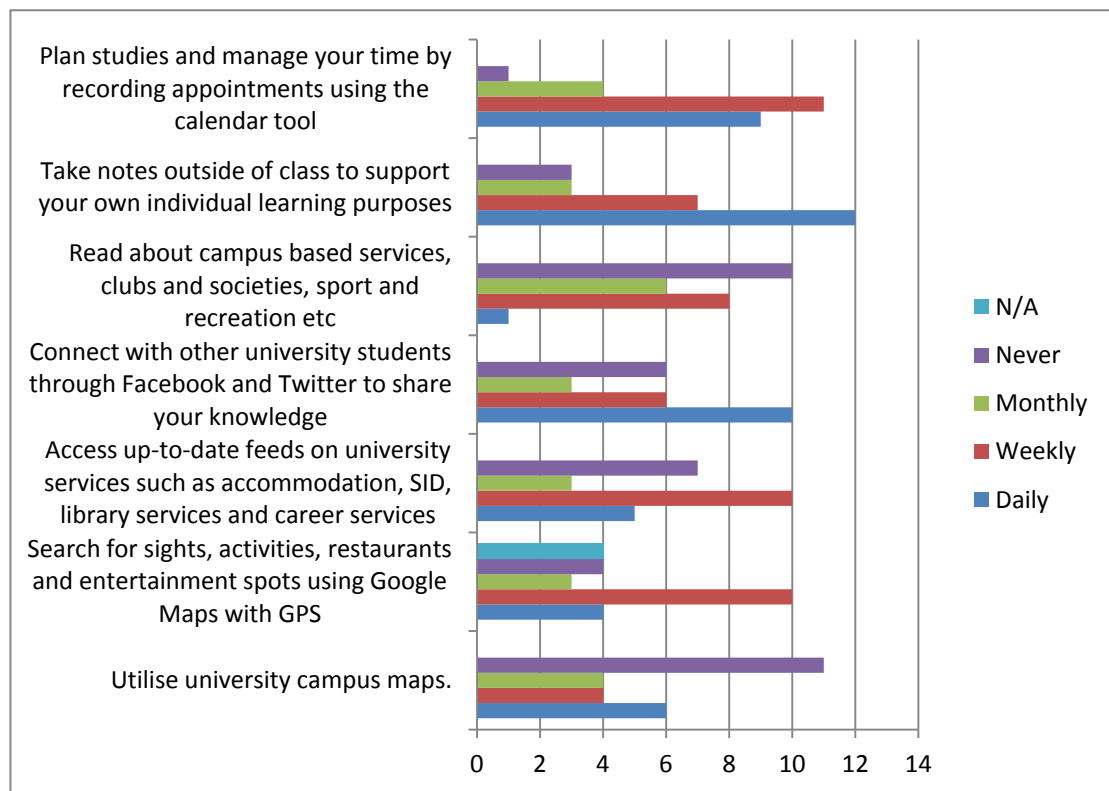


Figure 7: Informal mobile learning activities amongst participants

Usefulness of informal mobile learning activities as according to participants

Participants found a number of informal mobile learning activities that were drafted in the features of the artefact within the proposal to be very useful as shown in **Figure 8**. These included: - planning studies and manage their time by recording appointments using the calendar tool, taking notes outside of class to support their own individual learning purposes, connecting with other university students through Facebook and Twitter to share their knowledge and access up-to-date feeds on university services.

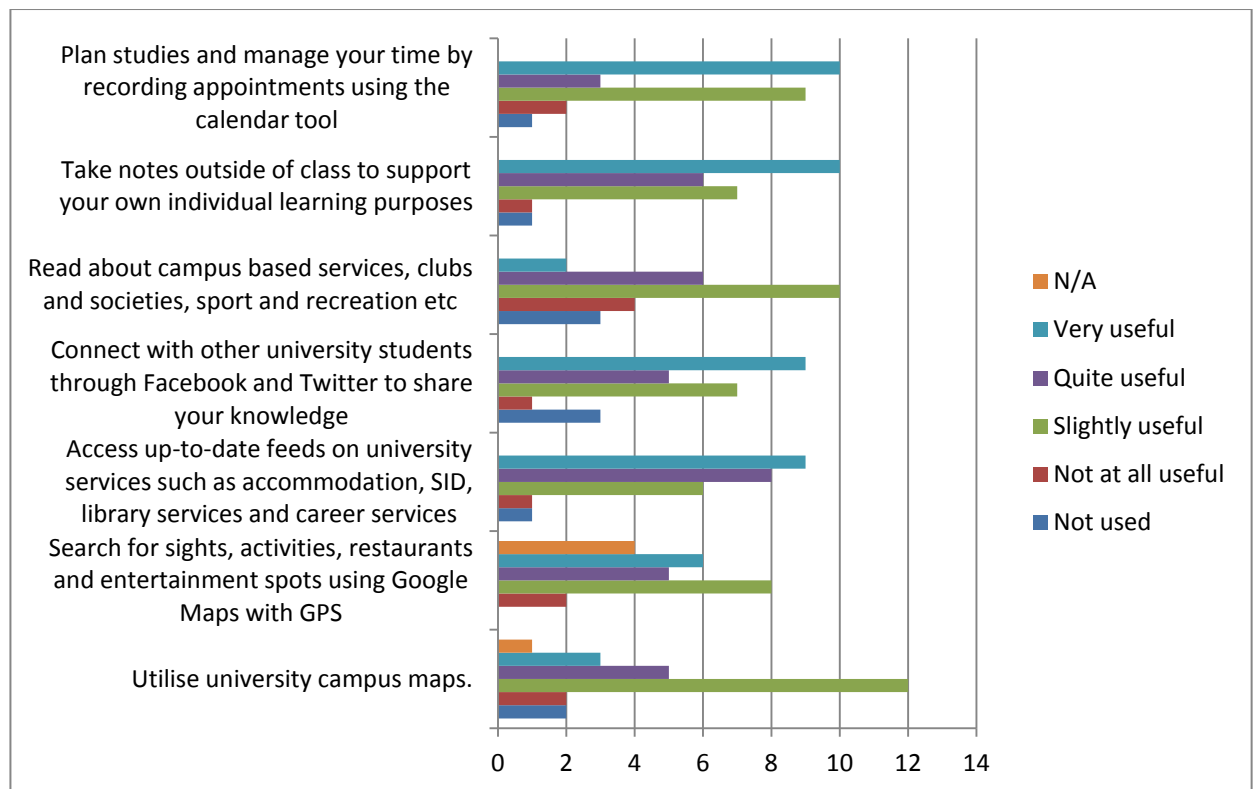


Figure 8

Perceived advantages of using the proposed system

The following perceived advantages as highlighted in **Figure 9** scored highest amongst the participants.

These included:

- Staying up-to-date on recruitment events and services to assist with their employability.
- Ability to make better choices to improve management of student life.
- Keeping informed within the university community to help adjust quickly to university life.

Participants also felt that the proposed system could help them be familiarized with the university and the Luton community.

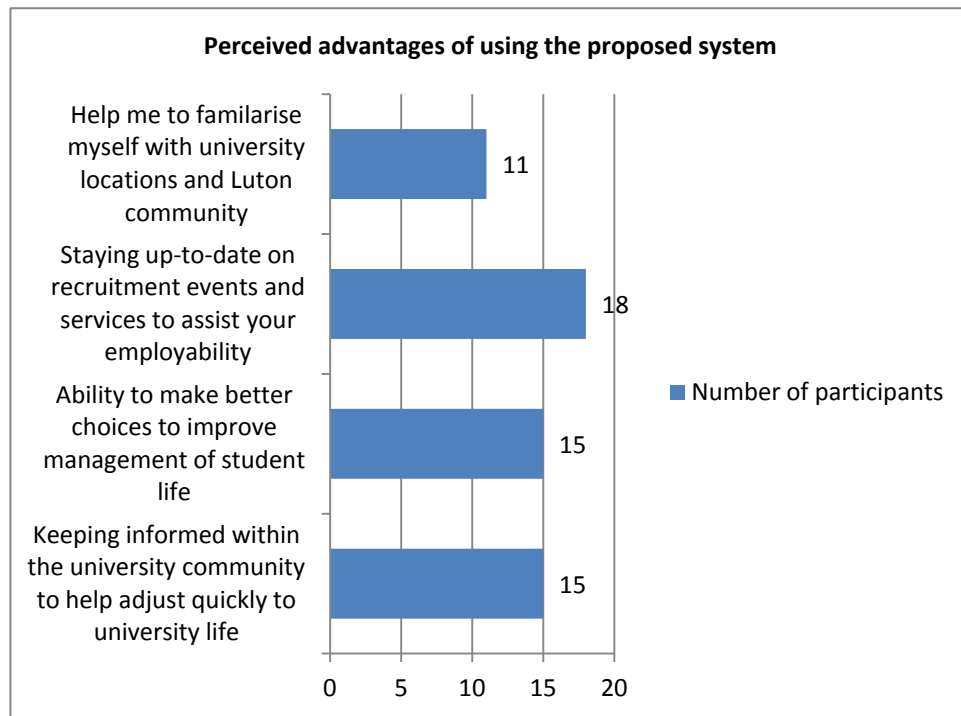


Figure 9

Assessing how useful respondents would find the implementation of the proposed system

The nature of the responses to this question was positive. 36% of participants felt the proposed system would be extremely useful. This was followed by 28% who felt it would be very useful and 24% who felt it would be useful. These results are clearly indicated in **Figure 10**.

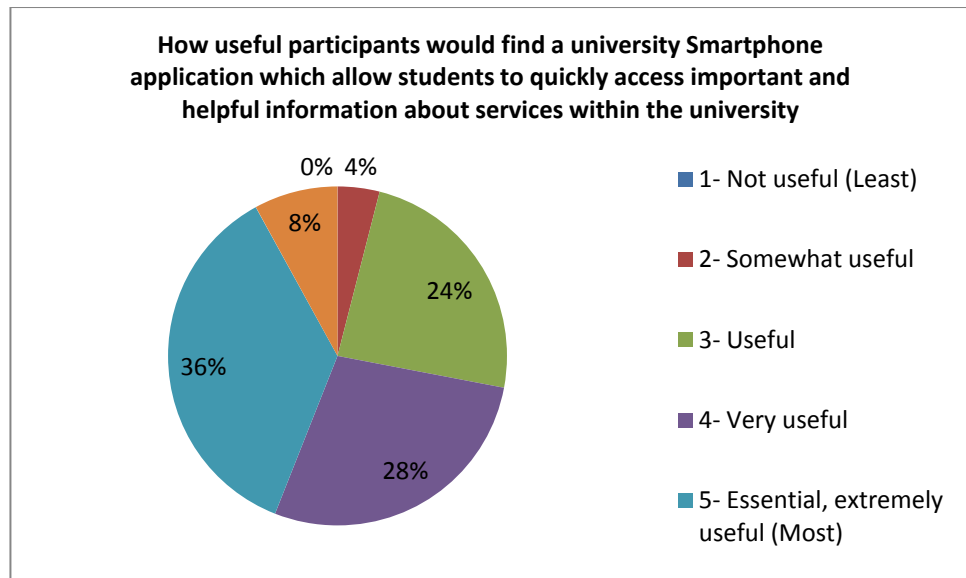


Figure 10

Assessing the need for the proposed system

72% of respondents said they would prefer to regularly use the proposed system in the future rather than any other means available based on its description. A minority of students (24%) disagreed as they stated they would not prefer to regularly use the proposed system in the future rather than any other means available based on its description. This is clearly shown in **Figure 11**.

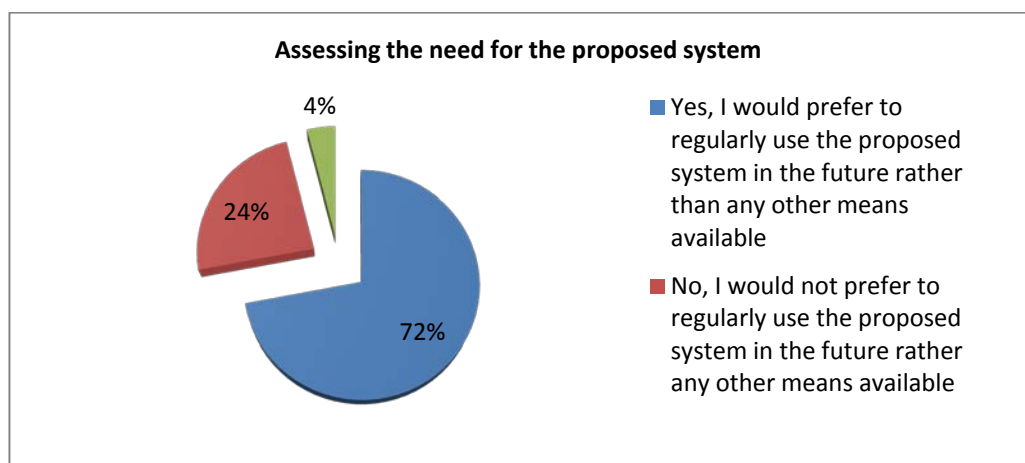


Figure 11

Assessing the level of recommendation intention for the proposed system

The majority of participants (92%) indicated that they predict that they would recommend the proposed system to other students based on its description. This is illustrated below in **Figure 12**.

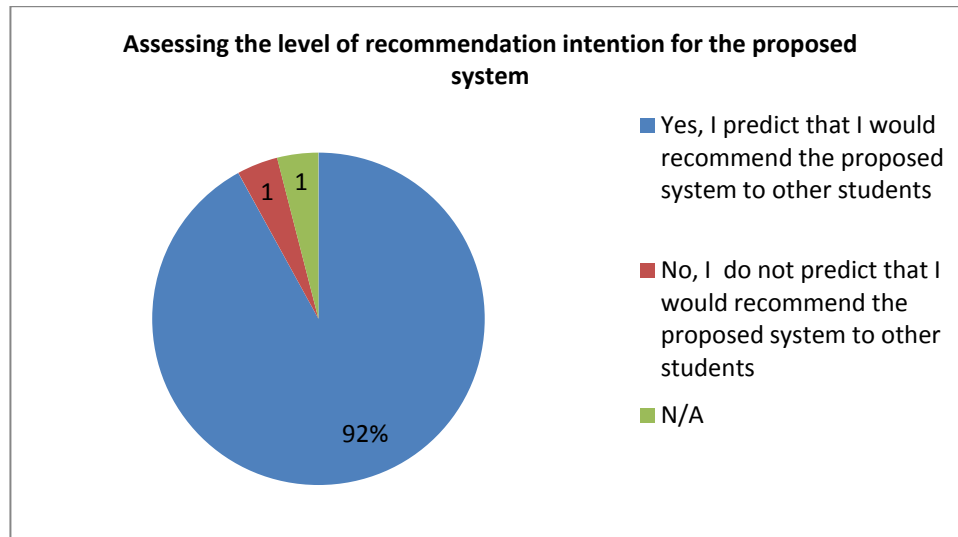


Figure 12

APPENDIX C: SURVEYS CONDUCTED

C-1: User Acceptance Survey

Informed Consent Form

Participant Information: My name is Suzanne Gilkes and I am in my final year completing a BSc. in Computer Science and Software Engineering. I am conducting a research study as part of my undergraduate project. This project is expected to last seven months. The title of this research project is the development of a university mobile application which will allow students to quickly access important and helpful information about services within the university. Twenty-five participants have been chosen at random for inclusion in this study.

Purpose of this study: The goal of this project is to develop a university mobile application which supports informal learning whilst bridging the gap between formal and informal learning.

Participant Involvement: You will be required to answer a survey containing ten questions in this User Acceptance Survey. Your participation is voluntary and you can stop at anytime. Your participation in this study will take approximately ten minutes. Your answers will be tabulated with those of others and will be treated as anonymous ensuring confidentiality.

Data Usage: The data collected in this survey will be used to determine your acceptance of the proposed system.

Confidentiality: To prevent unauthorized physical access to these surveys, these materials will be stored in a locked cabinet in my supervisor's office. All data collected will be computerized and encrypted to prevent unauthorized electronic access. All data collected will be stored for a minimum of three years before it is destroyed. In the event of a publication or presentation resulting from the research, no personally identifiable information will be shared.

Questions about the research and rights of research participants: Please contact me via my email at susannah_lily@hotmail.com with questions, complaints or concerns about this research.

Benefits to the Participant: You may find out what sources of information are important to your learning experience outside of the traditional classroom environment.

Right to Withdraw from the Study: You do not have to answer any questions you do not want to answer. Refusal to take part in or withdrawing from this study will involve no penalty or loss of benefits you would receive otherwise.

Dissemination of Research Outcomes: Dissemination of the research outcomes will be released to participants who are willing to provide their email for this purpose and have an interest in the outcomes of the study.

You must be 18 years of age or older to take part in this research study. Please fill out and sign the consent form below to indicate whether you agree to participate in this research. You will be given a copy of this form to keep after you have signed it.

Have you read this consent form or it has it been read to you? Yes _____ No _____

Have all of your questions about the form and the research been answered? Yes _____ No _____

Do you understand the nature of the research in which you are about to participate? Yes _____ No _____

Do you agree to participate in this study? Yes _____ No _____

Name of Participant (please print): _____

Signature of Participant: _____ Date: _____

Researcher's Signature _____ Date: _____

Demographics and Background

Gender: female ☐ male ☐

Age Group: 18-25 ☐ 26-35 ☐ 36-55 ☐ 55+ ☐

Home student ☐ Overseas student ☐

Level: Undergraduate ☐ Postgraduate (masters) ☐ Doctorate ☐

Subject Area:

Advertising ☐ Marketing and Public Relations ☐ Art and Design ☐

Computer Science and Technology ☐ Sports, P.E., Leisure and Tourism ☐ Psychology ☐

Biological and ☐ Education and Teaching ☐ Social Sciences ☐

Biomedical Sciences ☐ Media, Creative and Performing Arts ☐ Law ☐

Nursing, Midwifery and Health ☐

Assessing current technology used

1. Do you currently own a Smartphone? Yes ☐ No ☐

2. If yes, please select the brand of Smartphone you use:

iPhone ☐ Android ☐

Windows Phone ☐ Blackberry ☐

Nokia ☐ Other _____

Assessing your current practices

3. Please tick the types of informal mobile learning activities you conduct based on the frequencies below. Tick **ONE** box where applicable for each activity type.

Location Aware

Daily Weekly Monthly Never

Utilise university campus maps.

Search for sights and activities, restaurants and

entertainment spots using Google Maps with GPS.

☐ ☐ ☐ ☐

Referential

Access up-to-date feeds on university services

such as accommodation, SID, library services and

career services.

□ □ □ □

Constructive

Connect with other university students through

Facebook and Twitter to share your knowledge.

□ □ □ □

Reflective

Read about campus based services,

clubs and societies, sport and recreation etc.

□ □ □ □

Data Collection

Take notes outside of class to support your own

individual learning purposes.

□ □ □ □

Administrative

Plan studies and manage your time by recording

appointments using the calendar tool.

☐ ☐ ☐ ☐

Evaluating usefulness of informal mobile learning activities

4. Please rate your usefulness of these informal mobile learning activities.

From **1**(Not used), **2** (Not at all useful), **3** (Slightly useful), **4** (Quite useful), **5** (Very useful)

Location Aware

1 2 3 4 5

Utilise university campus maps.

☐ ☐ ☐ ☐ ☐

Search for sights and activities, restaurants and

entertainment spots using Google Maps with GPS.

☐ ☐ ☐ ☐ ☐

Referential

Access up-to-date feeds on university services such

as accommodation, SID, library services and career

services.

☐ ☐ ☐ ☐ ☐

Constructive

Connect with other university students through

Facebook and Twitter to share your knowledge.

☐ ☐ ☐ ☐ ☐

Reflective

Read about campus based services,

clubs and societies, sport and recreation etc.

☐ ☐ ☐ ☐ ☐

Data Collection

Take notes outside of class to support your own

individual learning purposes.

☐ ☐ ☐ ☐ ☐

Administrative

Plan studies and manage your time by recording

appointments using the calendar tool.

☐ ☐ ☐ ☐ ☐

Assessing your acceptance of the proposed system

5. Personalization

Personalization is about customizing a product to satisfy your individual needs.

The proposed system will be fully personalized through the following features:

Personalising your information

Customised information such as viewing your selected university feeds etc.

Personalising your navigation

Ability to customise organisation and arrangement of content.

Personalising your presentation

Ability to choose customised backgrounds and colours.

Are there other features you would like to see implemented for personalization?

Yes ☐ No ☐

If yes, please state these features

6. Performance Expectancy

What are some of the derived advantages you think you can gain from the proposed system?

- ☐ Keeping informed within the university community to help adjust quickly to university life.
- ☐ Ability to make better choices to improve management of student life.
- ☐ Staying up-to-date on recruitment events and services to assist with your employability.
- ☐ Help me to familiarize myself with university locations and Luton community.
- ☐ Other _____

7. Effort Expectancy

How useful would you find a university mobile application which will allow students to quickly access important and helpful information about services within the university?

Please circle, rate from 1 (least) to 5 (most): **1 2 3 4 5**

8. Usage Intention

Based on the description of the proposed system, I would prefer to use the proposed system on a regular basis in the future rather than any other means available? Yes ☐ No ☐

9. **Recommendation Intention**

Based on the description of the proposed system, I predict that I would recommend it to other students. Yes ☐ No ☐

Thank you for completing this survey.

10. Would you be interested in the research outcomes of this project? Yes ☐ No ☐

If yes, please provide your email address here. Your email address will only be used to inform you about the research outcomes of this study.

C-2: Recommendation Level Survey

Thank you for participating in these focus group sessions in which your valuable input was utilized for the development of the mobile application.

1. Gender: female ☐ male ☐

2. Please let me know of your level of recommendation about the mobile application based on your user experience.

Select **ONE** of the following:-

Not Recommended ☐ Slightly Recommended ☐
Quite Recommended ☐ Highly Recommended ☐

3. Are there any additional desired features you would like to see implemented in future developments for the mobile application? Yes ☐ No ☐

If yes, please state these desired features below:

APPENDIX D: FOCUS GROUP SESSIONS CONDUCTED

D-1: Analysis of the first focus group session conducted

Purpose

The main purpose of the first focus group session was to derive the requirements for the mobile application based on the feedback of the students who participated in the Peer Assisted Learning Scheme. To achieve this, an evaluation of:- what kind of information they would like access on university services, their attitudes to using their Smartphones for learning through the use of user scenarios and how they use their Smartphones for learning was performed as these were critical to the scope of the project.

Analysis of Transcript

A transcript of the tape-recording from the focus group session was created and the findings are presented in the next section.

Findings

Selection of university services and information required

In response to the first question, out of the fifteen students who participated in the focus group, the majority of the students, thirteen in all, were aware of, used and found useful the following university services in the context of the mobile application: Library Services, Careers and Employment Services, International Student Support Services and Student Information Desk.

In re-assessing the project scope, based on the User Acceptance Survey, it was thought that students would also want to access information on community services based on their feedback on the advantages of using the system. Students were aware of these services, all students felt that they would use or found them useful in the context of the mobile application.

Fourteen out of the fifteen students indicated they wanted to call the selected university services via the phone. Information such as an overview of the purpose of the university service, their location, opening times, location and phone number should be included.

Use of Smartphone to help with learning

Fourteen out of the fifteen students did use their Smartphone to help with their learning engaging in referential, constructive, reflective and data collection informal mobile learning activities outlined in the User Acceptance Survey. With regards to location aware activities, students felt that the use of Google Maps was not accurate in pinpointing the locations of some places on campus. Moreover it could not be used for mapping directions inside the campus buildings. All fifteen students also utilized their Smartphone for administrative activities.

Creating User Stories

Overall throughout the creation of the user stories students' attitudes were positive towards flexible learning and therefore the potential for mobile learning. User stories were created based on the following user scenarios and corresponding questions which were presented to the students. In doing so students felt encouraged to provide user scenarios of their own and this helped them to create well formed user stories.

All fifteen students felt that being able to learn at any time in any place was extremely important to them and agreed with the corresponding user scenarios presented. Based on this the following high level user stories were created:

Access Student Support Services

Enables students to access important and helpful information about student support services in order to help them adjust quickly to university. Such services include:- Career and Employment Services, Library and Learning Resources, Student Information Desk, Student Union, International Student Support Services which are commonly accessed. Viewing a list of events, clubs and societies provided by the Student Union and information on services within the community was also important to participants. Important information include opening times, description of services offered, location and contact numbers to dial directly via the mobile application.

View Events

View a list of up-to-date university events provided by the Student Union via Twitter Feeds.

View Club and Societies

View a list of clubs and societies provided by the Student Union.

Access Services within the Community

Enables students to access important and helpful information about services within the community such as food and retail, financial services, sights and activities and arts and culture which are commonly accessed, help students adjust quickly to university life. Such information includes opening times, description of services offered, location and contact numbers to dial directly via the mobile application.

View Local Maps

View local maps containing locations of services within community and the university to allow students to familiarize themselves with these locations in the community and university.

Twelve out of the fifteen students did not agree with the first user scenario presented in my attempt to gauge what the university could do to make it easier to engage in mobile learning. They felt that receiving alerts on deadlines for reminders on assignments would be annoying. However all students were in agreement for having the ability to view campus alerts, personalize which alerts they can view as well as using a student profile in accessing the mobile application.

The following user stories were devised:

View Campus Alerts

View alerts on library services, career and employment services, international student support services, and student information desk services for improved management of student life.

Personalize Alerts

Enables students to select campus alerts they would like to view for improved management of student life.

Access Student Profile

Allows students to create a profile which enables them to use the mobile application and enables additional personalization features.

All students agreed with the user scenario reflecting using the Smartphone to organise student life and created the following high level user story.

Access Calendar

Plan studies and manage your time by recording appointments using the calendar tool.

All students also agreed to the user scenario demonstrating the use of Social Media to connect with other students and engage in a collaborative environment to share information regarding their coursework.

The following user story was created:

Access UoB MOBILE on Facebook

Students can access the Facebook group page via the mobile application in order to connect with other university students to share knowledge regarding course work.

All students felt that it was very important to access learning material via their Smartphone and with full agreement with the user scenario presented established the following user story:

Access Note Taker

Create notes for your lectures on the go for improved management of student life.

D-2: First focus group session conducted

DATE HELD: 17 January 2013

INTRODUCTIONS

Hello, my name is Suzanne Gilkes and I am a final year student completing my undergraduate project.

PURPOSE OF FOCUS GROUP

The title of this research project is the development of a university mobile application for the Android Smartphone which will allow students to quickly access important and helpful information about services within the university. I would greatly appreciate your assistance in participating in a focus group which will require periodic visits to yourselves, in order to develop the mobile application. The short focus group session will take approximately 30 minutes. I kindly request your input and want you to share your honest and open thoughts with me. Thank you for your willingness to participate in this focus group. Your feedback during this focus group session is vital for the development of the mobile application.

OVERVIEW OF TOPIC AND SESSION

This first focus group session will be used to create user stories which identify what you want to achieve from using the mobile application. The PAL scheme has several aims and is intended “to help students adjust quickly to university life, improve their study skills and adjust their study habits to meet the requirements of higher education and enhance their understanding of the subject matter of their course through collaborative group discussion” (University of Bedfordshire, no date c, no page). The proposed mobile application will not only address these students’ needs but capitalize on the benefits of mobile learning which are in direct alignment of the aims of the PAL scheme.

GROUND RULES

I would greatly value if everyone participated in the session. There are no wrong answers but rather differing points of view. Please feel free to share your point of view even if it differs from what others have said. Keep in mind that I am just as interested in negative comments as positive comments, and at times the negative comments are the most helpful. What is said in this room stays here. I want you to feel comfortable sharing when sensitive issues come up. I will be taping the focus group in order to capture everything you have to say. I will not identify anyone by name in my report. You will remain anonymous. I kindly ask that you turn off your phones. If you cannot and if you must respond to a call, please do so as quietly as possible and rejoin us as quickly as you can. Kindly let me know when you have completed the informed consent form and I will proceed to conduct the focus group. Before I begin, let's find out some more about each other by providing brief introductions of ourselves.

Preliminary Questions

Can I have a show of hands for the following questions, in the context of the mobile application?

1. How many of you are aware of, use and find useful the following university services?
 - Career and Employment Services
 - Library Services
 - Student Information Desk
 - Student Union
 - International Student Support Services

2. How many of you are aware of, use and find useful the following services within the community?
 - Food and Retail
 - Financial Services

- Sights and Activities
 - Arts and Culture
3. What kind of information would you find important about these services?
 4. Do you currently use your Smartphone to help with your learning, and if so, what do you use it for? The following informal mobile learning activities can assist with assessing your current practices.

Location Aware

Search for campus locations, sights and activities, restaurants and entertainment spots using Google Maps with GPS.

Referential

Access up-to-date feeds on university services such as accommodation, SID, library services and career services.

Constructive

Connect with other university students through Facebook and Twitter to share your knowledge.

Reflective

Read about campus based services, clubs and societies, sport and recreation etc.

Data Collection

Take notes outside of class to support your own individual learning purposes.

Administrative

Plan studies and manage your time by recording appointments using the calendar tool.

Creating User Stories

5. How much is the ability to learn at any time and in any place important to you?

User Scenario

“Mary uses her phone for learning because it is easier, it is accessible (always connected to the Internet and other people), you can use it anywhere and everywhere, and you don’t have to carry a heavy laptop around with you. She said “it really helps you because it saves on time and money. For example if you’re waiting for someone you can check on the latest events in university as well as updates on university services” (Bradley and Holley, 2010, no page).

User Scenario presented by two students

Two participants felt that the university could provide videos on using the universities services such as the Library Service as this would be very helpful in getting accustomed to using services provided by the university. This will be evaluated in the next focus group to determine if it would be required by the majority of participants.

6. Can the University do anything to make it easier to engage in mobile learning?

User Scenario-

“Mark is a first year student studying Computer Science. He is between 18-20 years old and has a Black Berry. About 24% of students have a BlackBerry because it is free to do Black Berry Messenger and you can access Facebook from it. He also believes that mobile phones are no longer just for calling and texting, and have become a fashion accessory, and that everyone wants to have the latest phone.

He reckons that he uses his phone more now than other students (because he’s just got the new, more sophisticated one). As an example, he mentioned that one student listens to music on his phone when he draws to help him concentrate. He thinks that all the students on his course will have used their mobiles at least once during the year for learning purposes. When asked if the University could do anything to make it easier to engage in mobile learning, he said maybe send texts about deadlines as reminders, such as have you done the work, do you need any help, if so contact this number” (Bradley and Holley, 2010, no page).

User Scenario-

John sets reminders in the calendar and using the clock and alarm to help organise his studying and his schedule (Bradley and Holley, 2010).

User Scenario-

“Shriya also uses her phone to access Facebook. One of her tutorial groups set up a Facebook group before Easter to facilitate the exchange of work and ideas for a group project. Over the Easter holidays, many of the students travelled abroad, and 3 of them were unable to fly back to the UK in time for the completion of the project because of the Volcanic ash cloud affecting European airspace. The group was therefore able to continue with the project, and those unable to be physically present at the University were able to participate through their participation in the Facebook group. What was started as a means of communicating turned into a “life-saver” for these students” (Bradley and Holley, 2010, no page).

7. How useful would it be to access learning materials via your mobile?

User Scenario-

“Shriya is a first-year International student from India, studying Computer Games and Design. She is female, aged between 18 to 20, and has a Blackberry Curve which she has had for about 2 months. During a course one of her tutors encouraged the class to use their phones to take photographs and exchange ideas. Everyone was very enthusiastic to use their mobiles, and it was because of this course that she bought a good mobile phone (she wasn’t interested in them previously because she didn’t realise the benefits). She also realised that using mobile phones in such a situation had the power to “hook” you into the learning activities. She bought a Blackberry because it is equipped with all the Windows software. She has an Apple MacBook laptop, and finds it difficult to connect with some of the University systems (such as Webmail, the email system), whereas on her Blackberry, she has no problems and can access anything she wants. She said “I’m forced to use my phone as my main source of

communication. She has started to use her Blackberry for so much now, that it has overtaken the use of her laptop. Having the ability to learn at any time and in any place is important to her, and her Blackberry facilitates this. She thinks it's extremely important to be able to access learning materials via her mobile. She believes that students are used to their phones and because they can now do such things, it is important.

She uses her phone to make notes using the 'Memopad' in which you can write quick notes and attach alarms to give reminders, which can also link through to the calendar. She also uses 'Word to go' which is Blackberry's cut-down version of Microsoft Word to write reports. She will start typing up notes in lectures, and then use these as the basis of her notes and reports. She thinks that the University could give students more encouragement to use their mobiles, for example give interactive learning sessions on how they could use their mobiles for learning. She believes that using mobile phones can get students interested in the subject more, and they are fun and help to create enthusiasm for learning" (Bradley and Holley, 2010, no page).

Prototyping with user stories

Let us take some time now to clarify your vision of the mobile application and its functionalities by creating a User Experience prototype and a use case diagram of the proposed mobile application presented via a website. This prototype is changeable and any changes in functionalities derived at the end of this focus group will be updated in the prototype. The creation of the use case diagram will help present a high level pictorial view of you, the user, and how you would interact with the system. This marks the end of the session. As a token of appreciation, please feel free to have some complimentary chocolates.

Thanks Again and See you soon.

Informed Consent Form

Participant Information: My name is Suzanne Gilkes and I am in my final year completing a BSc. in Computer Science and Software Engineering. I am conducting a research study as part of my undergraduate project. This project is expected to last seven months. The title of this research project is the development of a university mobile application which will allow students to quickly access important and helpful information about services within the university. Participants who attend PAL sessions have been chosen for inclusion in this study.

Purpose of this study: The goal of this project is to develop a university mobile application which supports informal learning whilst bridging the gap between formal and informal learning.

Participant Involvement: You will be required to participate in a focus group. Your participation is voluntary and you can stop at anytime. Your participation in this study will take approximately fifteen minutes. Your answers will be tabulated with those of others and will be treated as anonymous ensuring confidentiality.

Data Usage: The data collected in this focus group will be used to develop the mobile application in accordance to the Extreme Programming methodology.

Confidentiality: In the event of a publication or presentation resulting from the research, no personally identifiable information will be shared. Although the focus group will be tape recorded, your responses will remain anonymous and no names will be mentioned in the report.

Questions about the research and rights of research participants: Please contact me via my email at susannah_lily@hotmail.com with questions, complaints or concerns about this research.

Benefits to the Participant: You will assist in the development of a new mobile application for the University of Bedfordshire which supports informal learning that can support your mobile lifestyle and needs as a student. Moreover this mobile

application can also help the university to better follow its strategic plan which is designed to enable students to benefit from a transformative learning experience.

Right to Withdraw from the Study: You do not have to answer any questions you do not want to answer. Refusal to take part in or withdrawing from this study will involve no penalty or loss of benefits you would receive otherwise.

Dissemination of Research Outcomes: Dissemination of the research outcomes will be released to participants who are willing to release their email for this purpose and have an interest in the outcomes of the study.

You must be 18 years of age or older to take part in this research study. Please fill out and sign the consent form below to indicate whether you agree to participate in this research. You will be given a copy of this form to keep after you have signed it.

Have you read this consent form or it has it been read to you? Yes _____ No _____

Have all of your questions about the form and the research been answered? Yes _____ No _____

Do you understand the nature of the research in which you are about to participate? Yes _____ No _____

Do you agree to participate in this study? Yes _____ No _____

Name of Participant (please print): _____

Signature of Participant: _____ Date: _____

Researcher's Signature _____ Date: _____

D-3: Analysis of the second focus group session conducted

Purpose

The purpose of the second focus group session was to create the release plan and have the customer write the user stories at a finer level of detail with the use of User Interface prototypes.

Findings

The following user scenarios were developed by the customer:

Access Help

Allows students to access the help feature if they have trouble creating a profile or logging in to their profile.

Access Student Support Services

Enables students to access important and helpful information about student support services such as a summary on the university service and the ability to call them directly, which help students to adjust quickly to university life.

View Events

View a list of events provided by the Student Union in order to adjust quickly to university life.

View Campus Alerts

View alerts on university services for improved management of student life.

Access Note Taker

Allows students to create notes on the go for their lectures to assist with their learning.

Access Camera

Enables students to take pictures of learning materials to assist with their learning.

Access Pocket Guide

Students can access common videos on support services to help them adjust quickly to university life and for improved management of student life.

Access UoB MOBILE on Facebook

Students can share information in a collaborative environment regarding their course work via a Facebook group page.

Personalize Alerts

Enables students to select alerts they would like to be notified of for improved management of student life.

Access About Us

Students can access the purpose and functionalities of the mobile application and submit ratings based on their user experience as well as provide optional feedback.

Towards the end of the session, students felt they would prefer logging into the mobile application with their student identification number and password. However based on the inability for the mobile application to connect with internal systems, this could not be implemented. As a result the Access Student Profile requirement was removed which was agreed to by participants.

Stemming from this the Access Help story would only be present for the user to access help features before using the mobile application. Moreover, the requirements for students to: - access important and helpful information about services within the community, view clubs and societies, view local maps and access calendar were also no longer required.

D-4: Second focus group session conducted

DATE HELD: 24 January 2013

INTRODUCTIONS

For those of you who may not know me, Hello, my name is Suzanne Gilkes and I am a final year student completing my undergraduate project.

PURPOSE OF FOCUS GROUP

The title of this research project is the development of a university mobile application for Android Smartphone which will allow students to quickly access important and helpful information about services within the university. I would greatly appreciate your assistance in participating in a focus group which will require periodic visits to yourselves, in order to develop the mobile application. The short focus group session will take approximately 30 minutes. I kindly request your input and want you to share your honest and open thoughts with me. Thank you for your willingness to participate in this focus group. Your feedback during this focus group session is vital for the development of the mobile application.

OVERVIEW OF SESSION

In the last focus group session, we formulated some high-level user stories for the mobile application. This second focus group session will be used to create the release plan containing the requirements in the form of user stories at a finer level for the mobile application with the use of User Interface prototypes. Also during this session, the iteration dates for the release plan will be finalized.

Activity 1

The following user stories collected from the last focus group session are as follows:

Access Student Support Services

View Events

View Club and Societies

Access Services within the Community

View Local Maps

View Campus Alerts

Personalize Alerts

Access Calendar

Access UoB MOBILE on Facebook

Access Note Taker

We will now use a program called FluidUI to create a non-working prototype for the mobile application in order to help refine the requirements already captured.

Activity 2

Our next activity involves creating a release plan with the user stories. In order to create a release plan that will deliver the most impact, please give your recommendations of the order for the user stories. The user stories which appear first will be given priority for development. After this, the release plan will be divided into segments or iterations and you will decide on the dates of delivery for each iteration.

This marks the completion of the second focus group session

Thanks Again and See you soon.

D-5: Third focus group session conducted

DATE HELD: 31 January 2013

INTRODUCTIONS

For those of you who may not know me, Hello, my name is Suzanne Gilkes and I am a final year student completing my undergraduate project.

PURPOSE OF FOCUS GROUP

The title of this research project is the development of a university mobile application for Android Smartphone which will allow students to quickly access important and helpful information about services within the university. I would greatly appreciate your assistance in participating in this focus group which will require periodic visits to yourselves, in order to develop the mobile application. The short focus group session which will take approximately 30 minutes. I kindly request your input and want you to share your honest and open thoughts with me. Thank you for your willingness to participate in this focus group. Your feedback during this focus group session is vital for the development of the mobile application.

OVERVIEW OF SESSION

This third focus group session will be used to write user acceptance tests for evaluating the mobile application.

Activity 1

During this activity we will create a test plan which will identify the areas of the mobile application to be tested. These test plans are broken down into testing for functionality and usability. Android Design Guidelines will be referred to in creating tests for evaluating usability. This marks the completion of the third focus group session.

Thanks Again and See you soon.

D-6: Fourth focus group session conducted

DATE HELD: 07 February 2013

INTRODUCTIONS

For those who may not know me, Hello, my name is Suzanne Gilkes and I am a final year student completing my undergraduate project.

PURPOSE OF FOCUS GROUP

The title of this research project is the development of a university mobile application for Android Smartphone which will allow students to quickly access important and helpful information about services within the university. I would greatly appreciate your assistance in participating in a focus group which will require periodic visits to yourselves, in order to develop the mobile application. The short focus group session which will take approximately 30 minutes. I kindly request your input and want you to share your honest and open thoughts with me. Thank you for your willingness to participate in this focus group. Your feedback during this focus group session is vital for the development of the mobile application.

OVERVIEW OF SESSION

This fourth focus group session will be used to run the acceptance tests applicable for this iteration.

Activity

Please find attached the acceptance test for this iteration. Please execute the following test and indicate whether it has passed or failed. Completion of this activity marks the end of the focus group session.

Thanks Again and See you soon.

D-7 to D-9: fifth, sixth and seventh focus group sessions conducted

These focus group sessions followed the same structure as the fourth focus group session conducted. They also retained the same scope which was to have the customer run the acceptance tests applicable per each iteration. After executing these tests the customer will indicate whether the test has passed or failed.

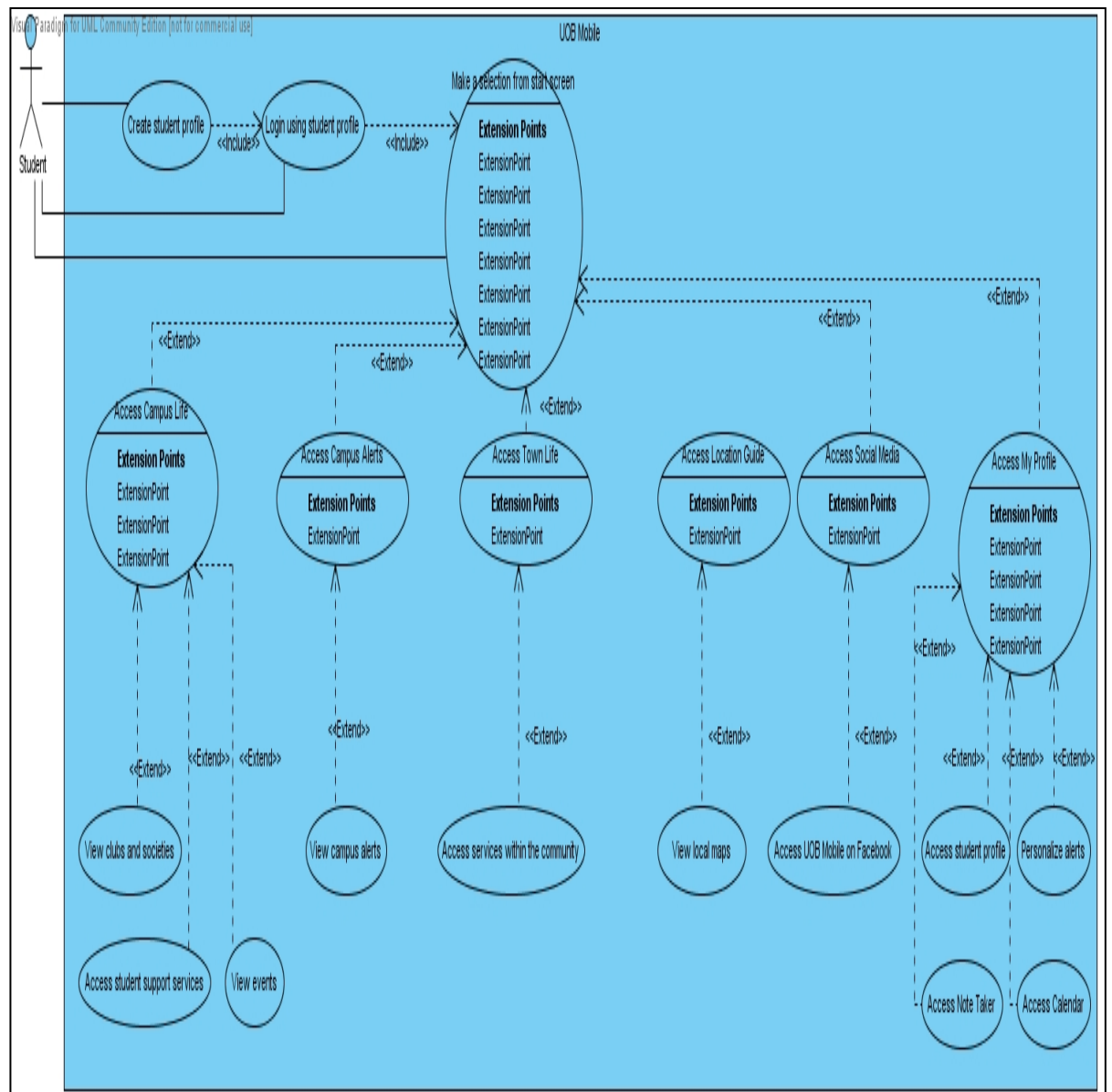
The focus group sessions were held on the following dates:

Fifth focus group session: 14 February 2013

Sixth focus group session: 21 February 2013

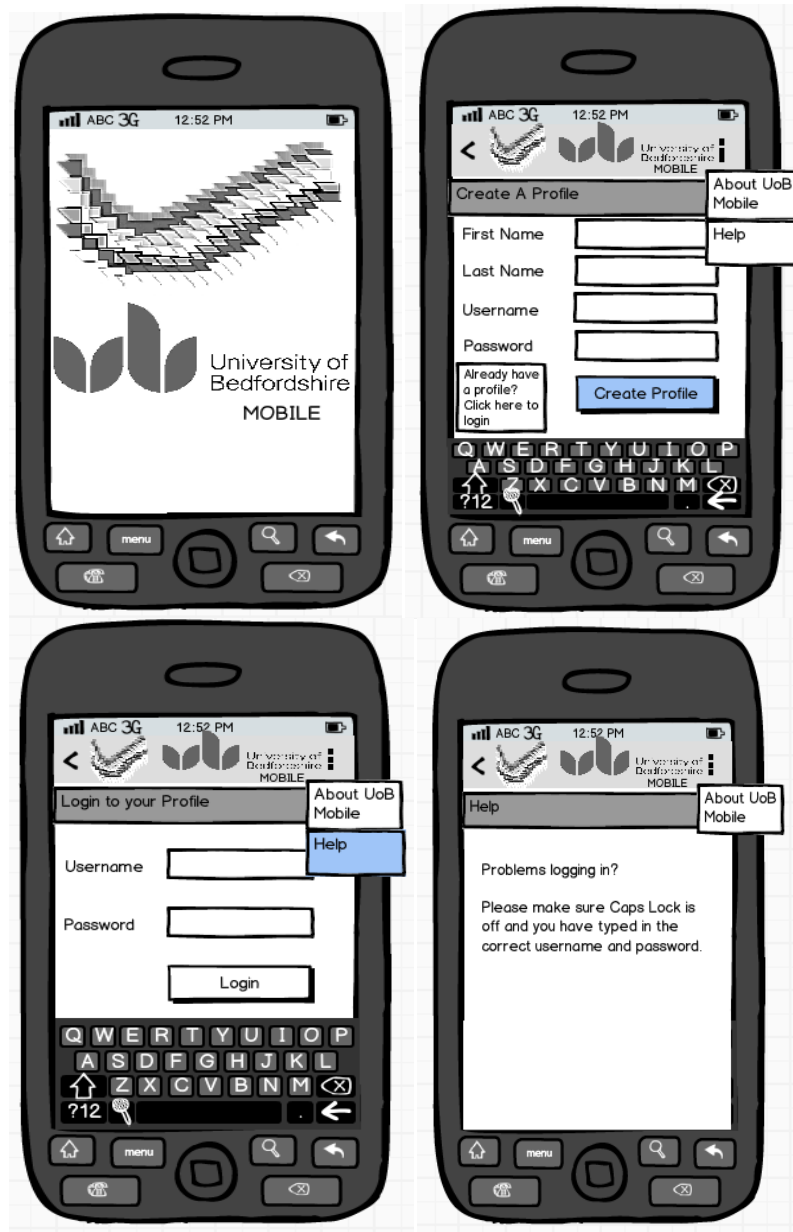
Seventh focus group session: 28 February 2013

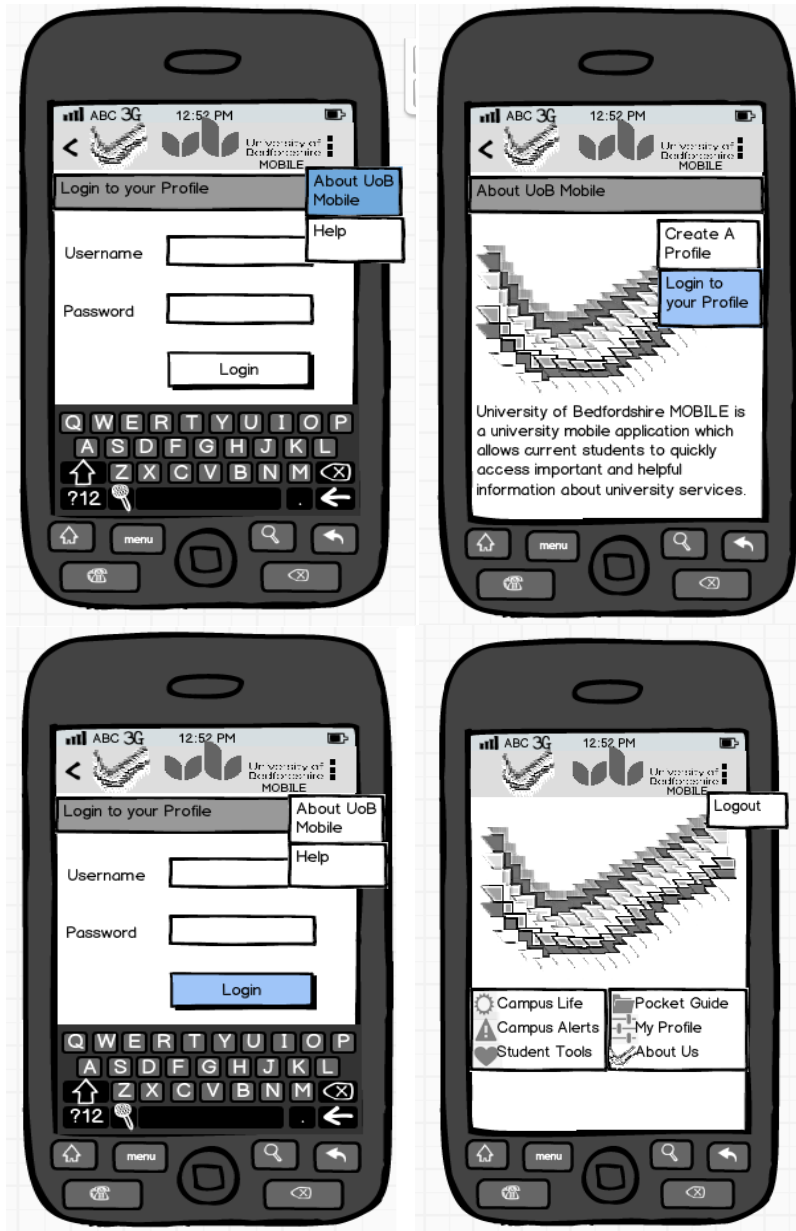
APPENDIX E: USE CASE DIAGRAM



APPENDIX F: USER EXPERIENCE DESIGNS

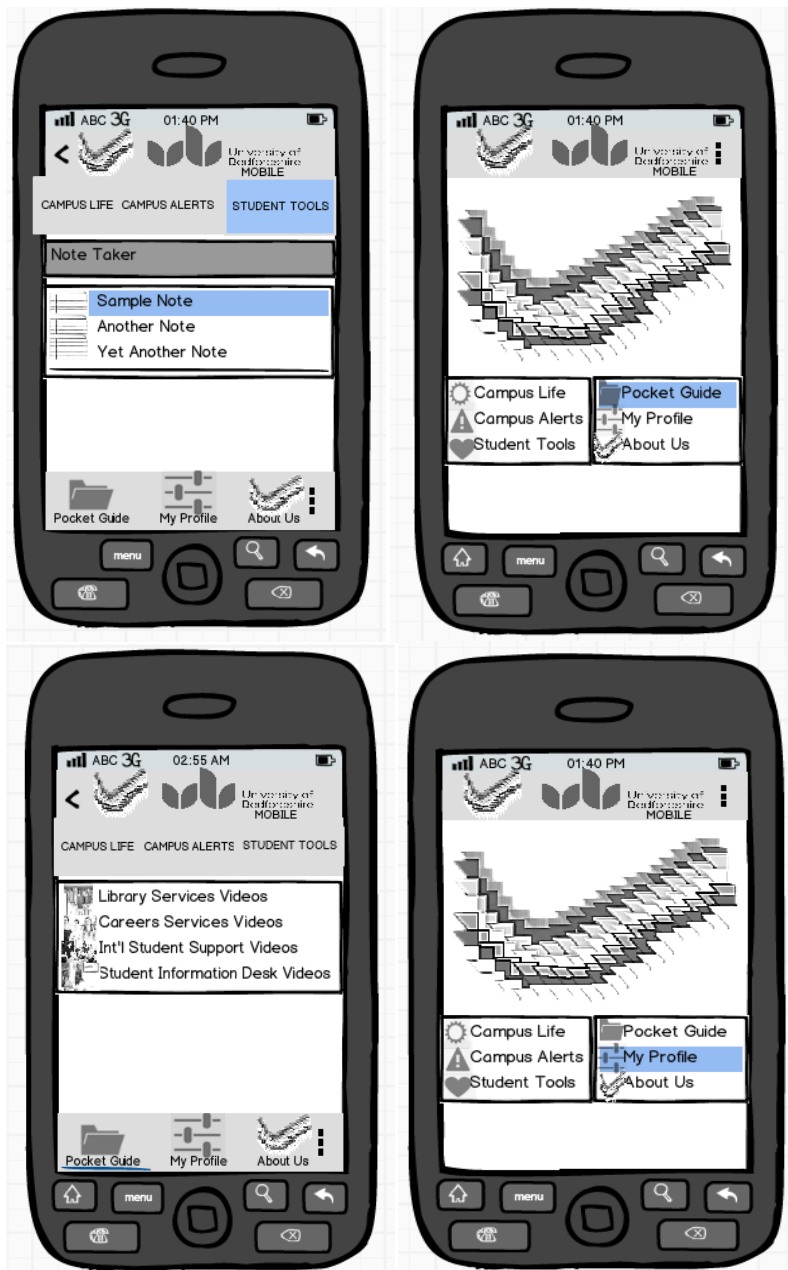
The following represents illustrations of the User Experience Design prototypes in which the screen flow order reads from left to right.



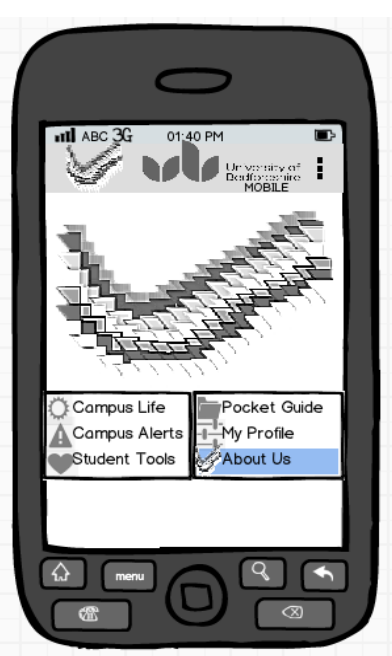












APPENDIX G: ADHERENCE TO THE ANDROID DESIGN GUIDELINES

There are Android design guidelines specified which help developers design exceptional Android apps (Creative Commons Attribution, 2.5). These guidelines have many different categories and it is advised for the developer to follow these design guidelines. Strict adherence to the guidelines is not imposed and the developer is encouraged to apply creativity and design thinking. However user's best interest must be kept in mind at all times (Creative Commons Attribution, 2.5, no date a). The guidelines were incorporated when designing the User Experience and User Interface Design prototypes.

Application Structure

UoB MOBILE, mobile application structure is aligned to the guidelines instituted for creating the general structure of a typical Android app. "A typical Android app consists of top level and detail/edit views" (Creative Commons Attribution, 2.5, no date b, no page) as seen in **Figure 1**. Emphasis on the structure of the UoB MOBILE was carried out during the design of the User Experience which affected the style of the mobile application.

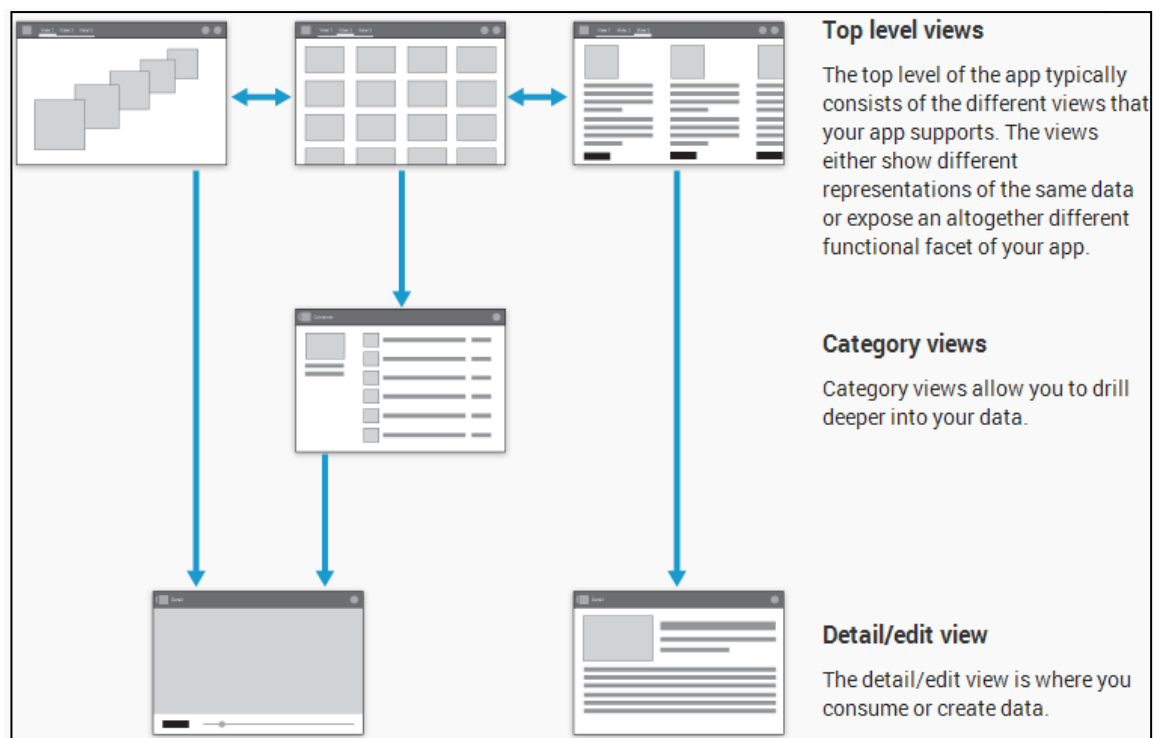


Figure 1: General Structure of a typical Android mobile application

Design Principles

The design principles were followed in order to ensure a highly usable mobile application (Creative Commons Attribution, 2.5, no date a) for both User Experience and User Interface Designs.

Common App User Interface (UI) Principles and elements

Common App UI principles and UI elements such as systems bars were integrated during both design prototypes as “Android's system UI provides the framework on top of which you build your app. Important aspects include the Home screen experience, global device navigation, and notifications. Your app will play an important part in keeping the overall Android experience consistent and enjoyable to use” (Creative Commons Attribution, 2.5, no date c, no page).

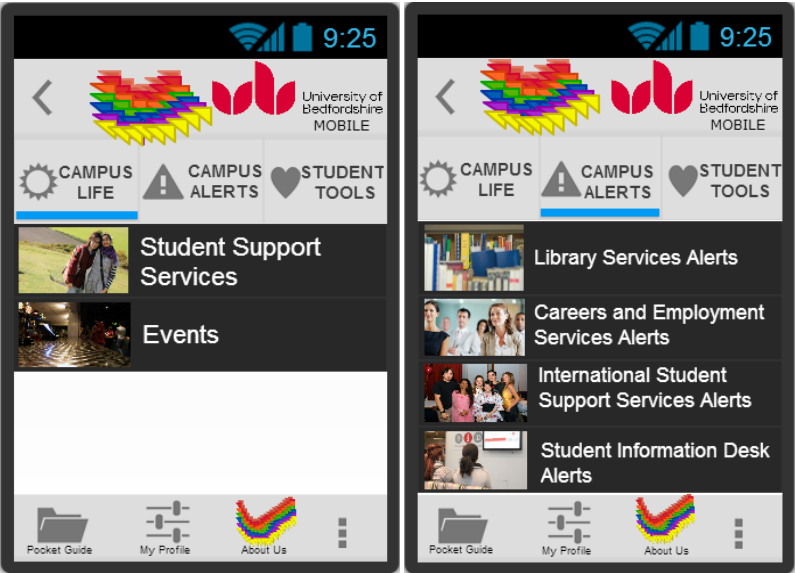
Style and Patterns

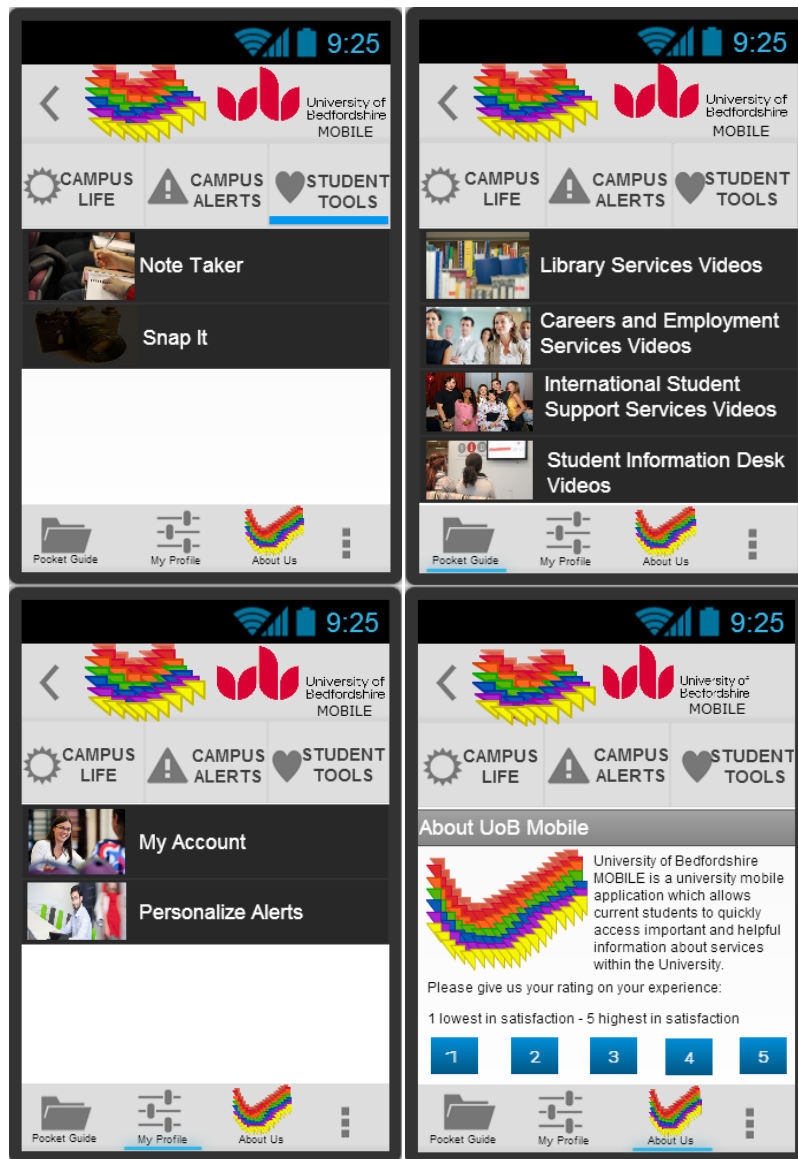
Style and pattern design elements such as style, action bar, split action bar and action bar general organization were also implemented during the design consideration for the user interface and user experience design prototypes in order to “build visually compelling apps that look great on any device” (Creative Commons Attribution, 2.5, no date d, no page) and “design apps that behave in a consistent, predictable fashion” (Creative Commons Attribution, 2.5, no date e, no page).

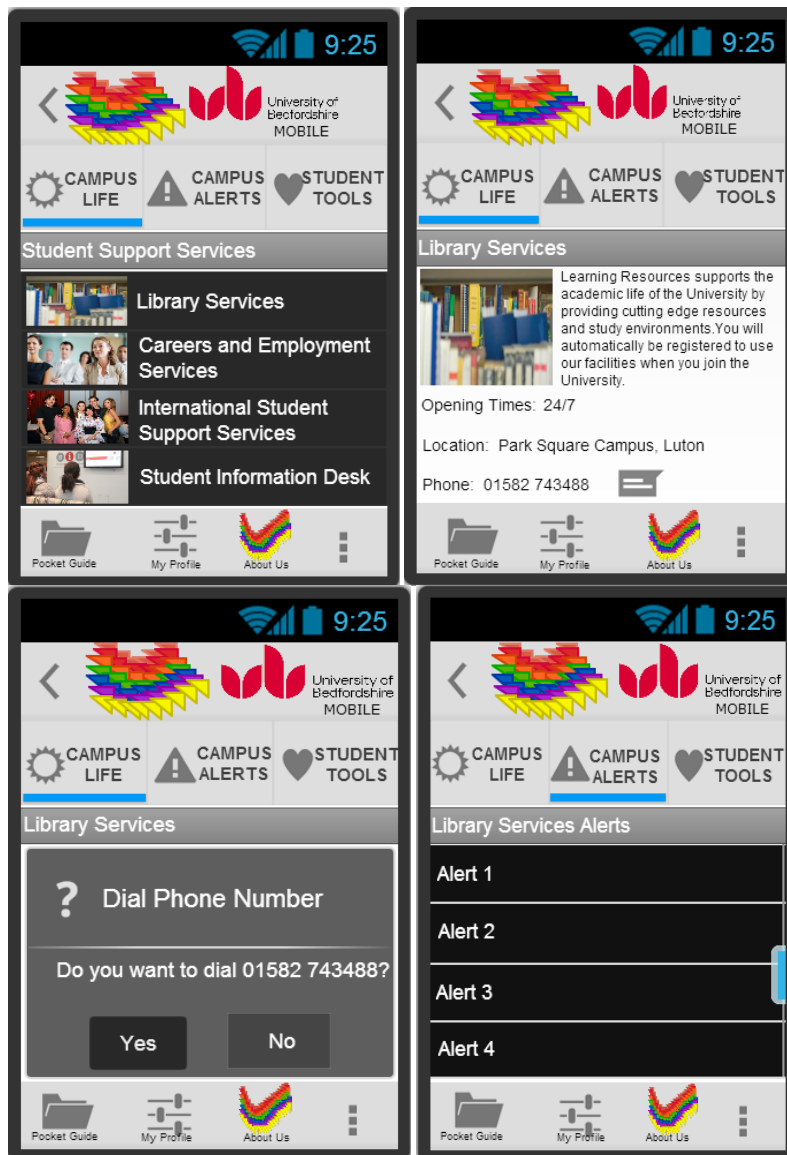
APPENDIX H: USER INTERFACE DESIGNS

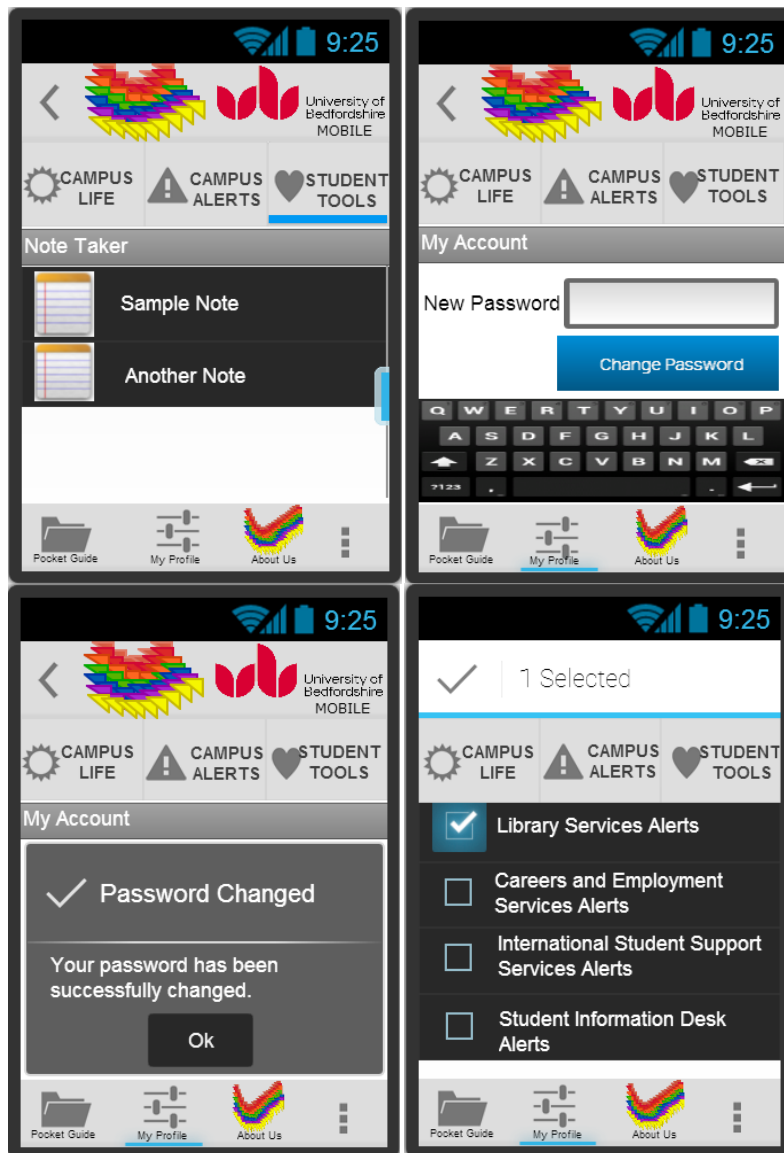
The following are illustrations of the User Interface Design prototypes in which the screen flow order reads from left to right.



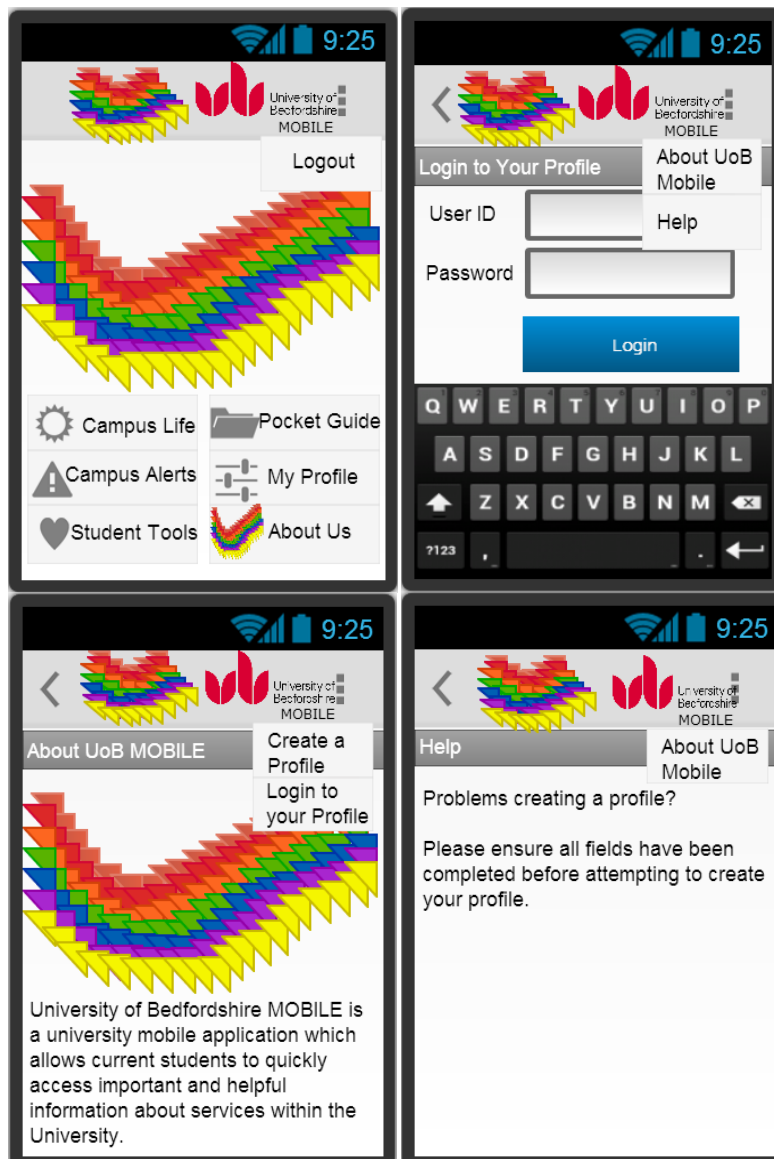


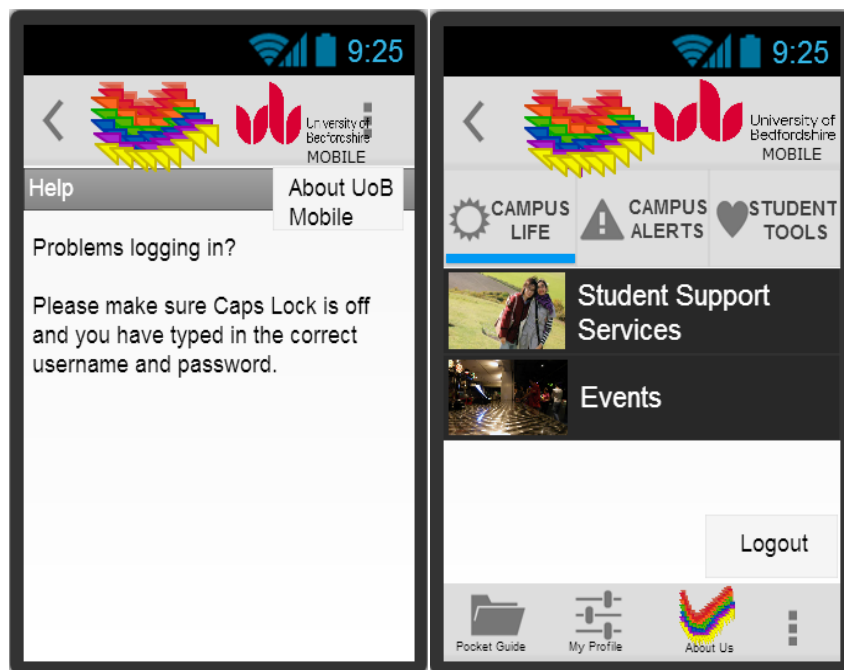












APPENDIX I: EXECUTED TEST PLANS FOR UNIT TESTING CONDUCTED

I-1: Executed test plan for the second, third, fourth, and fifth iterations

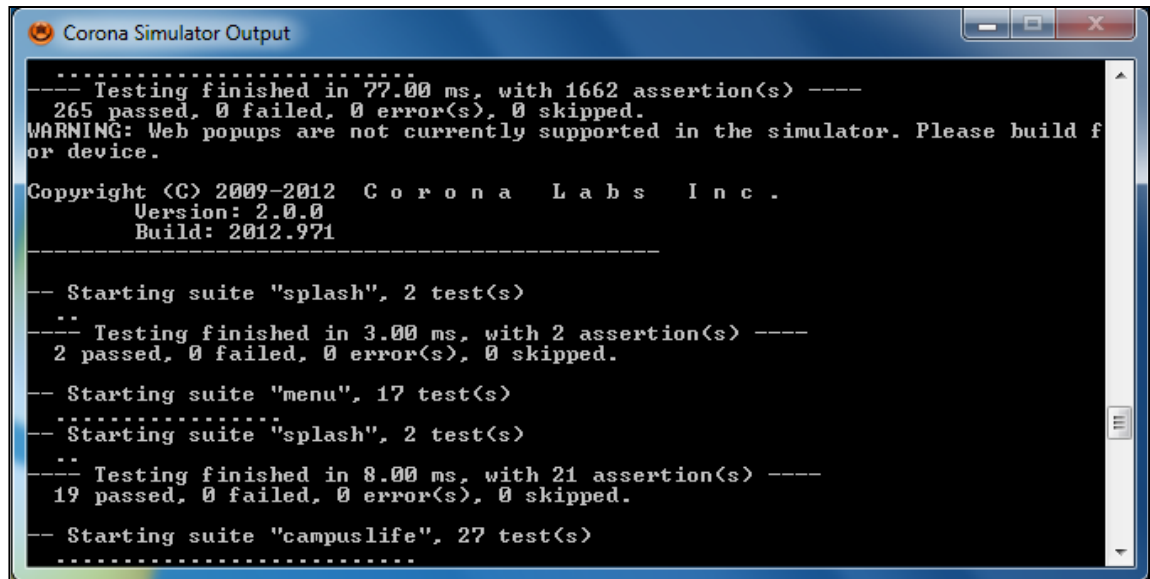
Figure 1 outlined below contains a sample of the unit tests executed for the second, third, fourth and fifth iterations.

Test Case	Description	Expected Result	Actual Result	Pass / Fail
test_forbackground	Background for the mobile application loads and is accessible	Background loads	Background loads	✓ Pass
test_forlogo	Logo for the mobile application loads and is accessible	Logo loads	Logo loads	✓ Pass
test_foractionmenu1	Actionmenu for the mobile application loads and is accessible	Actionmenu loads	Actionmenu loads	✓ Pass
test_fortab	Tab for the mobile application loads and is accessible	Tab1 loads	Tab1 loads	✓ Pass
test_fortab2	Tab2 for the mobile application loads and is accessible	Tab2 loads	Tab2 loads	✓ Pass
test_fortab3	Tab3 for the mobile application loads and is accessible	Tab3 loads	Tab3 loads	✓ Pass

Figure 1: Unit tests performed during the second, third, fourth and fifth iterations

I-2: A screenshot of the number of passed unit tests for the second iteration

A total of 265 unit tests passed successfully for the second iteration as seen in the screenshot below in **Figure 2**.



```
Corona Simulator Output

-----
Testing finished in 77.00 ms, with 1662 assertion(s) ----
265 passed, 0 failed, 0 error(s), 0 skipped.
WARNING: Web popups are not currently supported in the simulator. Please build f
or device.

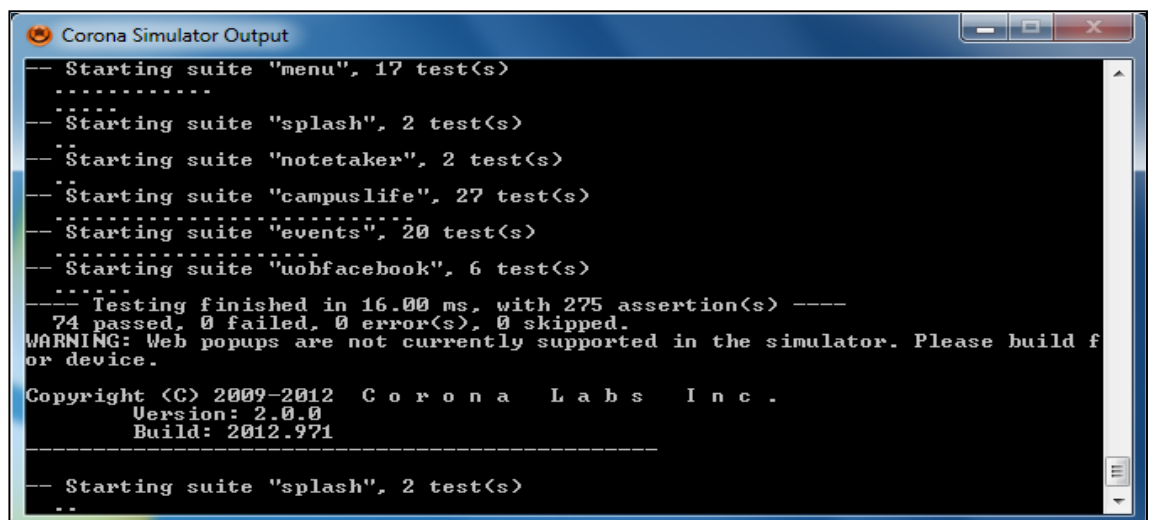
Copyright (C) 2009-2012 Corona Labs Inc.
Version: 2.0.0
Build: 2012.971

-----
-- Starting suite "splash", 2 test(s)
--
Testing finished in 3.00 ms, with 2 assertion(s) ----
2 passed, 0 failed, 0 error(s), 0 skipped.
-- Starting suite "menu", 17 test(s)
--
Starting suite "splash", 2 test(s)
--
Testing finished in 8.00 ms, with 21 assertion(s) ----
19 passed, 0 failed, 0 error(s), 0 skipped.
-- Starting suite "campuslife", 27 test(s)
-----
```

Figure 2: The number of passed unit tests for the second iteration

I-3: A screenshot of the number of passed unit tests for the third iteration

A total of 74 unit tests passed successfully for the third iteration as seen in the screenshot below in **Figure 3**.



```
Corona Simulator Output

-- Starting suite "menu", 17 test(s)
-----
-- Starting suite "splash", 2 test(s)
-- Starting suite "notetaker", 2 test(s)
-- Starting suite "campuslife", 27 test(s)
-- Starting suite "events", 20 test(s)
-- Starting suite "uobfacebook", 6 test(s)
-----
Testing finished in 16.00 ms, with 275 assertion(s) ----
74 passed, 0 failed, 0 error(s), 0 skipped.
WARNING: Web popups are not currently supported in the simulator. Please build f
or device.

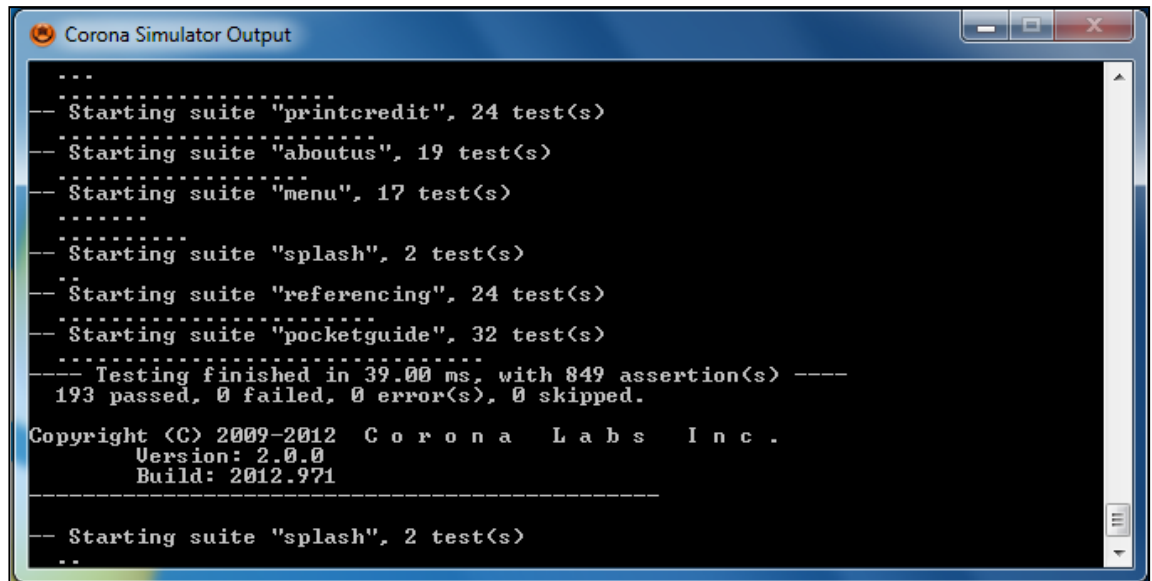
Copyright (C) 2009-2012 Corona Labs Inc.
Version: 2.0.0
Build: 2012.971

-----
-- Starting suite "splash", 2 test(s)
--
```

Figure 3: The number of passed unit tests for the third iteration

I-4: A screenshot of the number of passed unit tests for the fourth iteration

A total of 193 unit tests passed successfully for the fourth iteration as seen in the screenshot below in **Figure 3**.

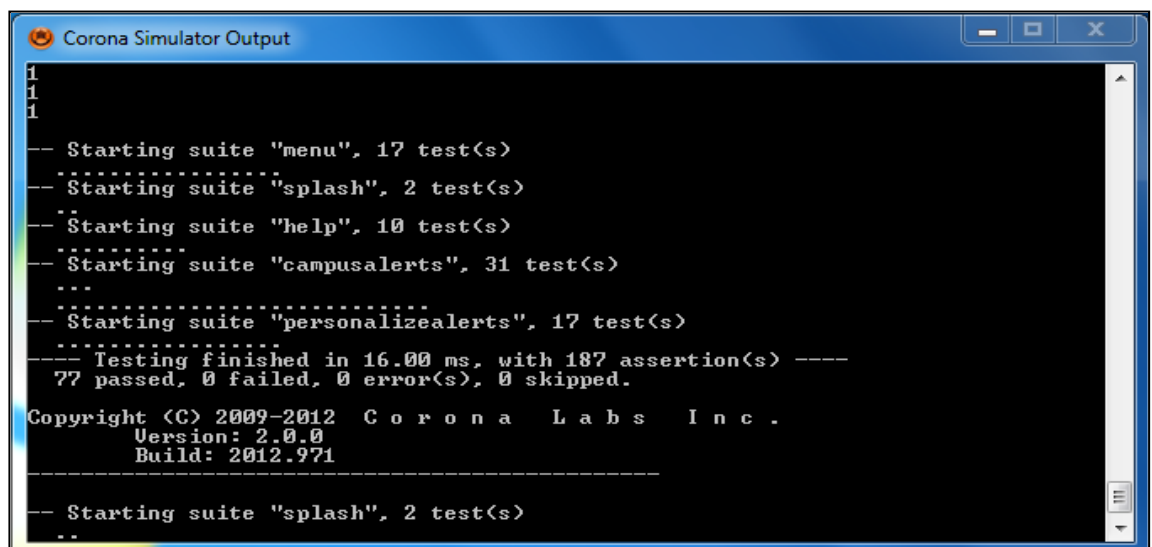


```
Corona Simulator Output
...
Starting suite "printcredit", 24 test(s)
Starting suite "aboutus", 19 test(s)
Starting suite "menu", 17 test(s)
Starting suite "splash", 2 test(s)
Starting suite "referencing", 24 test(s)
Starting suite "pocketguide", 32 test(s)
Testing finished in 39.00 ms, with 849 assertion(s) ----
193 passed, 0 failed, 0 error(s), 0 skipped.
Copyright (C) 2009-2012 Corona Labs Inc.
Version: 2.0.0
Build: 2012.971
Starting suite "splash", 2 test(s)
..
```

Figure 4: The number of passed unit tests for the fourth iteration

I-5: A screenshot of the number of passed unit tests for the fifth iteration

A total of 77 unit tests passed successfully for the fifth iteration as seen in the screenshot below in **Figure 3**.



```
Corona Simulator Output
1
1
1
Starting suite "menu", 17 test(s)
Starting suite "splash", 2 test(s)
Starting suite "help", 10 test(s)
Starting suite "campusalerts", 31 test(s)
Starting suite "personalizealerts", 17 test(s)
Testing finished in 16.00 ms, with 187 assertion(s) ----
77 passed, 0 failed, 0 error(s), 0 skipped.
Copyright (C) 2009-2012 Corona Labs Inc.
Version: 2.0.0
Build: 2012.971
Starting suite "splash", 2 test(s)
..
```

Figure 4: The number of passed unit tests for the fifth iteration

APPENDIX J: EXECUTED TEST PLANS FOR FUNCTIONAL TESTING CONDUCTED

J-1: Executed test plan for the second iteration

Section	Test Input	Expected Result	Actual Result
Main Menu Screen	Campus Life Menu Option selected	Displays the Campus Life Screen	Correct
All menu screens	Split action bar tab selected	Displays the appropriate menu screen	Correct
Campus Life Screen	Student Support Services Menu Option selected	Displays the Student Support Services Screen	Correct
Campus Life Screen	Caret button selected	Returns to the Main Menu	Correct
Student Support Services Screen	All Menu Option buttons	Displays appropriate services screen	Correct
Student Support Services Screen	Caret button selected	Returns to the Campus Life Screen	Correct
Services Screen	Phone icon selected	The built in phone dialler dials the service number	Correct
Services Screen	Caret button selected	Returns to the Student Support Services Screen	Correct
Main Menu Screen	Campus Alerts Menu Option selected	Displays the Campus Alerts Screen	Correct
Campus Alerts Screen	All Menu Options	Displays the appropriate alerts screen	Correct
Campus Alerts Screen	Caret button selected	Displays the Campus Life Screen	Correct
Alerts screen	Caret button selected	Returns to the Campus Alerts Screen	Correct

J-2: Executed test plan for the third iteration

Section	Test Input	Expected Result	Actual Result
Campus Life Screen	Events Menu Option selected	Displays events by the Student Union	Correct
Events Screen	Caret button selected	Campus Life Screen	Correct
Main Menu Screen	Student Tools Menu Option selected	Displays the Student Tools Screen	Correct
Student Tools Screen	Note Taker Menu Option selected	Displays the Note Taker Screen	Correct
Note Taker Screen	Take note with the On-screen keyboard	Text is display in textbox for note title and body of note	Correct

Note Taker Screen	Back button selected	Returns to the Student Tools Screen	Correct
Note Taker Screen	Email button selected	Sends the note as an email.	Correct
Student Tools Screen	Camera Menu Option selected	Displays the built-in camera	Correct
Student Tools Screen	Caret button selected	Displays the Campus Alerts Screen	Correct
Built-in Camera	Take picture with the camera	Takes a picture and saves it in the Photo Gallery	Correct
Built-in Camera	Back navigation button on phone	Returns to the Student Tools Screen	Correct
Student Tools Screen	UoB MOBILE on Facebook Menu Option selected	Displays the UoB Facebook group page and allows user to make a post	Correct
UoB MOBILE on Facebook Screen	Back button selected	Returns to the Student Tools Screen	Correct

J-3: Executed test plan for the fourth iteration

Section	Test Input	Expected Result	Actual Result
Campus Life Screen	Pocket Guide Menu Option selected	Displays the Pocket Guide Screen	Correct
Pocket Guide Screen	All Menu Options	Displays the appropriate guide screen	Correct
Pocket Guide Screen	Caret button selected	Returns to the Campus Life Screen	Correct
Guide screen	Play icon selected	Plays the relevant guide video	Correct
Guide screen	Scrollable transcript selected	Allows user to view the full transcript	Correct
Guide screen	Caret button selected	Returns to the Pocket Guide Screen	Correct
Main Menu Screen	Screen overflow button selected	Displays the menu option for About Us	Correct
Main Menu Screen	About Us Menu Option selected	Displays the About Us Screen	Correct
About Us Screen	Caret button selected	Returns to the Main Menu Screen	Correct
About Us Screen	Recommendation button selected	Allows user to send recommendation via email	Correct

J-4: Executed test plan for the fifth iteration

Section	Test Input	Expected Result	Actual Result
Campus Alerts Screen	Screen overflow button selected	Displays the menu option for Personalize Alerts	Correct
Personalize Alerts Screen	Allows user to select via checkbox with alerts to view	Checkbox shows selection or de-selection based on this action	Correct
Personalize Alerts	Caret button selected	Display the relevant alert based on personalized alerts selected on the Campus Alerts Screen	Correct
Main Menu Screen	Screen overflow button selected	Displays the menu option for Help	Correct
Main Menu Screen	Help menu option selected	Displays the Help Screen	Correct
Help Screen	Caret button selected	Returns to the Main Menu screen	Correct

APPENDIX K: INTEGRATION OF THE SYSTEM THROUGHOUT THE ITERATIONS

K-1: Build process for the second, third, fourth and fifth iterations

The following screenshots outline the process for daily integration of the code for the system on the local machine per each iteration.

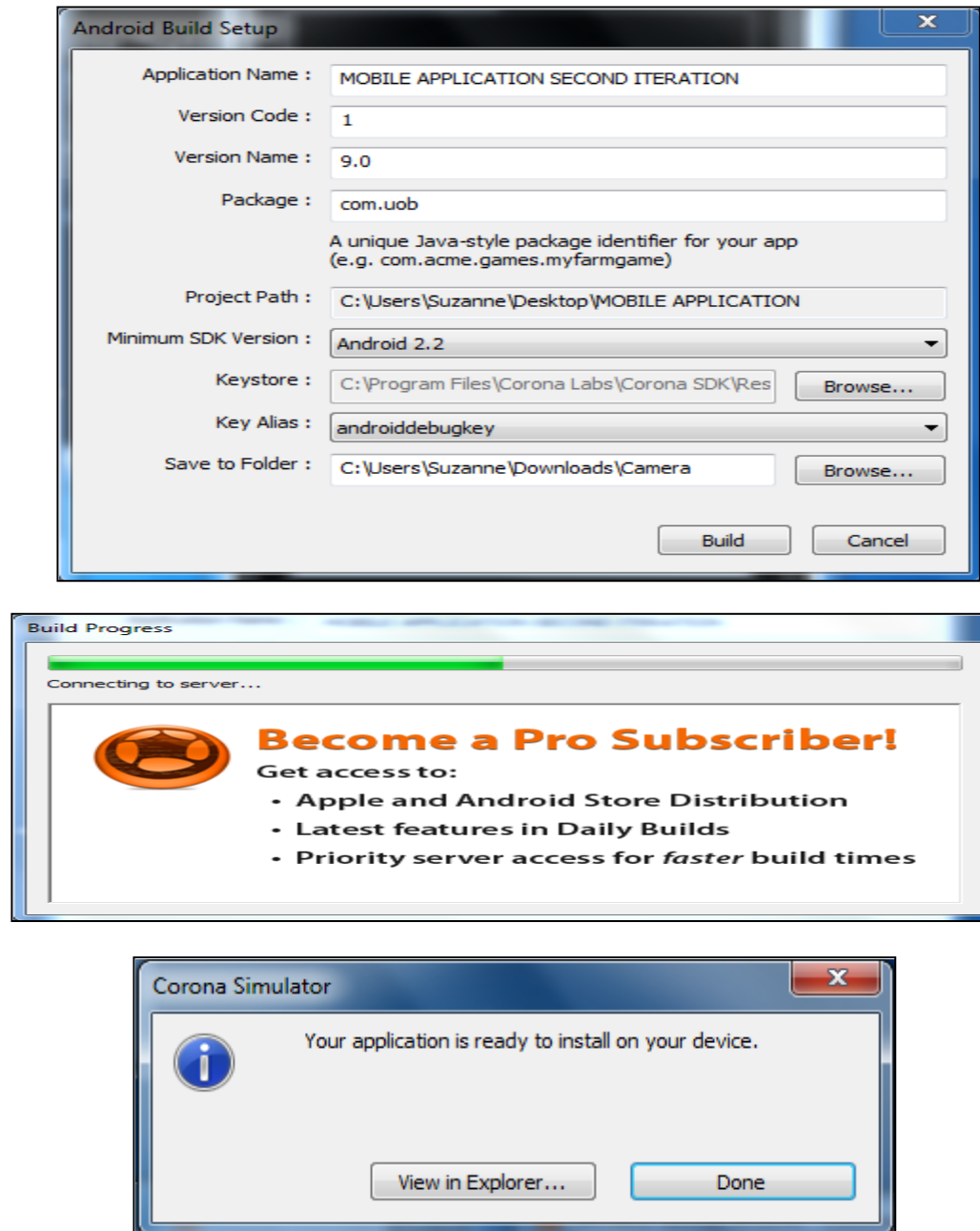


Figure 1: The successful build message

APPENDIX L: TEST RESULTS FOR THE USER ACCEPTANCE TESTS

L-1: Test results for the second iteration

Within the second iteration, the test results for acceptance tests 12-18 which evaluated usability indicated success for all of these tests.

Acceptance Test	Expected Result	Actual Result	✓ Pass/x Fail
Test 1	The customer can access important and helpful information about student support services. The mobile application should enable the users to easily read the information and quickly make calls to access these services.	The customer can access important and helpful information about student support services. The mobile application should enable the users to easily read the information and quickly make calls to access these services.	✓
Test 2	The customer can easily read the alerts displayed from the university service's Twitter feed and has access to the screen via a scroll object to read the information comfortably.	The customer can easily read the alerts displayed from the university service's Twitter feed and has access to the screen via a scroll object to read the information comfortably.	✓

Figure 1: Acceptance tests results which evaluated system functionality for the second iteration

Acceptance Test	Expected Result	Actual Result	✓ Pass/x Fail
Test 11	Navigation between screens is consistent. Transitions and tabs are used to show relationships among screens so the user always knows their way around. The ability to scroll through any of the menu options is clearly made aware to the user.	Navigation between screens is consistent. Transitions and tabs are used to show relationships among screens so the user always knows their way around. The ability to scroll through any of the menu options is clearly made aware to the user.	x
Test 11	The ability to scroll through any of the menu options is clearly made aware to the user.	The ability to scroll through any of the menu options is clearly made aware to the user.	✓
Test 12	Only information relevant to the user needs is presented. Tasks and information is broken down into small digestible chunks to avoid the user being overwhelmed.	Only information relevant to the user needs is presented. Tasks and information is broken down into small digestible chunks to avoid the user being overwhelmed.	✓
Test 13	Complex tasks are broken into smaller steps that can be easily accomplished. Feedback on actions is	Complex tasks are broken into smaller steps that can be easily accomplished.	✓

	provided for all screens using a subtle glow.	Feedback on actions is provided for all screens using a subtle glow.	
Test 14	The mobile application is easy to learn as it leverages visual patterns such as the caret and screen overflow for navigation from other Android mobile applications.	The mobile application is easy to learn as it leverages visual patterns such as the caret and screen overflow for navigation from other Android mobile applications.	✓
Test 15	The user is presented with the activity, watching the video, rather than being asked for confirmation on whether they would like to execute the activity.	The user is presented with the activity, watching the video, rather than being asked for confirmation on whether they would like to execute the activity.	✓
Test 16	The user is presented with the activity, making a phone call, rather than being asked for confirmation on whether they would like to execute the activity.	The user is presented with the activity, making a phone call, rather than being asked for confirmation on whether they would like to execute the activity.	✓
Test 17	Personalized campus alerts settings the user created are stored and remembered by the	Personalized campus alerts settings the user created are stored and remembered by the	✓

	mobile application which adds a personal touch to the user experience.	mobile application which adds a personal touch to the user experience.	
Test 18	A well-timed sound effect in the splash screen has a positive effect on the user experience.	A well-timed sound effect in the splash screen has a positive effect on the user experience.	✓

Figure 2: Acceptance tests results which evaluated system usability for the second iteration

L-2: Test results for the third iteration

Within the third iteration, the test results for acceptance tests 11-18 which evaluated usability indicated success for all of these tests.

Acceptance Test	Expected Result	Actual Result	✓ Pass/x Fail
Test 11	Navigation between screens is consistent. Transitions and tabs are used to show relationships among screens so the user always knows their way around. The ability to scroll through any of the menu options is clearly made aware to the user.	Navigation between screens is consistent. Transitions and tabs are used to show relationships among screens so the user always knows their way around. The ability to scroll through any of the menu options is clearly made aware to the user.	✓

			✓
Test 3	The customer can view a list of up-to-date events provided by the Student Union which also allows for different views of these events.	The customer can view a list of up-to-date events provided by the Student Union which also allows for different views of these events.	✓
Test 4	Notes are easily taken, can be saved on the phone and emailed within the mobile application.	Notes are easily taken, can be saved on the phone and emailed within the mobile application.	✓
Test 5	Pictures of learning materials are saved on the phone and accessed via the Photo Gallery of the phone. Taking pictures using the mobile application does not close the mobile application.	Pictures of learning materials are saved on the phone and accessed via the Photo Gallery of the phone. Taking pictures using the mobile application closes the mobile application.	X
Test 6	Posts can be made in the interactive environment which allows the customer to collaborate on information regarding their coursework.	Posts can be made in the interactive environment which allows the customer to collaborate on information regarding their coursework.	✓

Figure 1: Acceptance tests results which evaluated system functionality for the third iteration

L-3: Test results for the fourth iteration

During this iteration, the test results for acceptance tests 11-18 which evaluated usability indicated success for all of these tests.

Acceptance Test	Expected Result	Actual Result	✓ Pass/x Fail
Test 5	Taking pictures using the mobile application does not close the mobile application.	Taking pictures using the mobile application does not close the mobile application.	✓
Test 7	Students can access a video on support services. Access to the video transcript is available for those who are hearing impaired.	Students can access a video on support services. Access to the video transcript is available for those who are hearing impaired.	✓
Test 8	The customer can view the purpose of the app and select a recommendation level which is then sent to the developer via an email.	The customer can view the purpose of the app and select a recommendation level which is then sent to the developer via an email.	✓

Figure 1: Acceptance tests results which evaluated system functionality for the fourth iteration

L-4: Test results for the fifth iteration

For the fifth iteration, the test results for acceptance tests 11-18 which evaluated usability indicated success for all of these tests.

Acceptance Test	Expected Result	Actual Result	✓ Pass/x Fail
Test 9	The customer can personalize the campus alerts they want to receive. Personalization is made easy with use of checkboxes, choices	The customer can personalize the campus alerts they want to receive. Personalization is made easy with use of checkboxes, choices	✓

	selected can be varied and are saved for future use.	selected can be varied and are saved for future use.	
Test 10	The customer can quickly view help information on how to use the mobile application.	The customer can quickly view help information on how to use the mobile application.	✓

Figure 1: Acceptance tests results which evaluated system functionality for the fifth iteration

APPENDIX M: USER GUIDE

M-1: What is University of Bedfordshire MOBILE

UoB MOBILE is a university mobile application which allows students to quickly access important and helpful information about services within the University of Bedfordshire. The educational mobile application also has bonus features which assist students with their academic learning.

M-2: Requirements for the installation of UoB MOBILE

- A Smartphone or tablet that runs the Android operating system with the minimum version of 2.2.
- Internet connection on the Android Smartphone.
- QR Code Reader.
- 24.25 MB for installing the mobile application.

M-3: How to install UoB MOBILE

There is a .apk file which is found in the CD in the folder UoB MOBILE\MOBILE APPLICATION. This CD is attached to the printed copy of the dissertation. When you have located this file, please access your web browser and go to <http://www.apkinstall.com/>. It will ask you to browse and upload this .apk file. Once completed, it will produce a QR Code like . Please do not use the QR Code in the user guide. Use the one created via the website.

Installing a QR Code Reader

Please note you will need a QR Code Reader such as QR Droid to install the UoB MOBILE. You can find an install a QR Code Reader by visiting App Store, such as Play Store on your Android Smartphone. QR Droid is free to download and install.

Once a QR Code Reader has been installed, open the QR Code Reader app and use this to scan the QR Code generated by the website. This will download .apk file on your phone. After the download is complete, access the downloaded .apk from the Downloads section and select it to install UoB MOBILE. Before installation the mobile application will request to have certain permissions, once this is approved UoB MOBILE will install on the Smartphone.

M-4: Starting UoB MOBILE

When you have successfully installed the mobile application, it will appear as an orange ball entitled UoB MOBILE in your listing of apps on the phone. Selecting this icon will open the mobile application.

M-5: Getting started with UoB MOBILE

Upon opening UoB MOBILE you will notice the following splash screen introducing you to the mobile application seen in **Figure 1**. By tapping this screen you will be shown the main menu screen which provides three menu options for you to choose from found in **Figure 2**.



Figure 1: Opening splash screen



Figure 2: Main menu screen

M-6: Accessing Help

To access help before exploring the various features of UoB MOBILE, access the screen overflow button (highlighted by the orange star) in **Figure 2**. This will display additional menu options one of which is Help. Pressing the Help menu option seen in

Figure 3, will display advice on using UoB MOBILE in the following screen as seen in **Figure 4**.

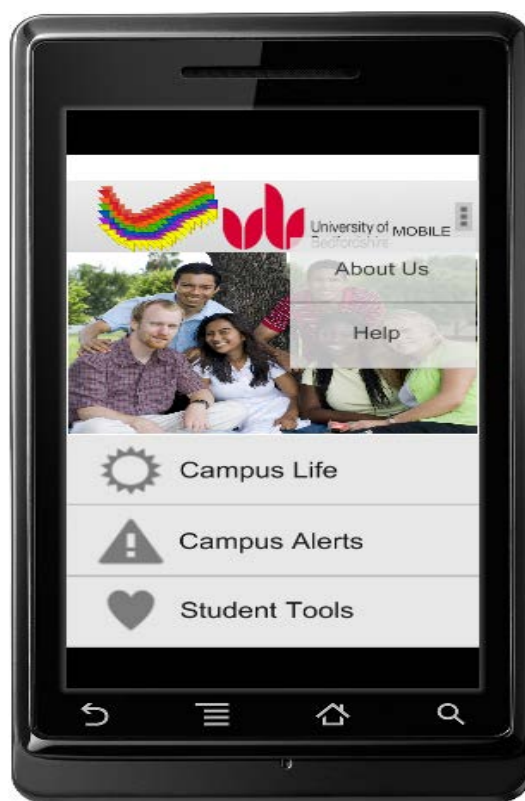


Figure 3: Accessing the Help Feature

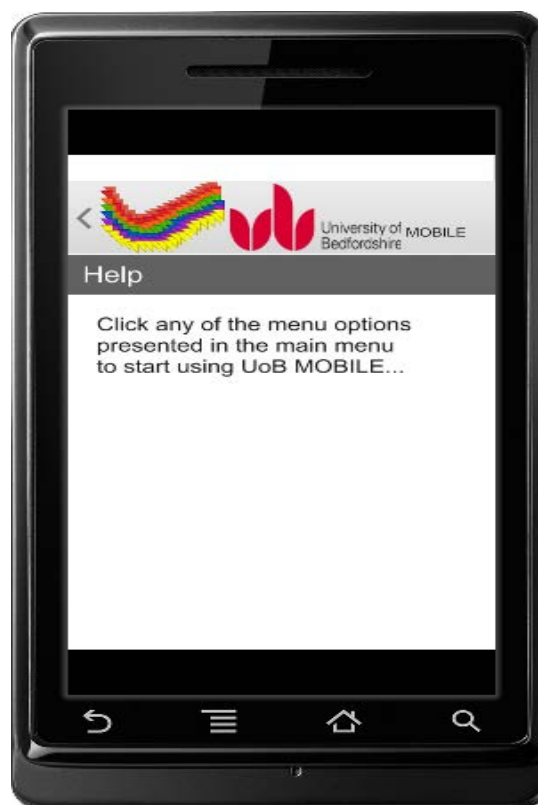


Figure 4: The Help Screen

M-7: Exploring the Campus Life features

This section provides information on university student support services, events and presents a list of videos on accessing Library services.

Selecting this menu option will display the Campus Life Screen with the following menu found in **Figure 5**.

Navigating the structure of UoB MOBILE

Using the caret button (highlighted by the purple star) in **Figure 6**, will allow you to navigate the hierarchical structure of the mobile application and move between screens.

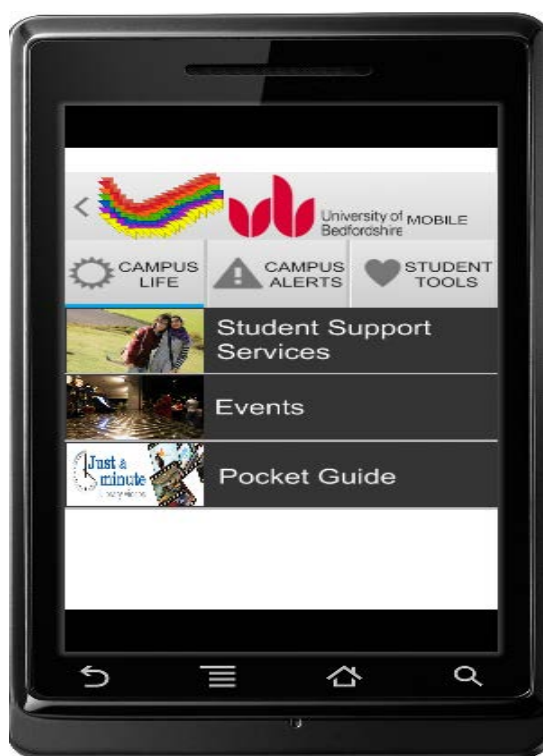


Figure 5: Campus Life Screen



Figure 6: Student Support Services Screen

M-8: Accessing Student Support Services

Selecting this menu option will allow you to view a summary of key university services commonly used by students seen in **Figure 6**. You can then select any one of the following student support services e.g. Library Services seen in **Figure 6**. Within this screen you can read an overview of the service.

You can also call the university service directly via the mobile application by pressing the phone icon (highlighted by the blue star in the screenshot) as seen in **Figure 7** which will display the following screen in **Figure 8**.

M-9: Accessing Events

From the Library Services screen, return to the Campus Life screen by pressing the caret button once to return to the Support Services screen and again to return to the Campus Life screen. Selecting the Events menu option will display a list of current events hosted by the Student Union as seen in **Figure 9**.



Figure 7: Library Services Screen



Figure 8: Dialling the selected service

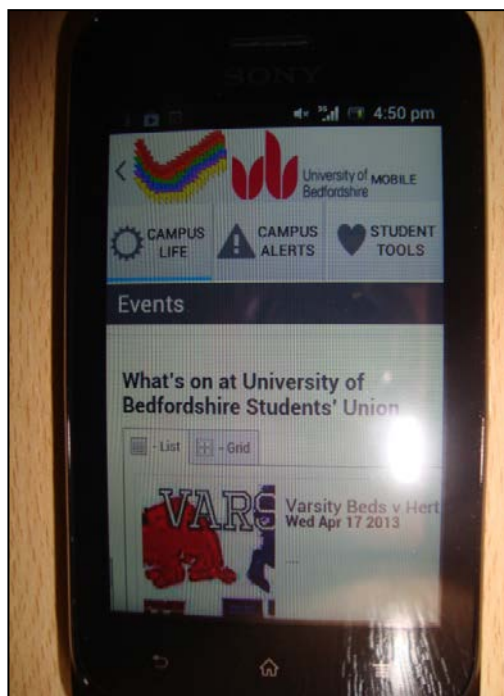


Figure 9: Events hosted by the Student Union



Figure 10: Pocket Guide Screen

M-10: Accessing Pocket Guide

From the Events screen, return to the Campus Life screen by pressing the caret button. Selecting the Pocket Guide menu option will display a list of short videos when using Library Services as seen in **Figure 10**. You can select any video of your choice to view e.g. selecting the Print Credit Machines will display the following screen as seen in **Figure 11**. A video transcript is provided for those who are hearing-impaired. This transcript is scrollable allowing the user to scroll the transcript upwards to view the full transcript and downwards to retract it. You can play the video by pressing the play icon button (highlighted by the red star in **Figure 11**). Pressing the play icon button starts the video as seen in **Figure 12**. You can stop the video at any time by pressing the back navigation button on the phone, which returns to the UoB MOBILE.

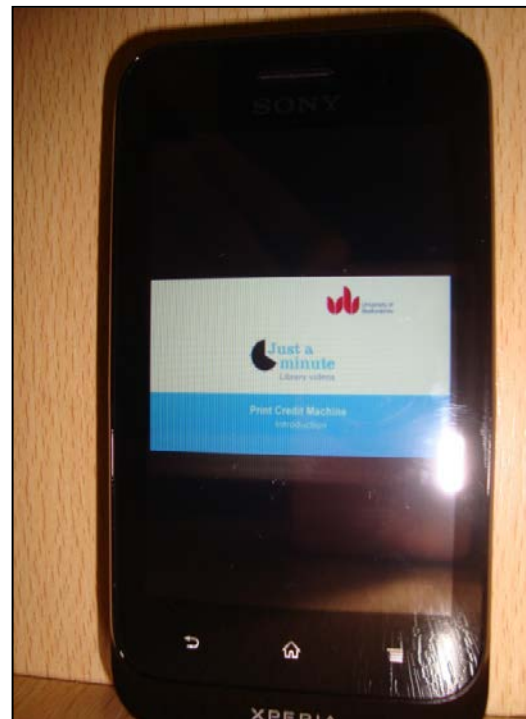


Figure 11: Print Credit Machines Screen **Figure 12: Print Credit Machines video**

M-11: Exploring the Campus Alerts features

From the Print Credit Machines Screen, you can access Campus Alerts by pressing the Campus Alerts tab button (highlighted by the green star in **Figure 11**). Campus Alerts displays up-to-date Twitter feeds from the university services. Pressing this tab button displays a list of campus alerts to choose from as seen in **Figure 13**. You can select any campus alert from the list e.g. Student Information Desk Alerts will display the screen in **Figure 14** which is scrollable allowing the user to scroll the feed upwards.

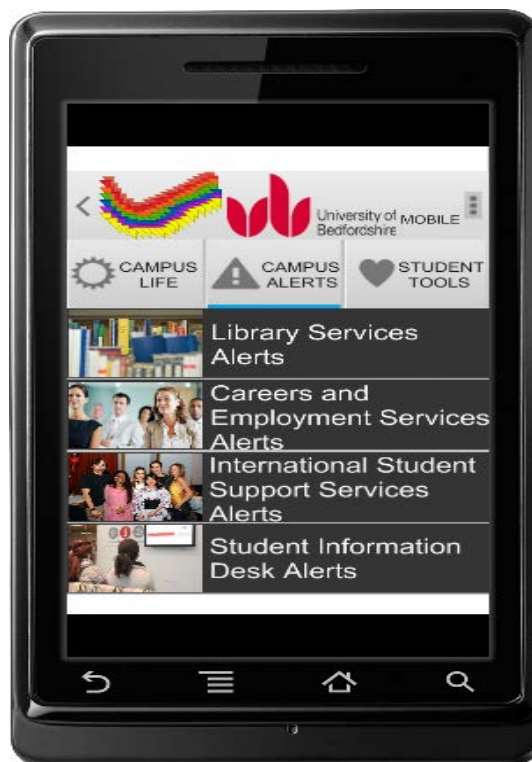


Figure 13: Campus Alerts Screen

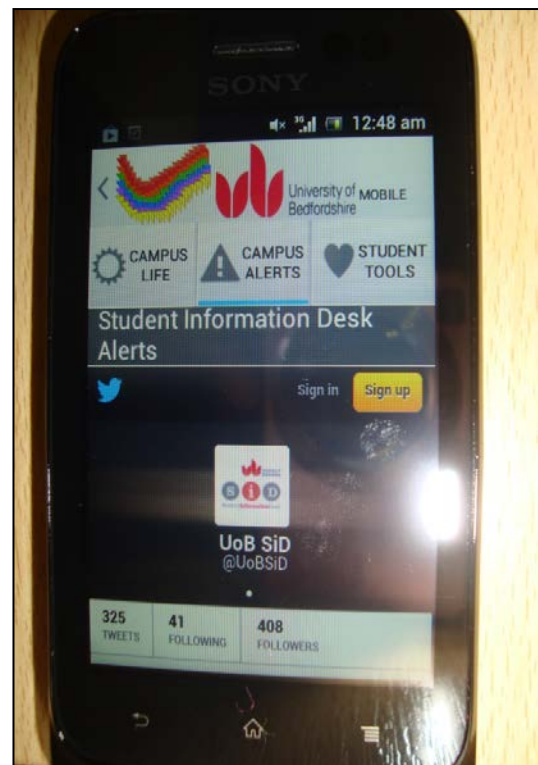


Figure 14: Student Information Desk Alerts

M-12: Personalizing Alerts

This feature allows the user to personalize which campus alerts they want to view each time they access the Campus Alerts Screen. To access this feature from the Campus Alerts Screen, select the screen overflow button, and select the menu option Personalize Alerts as seen in **Figure 15**. This will take you to the Personalize Alerts Screen seen in **Figure 16**. You can then select which campus alerts you want to view

in the Campus Alerts screen as shown in **Figure 17**. The updated menu for the Campus Alerts Screen based on the selections made in **Figure 17** can be seen in **Figure 18**.

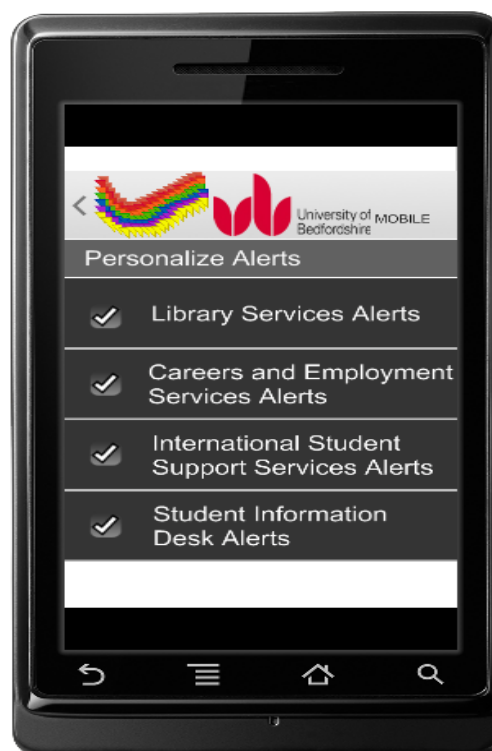
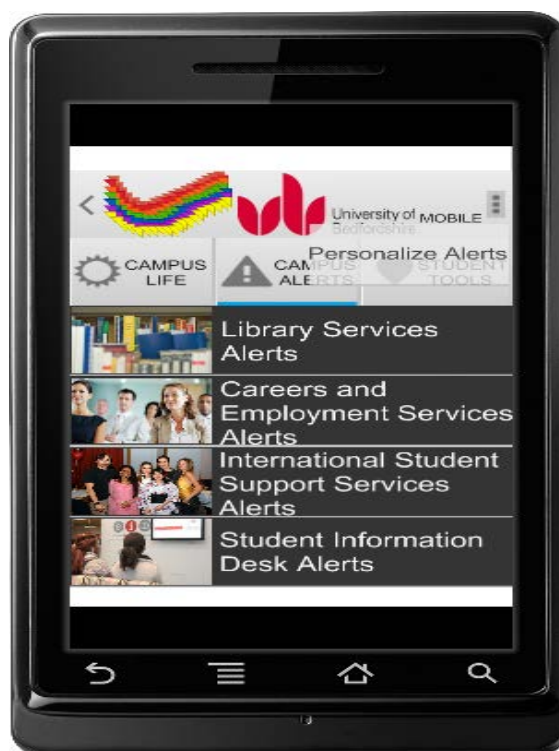


Figure 15: Accessing Personalize Alerts feature **Figure 16:** Personalize Alerts

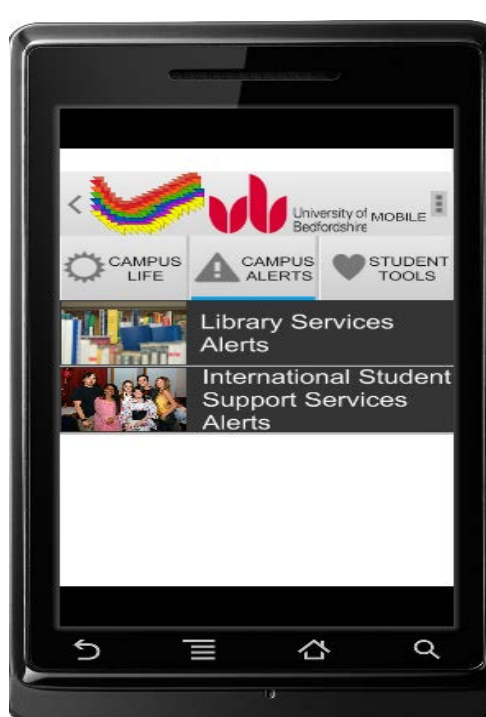
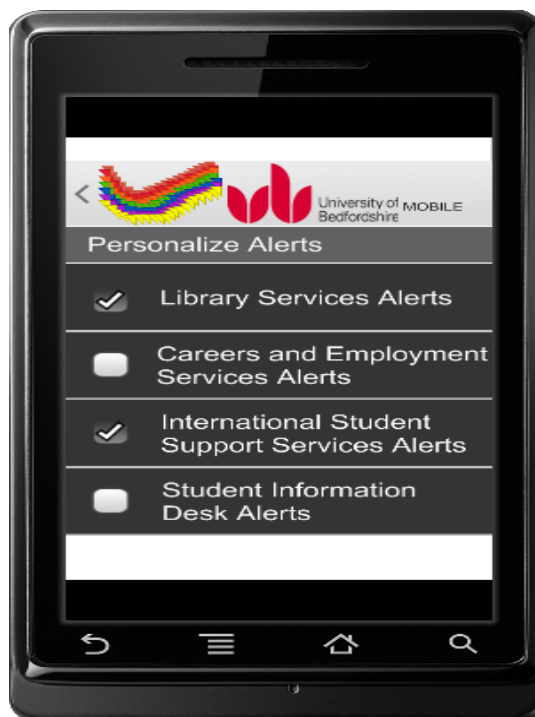


Figure 17: Personalizing campus alerts **Figure 18:** Updated Campus Alerts Screen

M-13: Exploring the Student Tools features

These tools are provided to help students with their academic learning.

M-14: Accessing the Camera

From the Campus Alerts Screen, access the Student Tools screen by pressing the Student Tools tab button. This presents the Student Tools Screen seen in **Figure 19**. Selecting the Camera menu option will call the built-in camera on the phone allowing the user to take a picture as seen in **Figure 20**. To view the picture taken, close the app by pressing the back navigation button on the phone and navigate to the Gallery on the phone which stores the images taken via the camera. This can be seen in **Figure 21**.

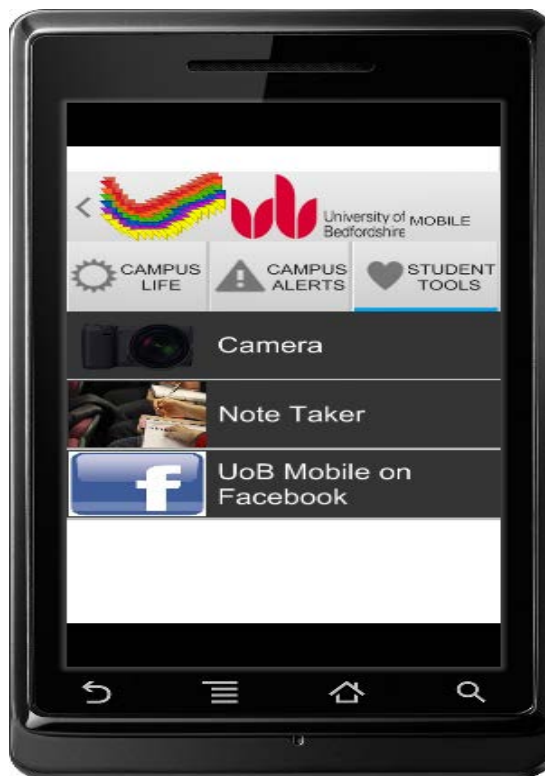


Figure 19: Student Tools Screen

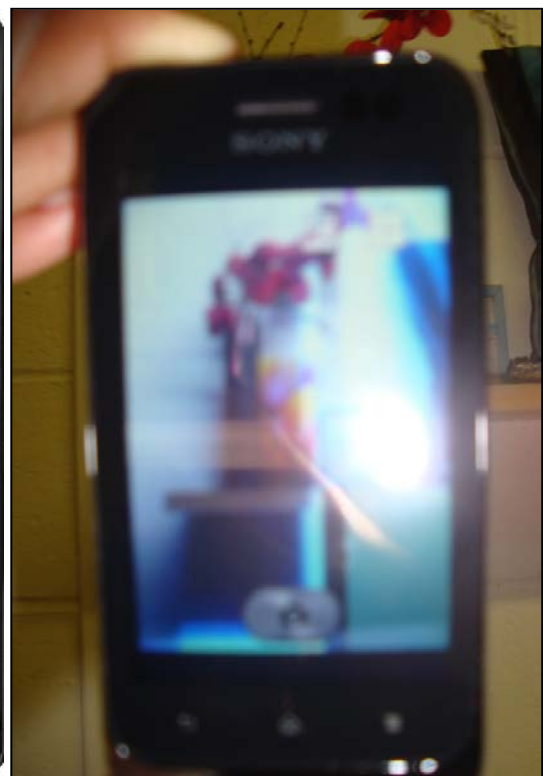


Figure 20: Taking a picture

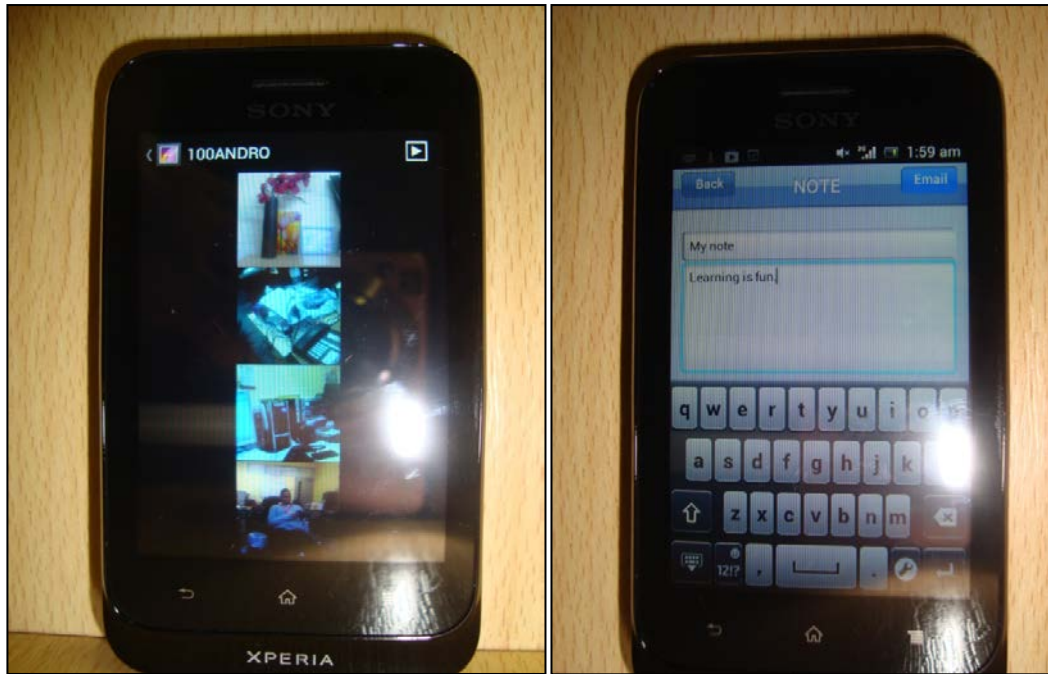


Figure 21: Photo Gallery with picture taken **Figure 22:** Creating a note

M-15: Accessing Note Taker

Selecting the Note Taker menu option from the Student Tools Screen seen in **Figure 19** will display the Note Taker screen seen in **Figure 22**. This feature allows students to take notes using the mobile application and email it as an attachment to an email address. These notes are also saved on the phone so that they are retrieved when returning to the Note Taker. The user can enter a note title and its contents into the text boxes using the on-screen keyboard provided seen in **Figure 22**.

This note can be emailed by pressing the Email button which will display the screen in **Figure 23** allowing the user to select which email service they want to use. Selecting Gmail will add the note as an attachment when sending the email as seen in **Figure 24**. You can return to UoB MOBILE at any time by pressing the back navigation button on the phone itself. Doing so without sending the message will save this message in your drafts folder.

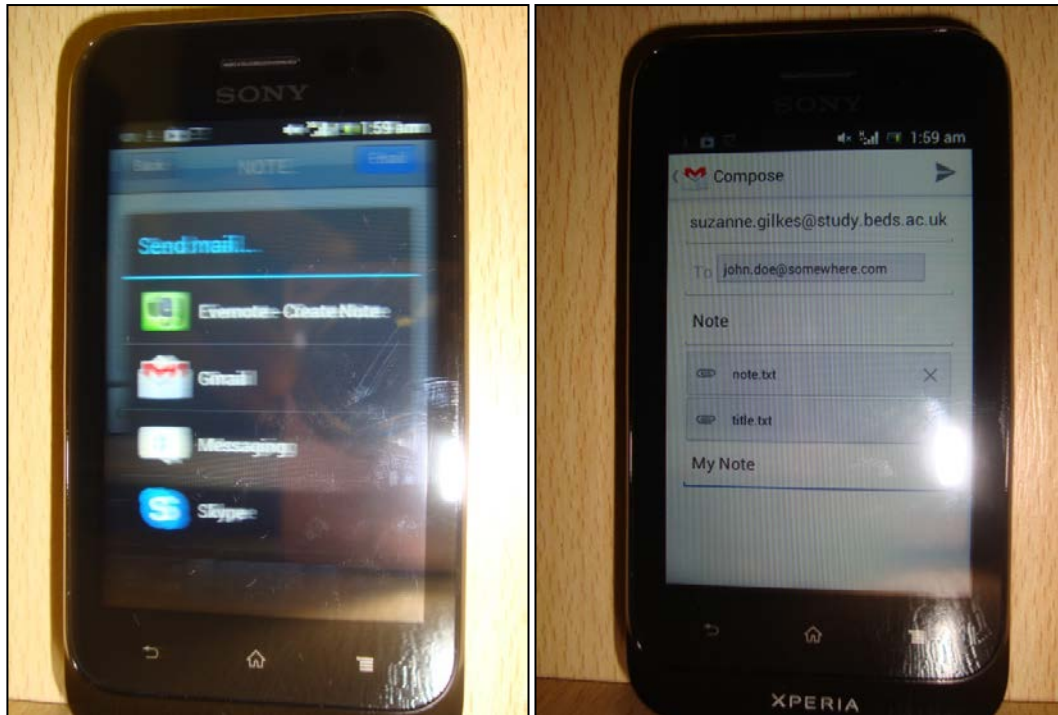


Figure 23: User selects an email service **Figure 24:** Sending the note via Gmail

M-16: Accessing UoB MOBILE on Facebook

UoB MOBILE on Facebook, is a Facebook group page that allows students to collaborate and share information regarding academics in a group environment. From the Note Taker Screen pressing the back button will take the user to the Student Tools Screen. Selecting the UoB MOBILE on Facebook menu option displays the UoB MOBILE Facebook group page which allows the user to make posts etc as seen in **Figure 25**.

M-17: Accessing About Us

The About Us feature gives a brief summary of the purpose of the mobile application and allows the user to rate the app based on their user experience as well as provide optional feedback. From the UoB MOBILE on Facebook Screen, the user can navigate to the Main menu screen by pressing the back button which will take the user to the Student Tools Screen. From the Student Tools Screen the user can press the Campus Life tab button which will take the user to the Campus Life Screen. At the Campus Life Screen pressing the caret button will return the user to the Main menu Screen as seen in **Figure 2**. To access the About Us feature press the screen

overflow button and select the About Us menu option as seen in **Figure 3**. Selecting the About Us menu option will display the About Us Screen as seen in **Figure 26**.

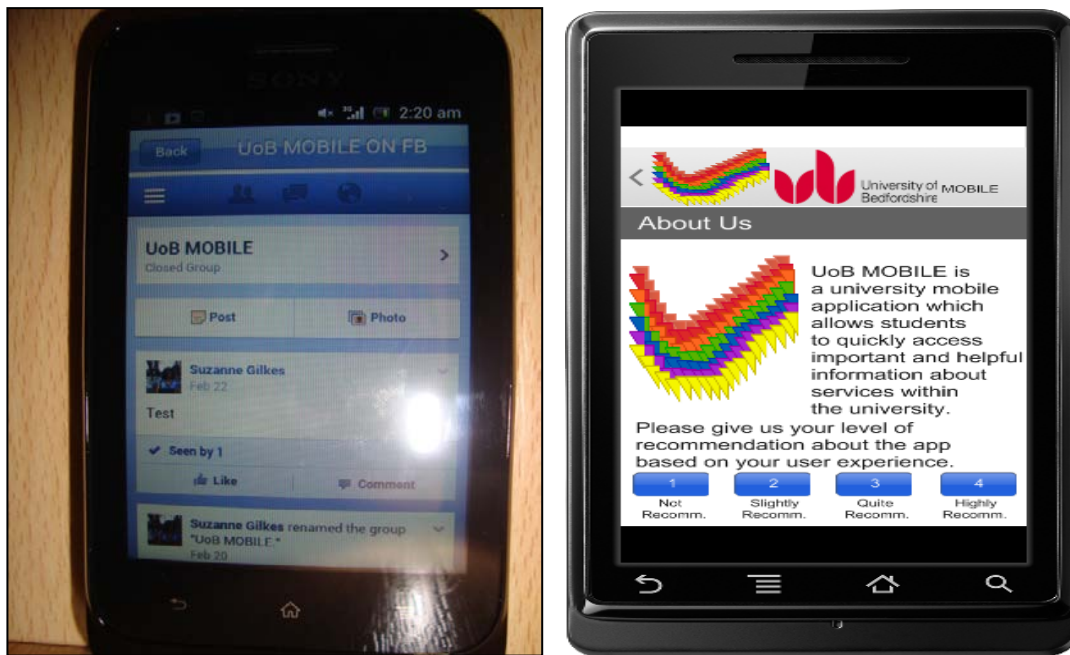


Figure 25: UoB MOBILE on Facebook Screen **Figure 26:** About Us Screen

The user can send a rating for the mobile application to the developer via email by pressing one of the buttons labelled with a rating. Selecting any of the buttons will display the screen allowing the user to select an email service as seen in **Figure 23**. Gmail was selected in this example which then enables the user to send the rating to the developer and include optional feedback as seen in **Figure 27**. You can return to UoB MOBILE at any time by pressing the back navigation button on the phone itself. Doing so without sending the message will save this message in your drafts folder.

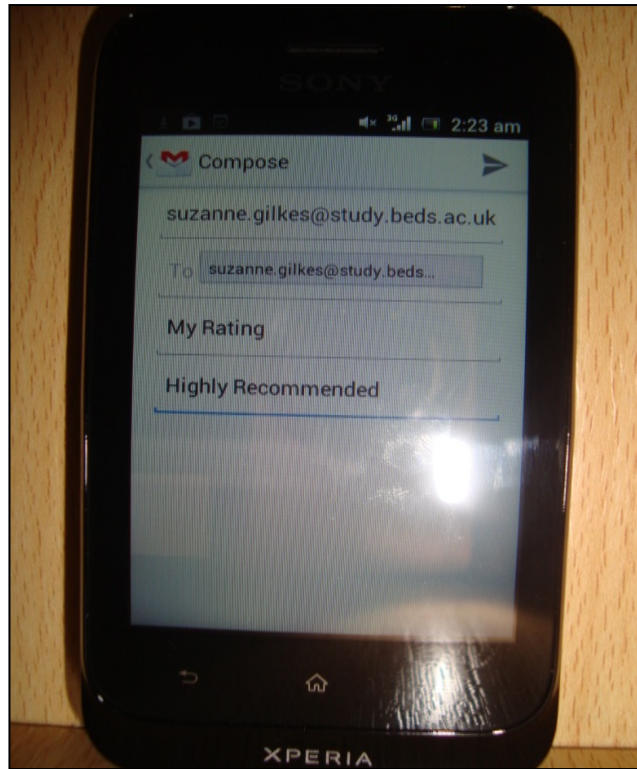


Figure 27: Sending the rating via Gmail

APPENDIX N: POSTER AND ARTEFACT CD



University of Bedfordshire

A university mobile application allowing students to quickly access important and helpful information about services within the university

Student Name: Suzanne Gilkes

Supervisor Name: Dr. Herbert Daly

Course: BSc (Hons) Computer Science and Software Engineering

Student Number: 1000985

Introduction

The University of Bedfordshire's strategic plan is built on enabling students to benefit from a transformative learning experience [5]. Mobile applications used for mobile learning creates a transformative learning experience [3]. Informal learning has a stronger leverage over formal learning as part of a mobile learning method in achieving this. The university has deployed "BREQ To Go" for formal learning. It also has mobile applications which support informal learning, but it lacks a mobile application which supports informal learning whilst bridging the gap between informal and formal learning.




Artefact Development

All development work was done in iterations based on XP because "the best indication the customer has as to whether the system is on track both with regards to delivery and features is to shorten the build cycle"[1]. With the exception of the first iteration, each iteration length was one week in which unit testing, coding, integration, acceptance testing and tracking of the iteration were conducted. Coding for the mobile application was done in the Lua programming language and compiled via the Corona SDK development platform.



Testing

Test Driven Development was one of XP's activities utilized where unit tests were written first using Lunatest before any code was developed. Unit tests were then executed after code was written for the test to pass successfully. Customers also wrote and executed acceptance tests which marked the closure of an iteration.

Evaluation

All acceptance tests which evaluated customer requirements on functionalities and usability passed successfully according to the customer, making the system fit-for-purpose. Development practices integrated have also compared favorably when benchmarked against best practices utilized in the industry. The system has a high degree of acceptance and could therefore be marketed by the university to its students.



Research

Mobile applications are designed to address a need. The university has several reasons to deploy a mobile application which addresses a need with the support of informal learning which include:

- It will address a student need to have access to university services on the go [2].
- It may also place the university at the forefront of the pedagogical practice [4].

Requirements Gathering

Requirements were gathered via user stories.

User Story	Effort	Iteration 0	Iteration 1	Iteration 2	Iteration 3	Iteration 4
Test framework setup	1 day	*				
Creation of database	1 day	*				
ER diagram						
Access Student Support Services	3 days		*			
View Campus Alerts	2 days		*			
View Events	2 days			*		
Access Note Taking	1 day			*		
Access Careers	1 day			*		
Access UoB MOBILE on Facebook	1 day			*		
Access Pocket Guide	3 days				*	
Access Alerts UoB	2 days				*	
Passionate Alerts	3 days					*
Access Help	2 days					*

Conclusions and Further Development

The project met its goal and all of its six objectives. Suggestions for further development include integrating the mobile application with internal systems in order to improve the genuinely useful functionality by incorporating additional desired features requested by students.

References

[1] Beale, L. (2009) *Don't Think You're a Programmer in 24 Hours*. Trans-Pix. [Online]. Available at: <http://www.transpix.co.uk/Research/Don'tThinkYoureAProgrammerIn24Hours.pdf> (Accessed 04 May 2013)

[2] Housman (2011) *Mobile IT in Higher Education*. 2011. Available at: <http://www.housman.co.uk/Research/ITinHigherEducation.pdf> (Accessed 04 May 2013)

[3] Gellies, L. (2004) 'Mobile learning in the 21st century: benefits for learners', *The Knowledge Times*, 6. [Online]. Available at: http://www.knowledge-times.com/issue6/issue6_gellies.pdf (Accessed 04 May 2013)

[4] Ockelton, P. and Evans, G. (2010) 'Opportunities for Mobile Applications to Enhance Informal Learning in University Environments', *Proceedings of World Conference on E-Learning in Corporate, Government, Healthcare, and Higher Education 2010*, Honolulu, Hawaii, 18-19 July. AACE. [Online]. Available at: http://www.aace.org/Research/WorldConference2010/Proceedings/WorldConference2010_OckeltonEvans.pdf (Accessed 04 May 2013)

[5] University of Bedfordshire (no date) *Values, Vision and Mission*. Available at: <http://www.beds.ac.uk/bedsabout/missionandvision/> (Accessed 04 May 2013)

The CD attached to the paper copy of the dissertation contains all development source code for the artefact and the executable file used to install UoB MOBILE on the Android Smartphone.